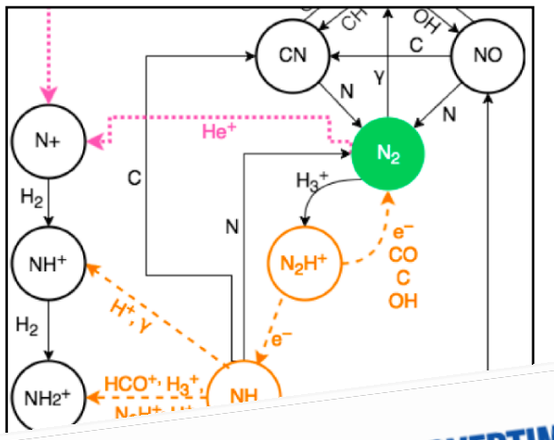
Effective Data Visualization

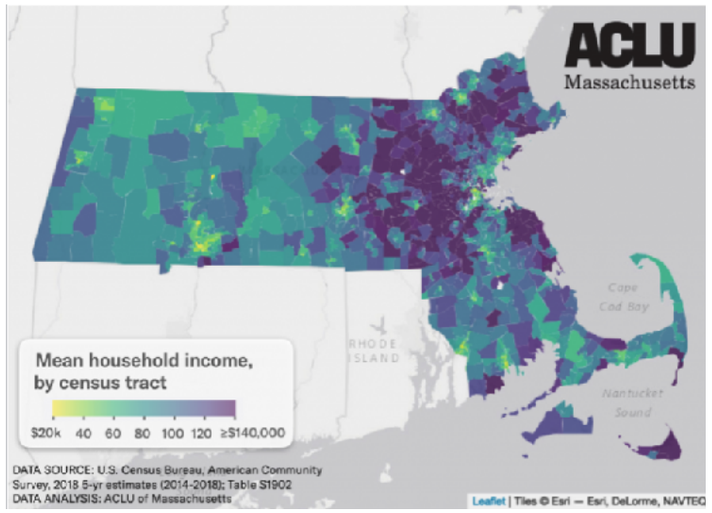
INFO 285 - Public Interest Technology Clinic

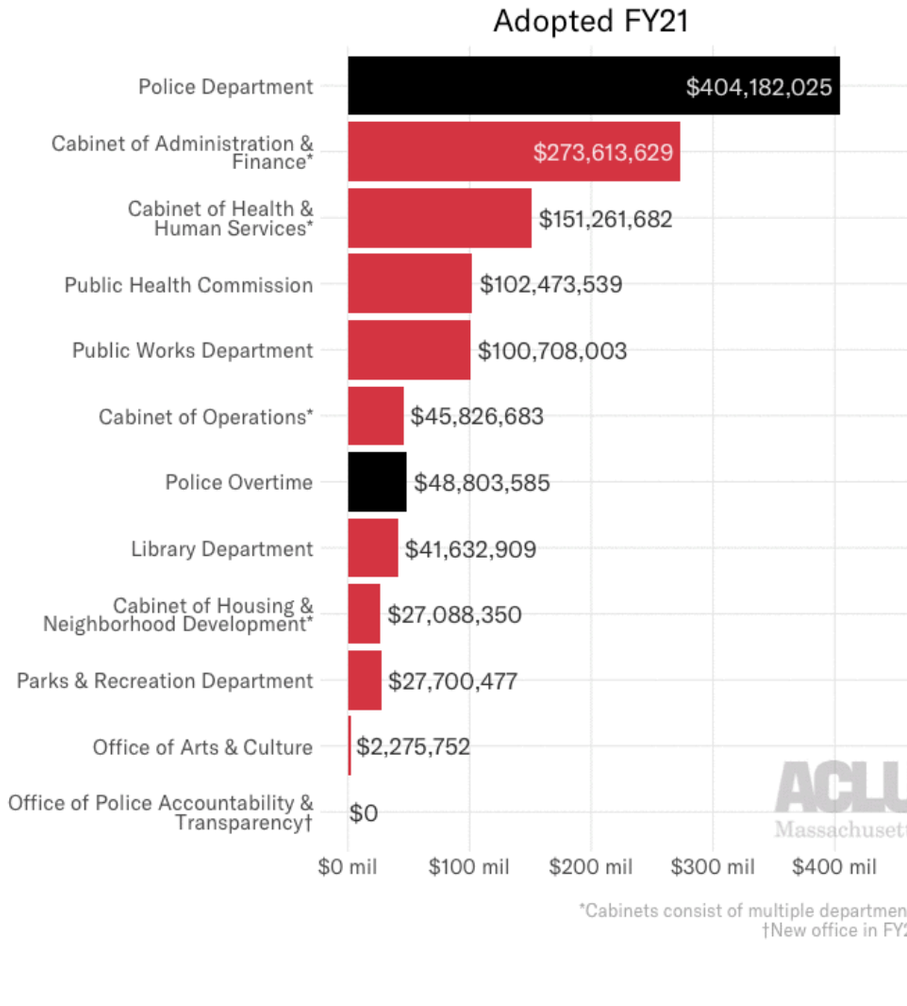
November 2, 2026

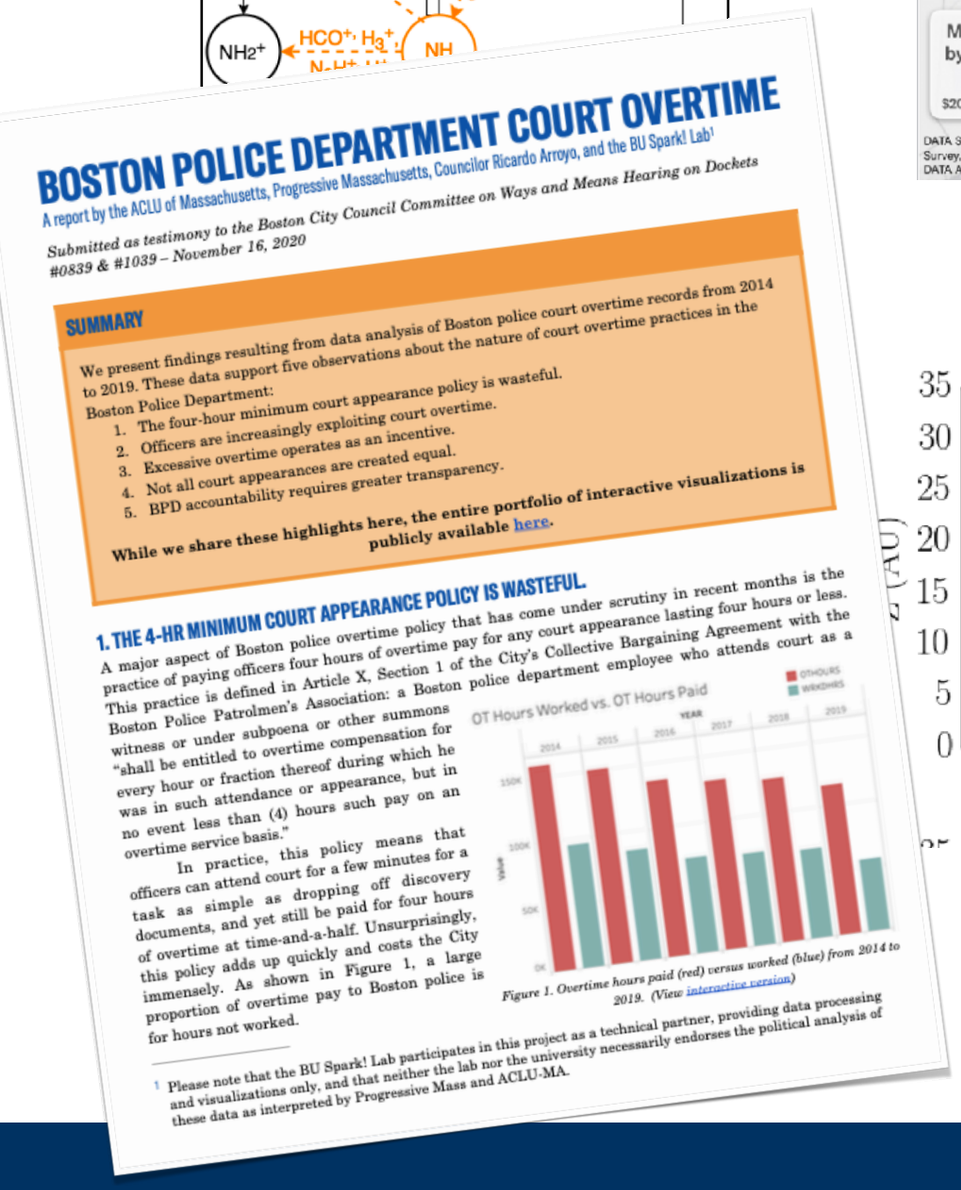
Berkeley SCHOOL OF INFORMATION

Data viz experience









BOSTON POLICE DEPARTMENT COURT OVERTIME

A report by the ACLU of Massachusetts, Progressive Massachusetts, Councilor Ricardo Arroyo, and the BU Spark! Lab[1]

Submitted as testimony to the Boston City Council Committee on Ways and Means Hearing on Dockets #0839 & #1039 – November 16, 2020

SUMMARY

We present findings resulting from data analysis of Boston police court overtime records from 2014 to 2019. These data support five observations about the nature of court overtime practices in the Boston Police Department:

1. The four-hour minimum court appearance policy is wasteful.
2. Officers are increasingly exploiting court overtime.
3. Excessive overtime operates as an incentive.
4. Not all court appearances are created equal.
5. BPD accountability requires greater transparency.

While we share these highlights here, the entire portfolio of interactive visualizations is publicly available here.

1. THE 4-HR MINIMUM COURT APPEARANCE POLICY IS WASTEFUL.

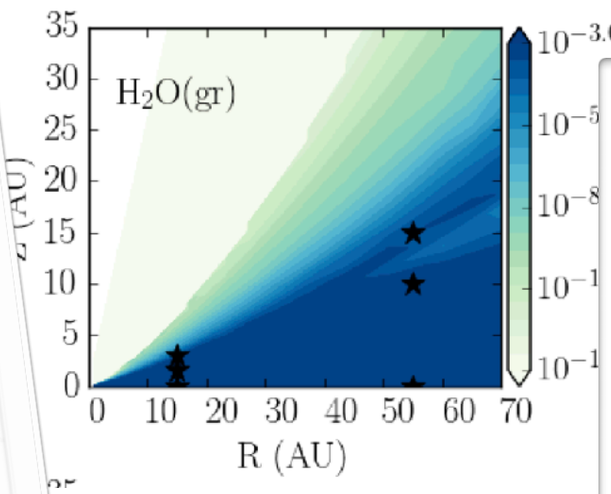
A major aspect of Boston police overtime policy that has come under scrutiny in recent months is the practice of paying officers four hours of overtime pay for any court appearance lasting four hours or less. This practice is defined in Article X, Section 1 of the City's Collective Bargaining Agreement with the Boston Police Patrolmen's Association: a Boston police department employee who attends court as a witness or under subpoena or other summons "shall be entitled to overtime compensation for every hour or fraction thereof during which he was in such attendance or appearance, but in no event less than (4) hours such pay on an overtime service basis."

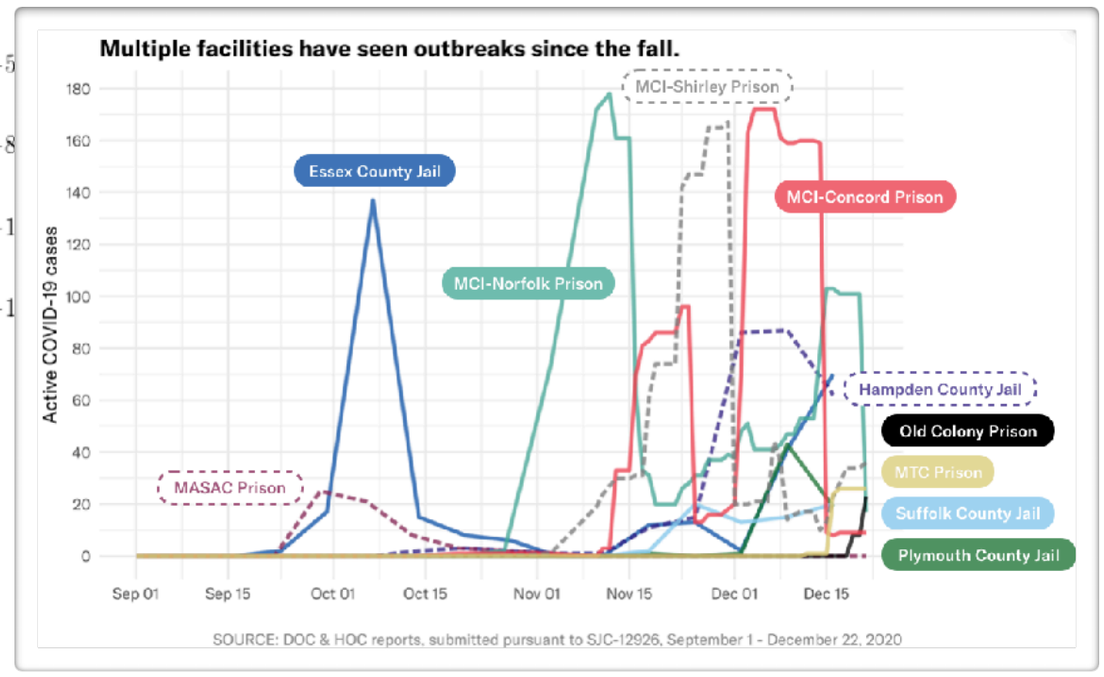
In practice, this policy means that officers can attend court for a few minutes for a task as simple as dropping off discovery documents, and yet still be paid for four hours of overtime at time-and-a-half. Unsurprisingly, this policy adds up quickly and costs the City immensely. As shown in Figure 1, a large proportion of overtime pay to Boston police is for hours not worked.

OT Hours Worked vs. OT Hours Paid

Figure 1. Overtime hours paid (red) versus worked (blue) from 2014 to 2019. (View interactive version)

[1] Please note that the BU Spark! Lab participates in this project as a technical partner, providing data processing and visualizations only, and that neither the lab nor the university necessarily endorses the political analysis of these data as interpreted by Progressive Mass and ACLU-MA.





Today we will explore...

- why data visualization is important
- how to identity the different components and types of data visualizations
- how to interpret visualizations critically
- how to intentionally tell a story by visualizing data

To learn more, consider taking INFO 247: Information Visualization & Presentation

All resources shared today are available at:

https://github.com/laurenmarietta/effective-data-viz

This presentation is adapted from a workshop designed by Kira Tebbe in February 2020



Workshop outline

- What is data visualization 🗣️
- Anatomy of a data visualization
- Types of data visualization
- Roast! This! Viz! 🗣️
- ✨ 10 Best Practices ✨
- Improve! This! Viz!
- *(maybe)* Exercises 🗣️
- Interactivity, inspiration, and learning more

What does the term "data visualization" mean to you? Why is it important?

Why is data visualization important?

data = information

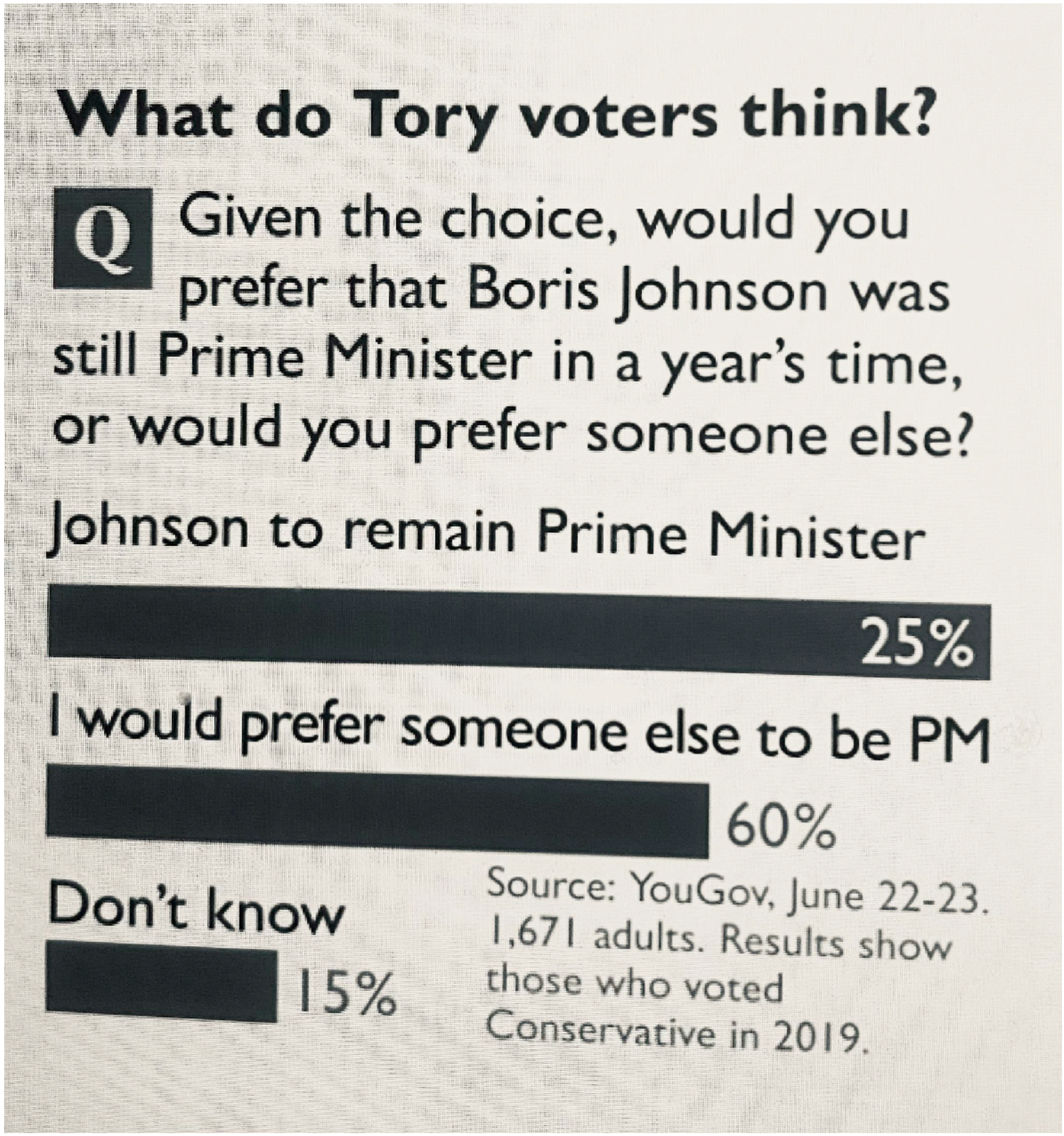
data visualization = communication

good data visualization = effective communication

bad data visualization = miscommunication

What do mediocre visualizations lead to?

- confusion and frustration
- loss of attention
- poor impression of work quality
- misinterpretation of findings



Source: Will Bailey-Watson on Twitter



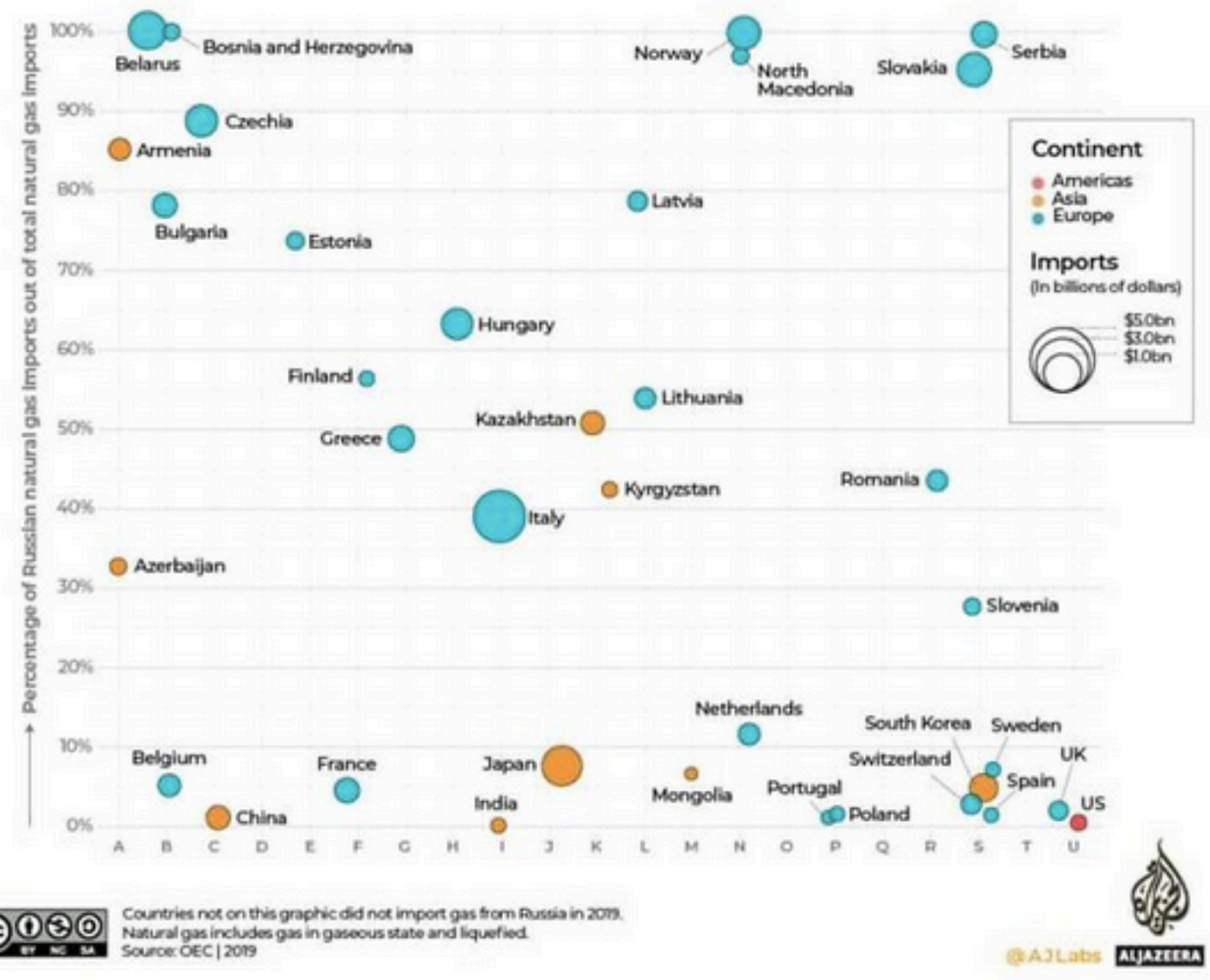
Ben Stanley
@BDStanley

Incredible x-axis here.

ENERGY

Which countries directly import the most Russian natural gas?

Russia is the world's third highest exporter of natural gas after Australia and Qatar.
In 2019, **37 countries bought Russian gas worth $24.5bn**.



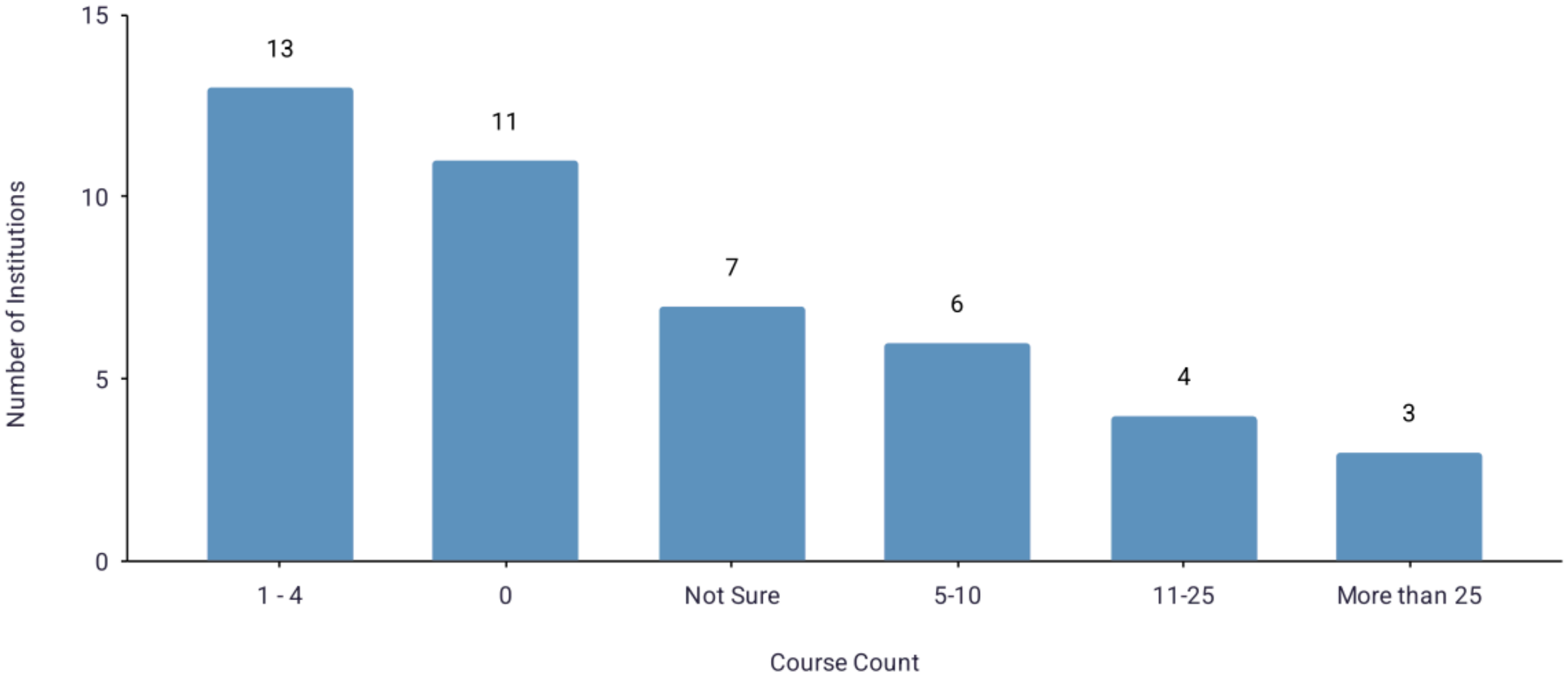
Countries not on this graphic did not import gas from Russia in 2019.
Natural gas includes gas in gaseous state and liquefied.
Source: OEC | 2019

@AJLabs ALJAZEERA

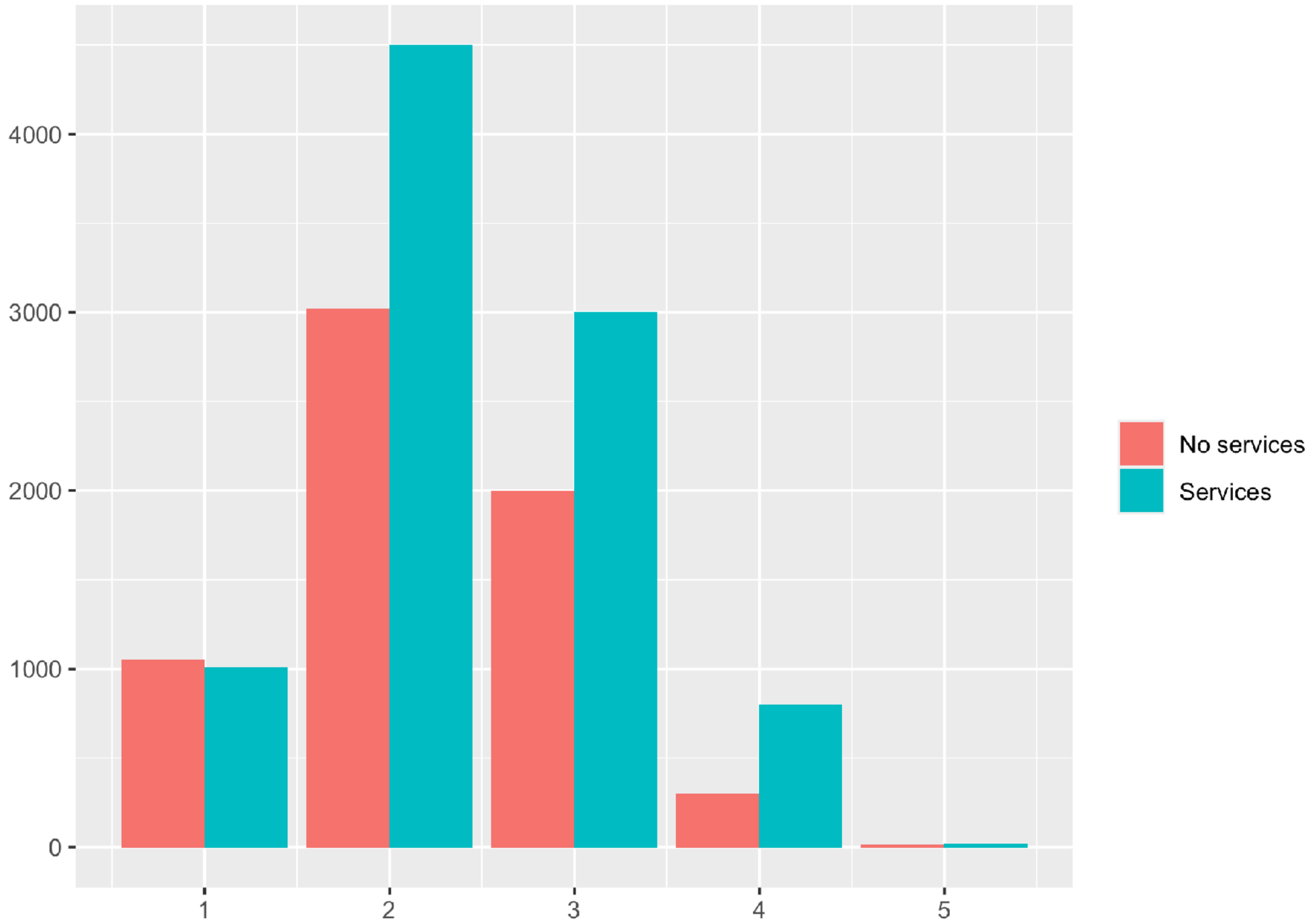
Source: Ben Stanley on Twitter

Figure 4.1 - AI-Related Course Availability Remains Limited Across Institutions

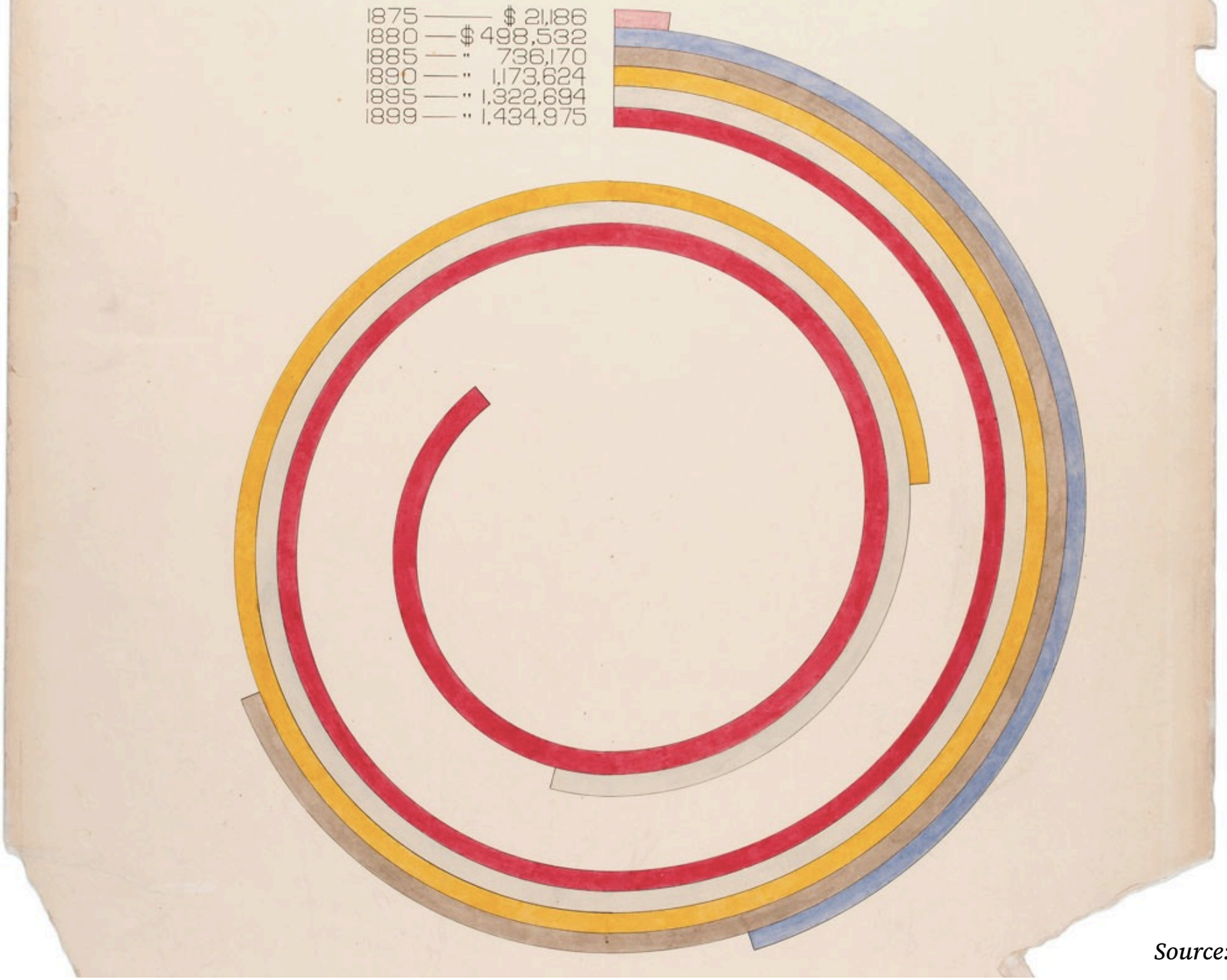
AI-Related Courses Offered as of April 2025



Source: "Landscape of AI across California HEIs"; SJSU 2026


4000
3000
2000
1000
0
1
2
3
4
5
No services
Services

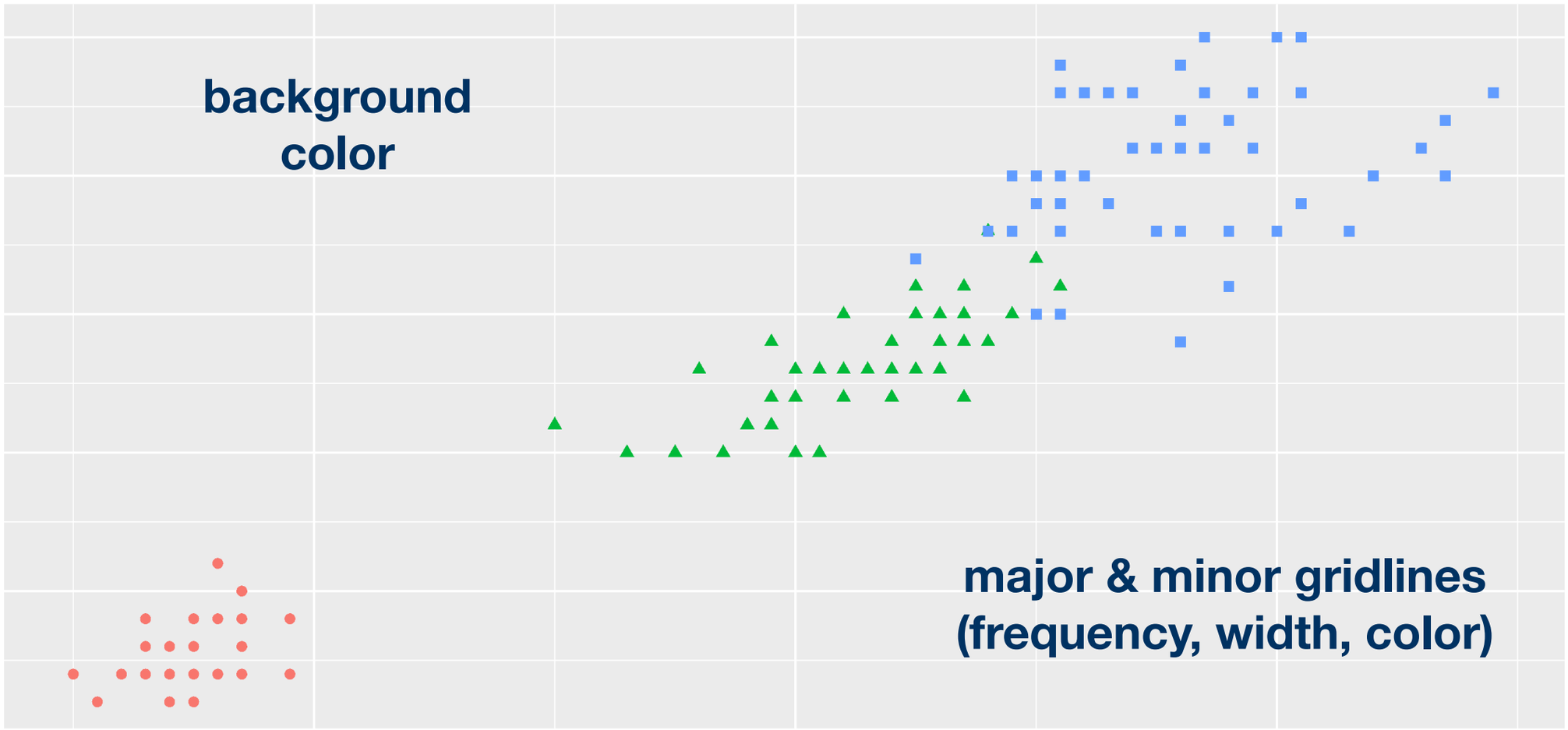
ASSESSED VALUE OF HOUSEHOLD AND KITCHEN FURNITURE OWNED BY GEORGIA NEGROES.



Source: W.E.B. Du Bois, ca. 1900

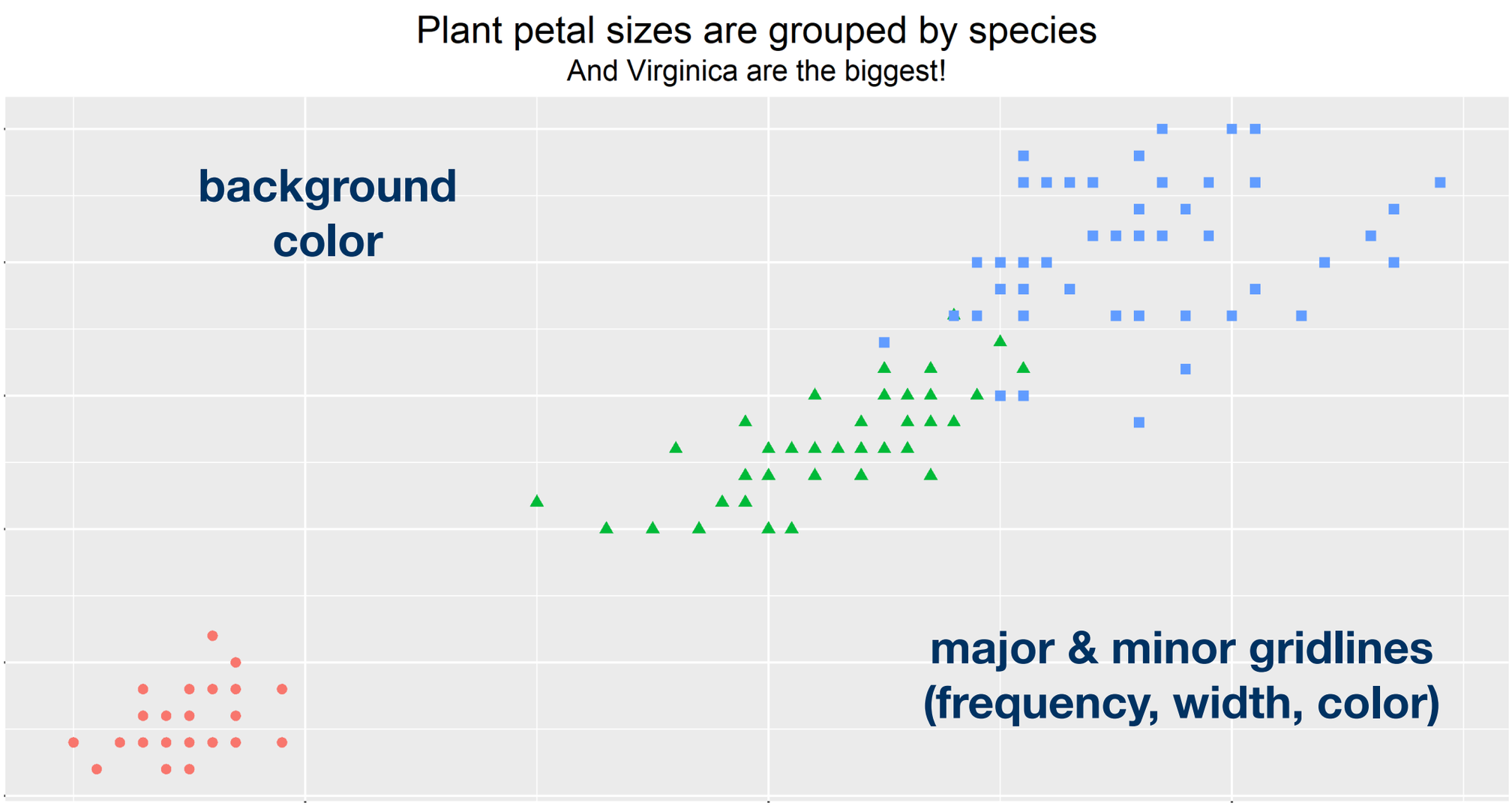
Anatomy of a data visualization

plot type & aesthetics
(colors, sizes, shapes, opacity)



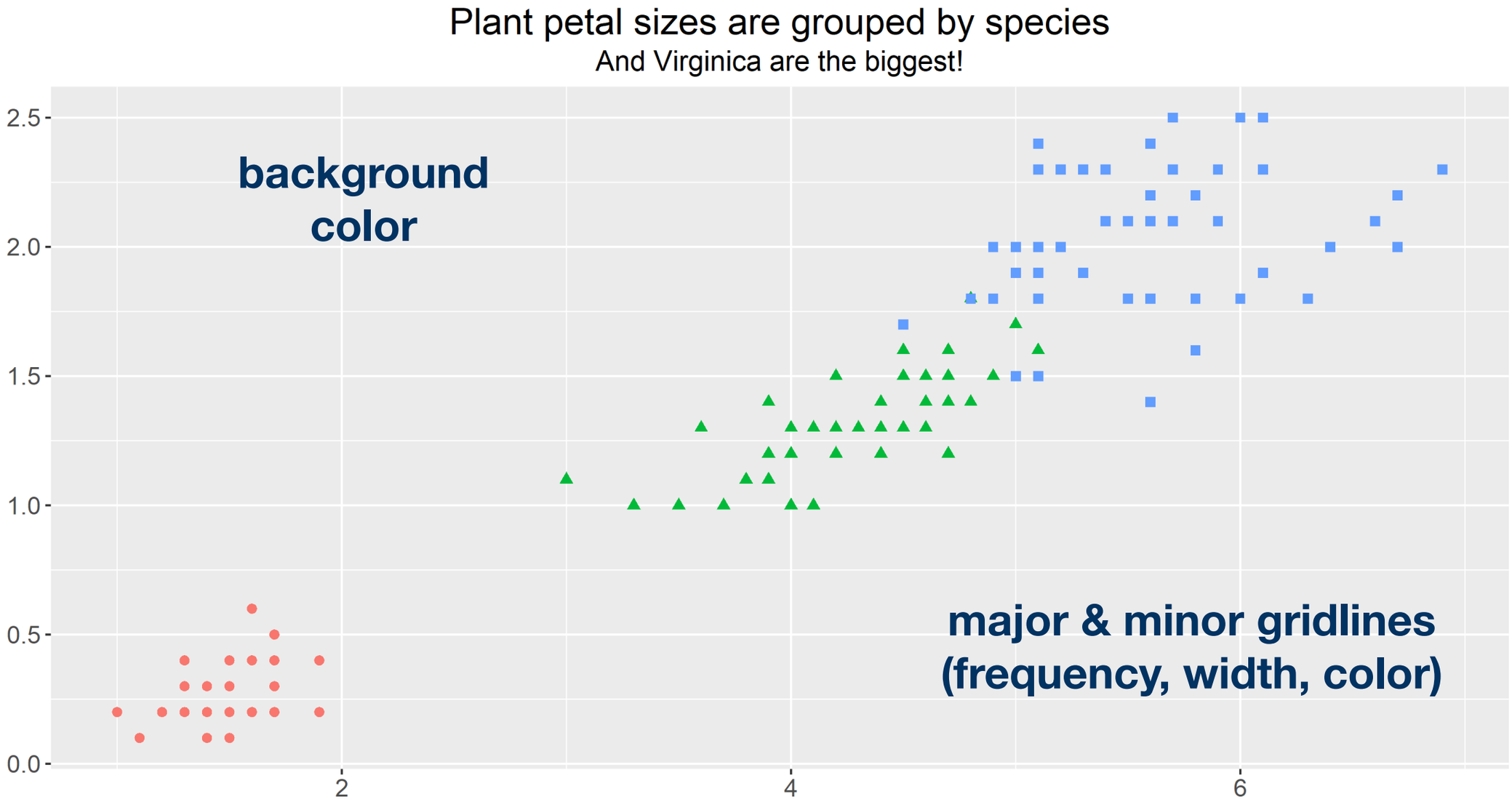
title & subtitle
(size, font, color, location)

plot type & aesthetics
(colors, sizes, shapes, opacity)



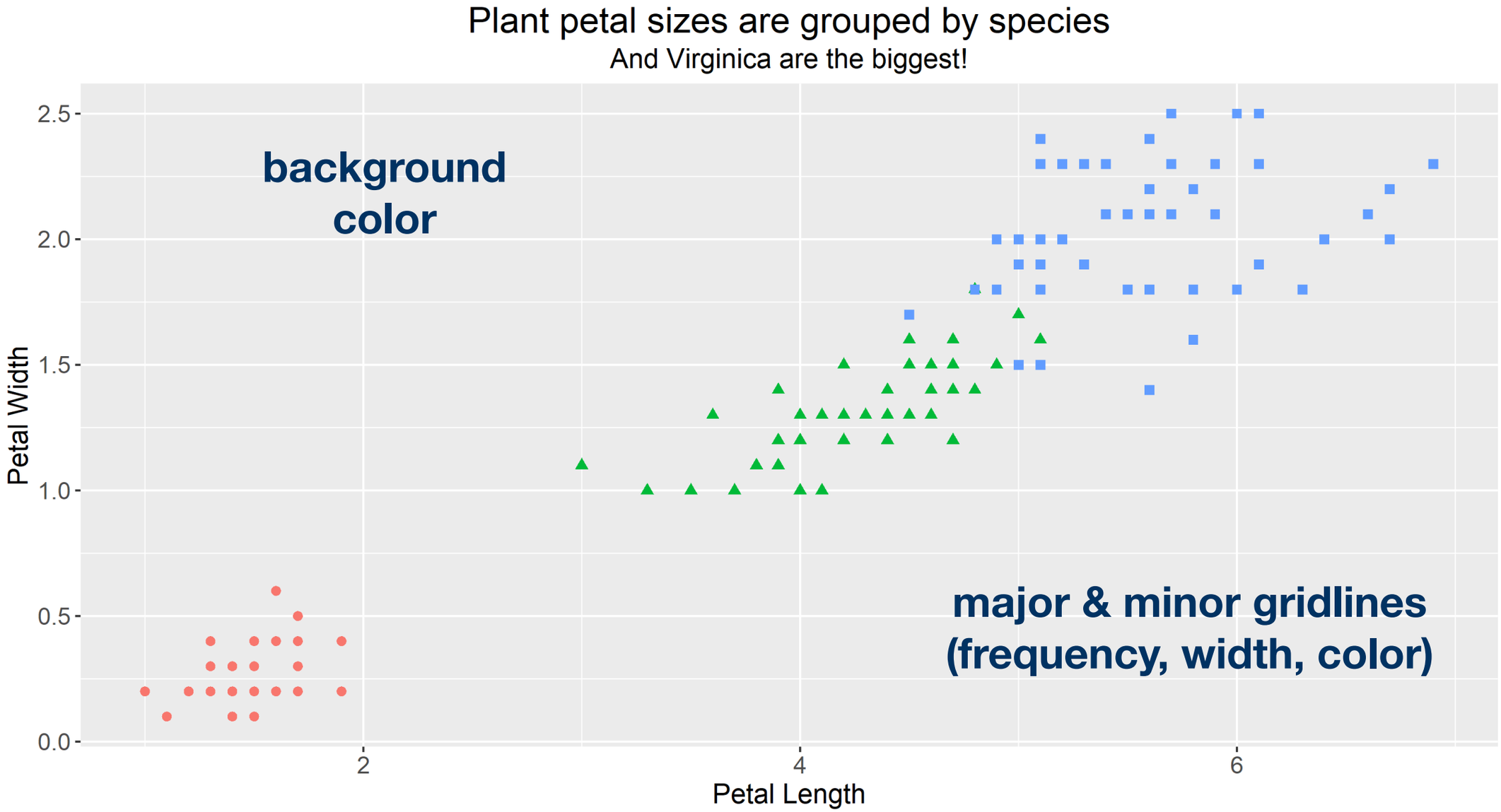
title & subtitle
(size, font, color, location)

plot type & aesthetics
(colors, sizes, shapes, opacity)



title & subtitle
(size, font, color, location)

plot type & aesthetics
(colors, sizes, shapes, opacity)

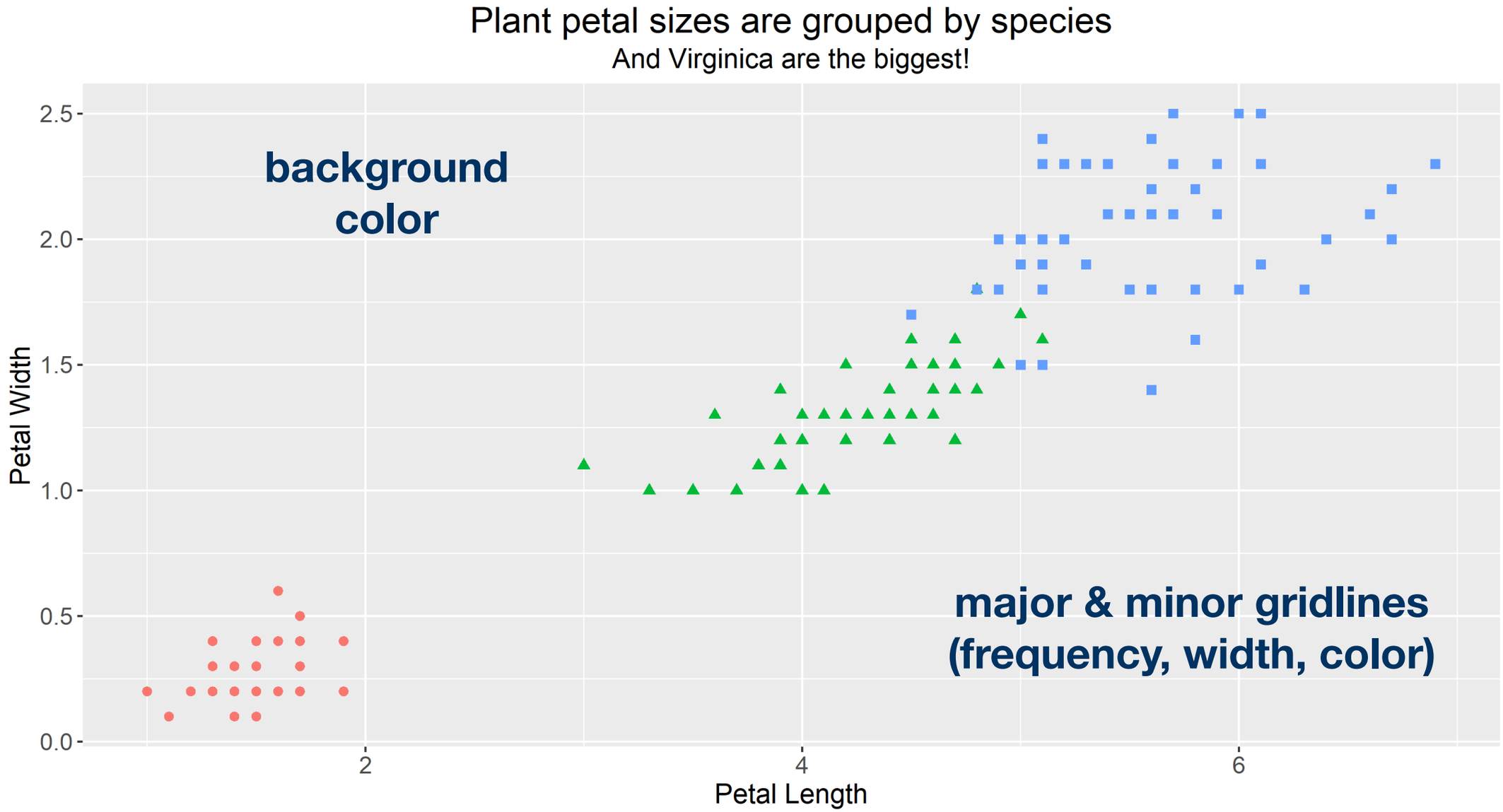


X & Y axis labels
(size, font, color, angle)

X & Y tick marks
(frequency, size, font, color, angle)

title & subtitle
(size, font, color, location)

plot type & aesthetics
(colors, sizes, shapes, opacity)



background color

legend & legend title (location, size, font, color, existence)

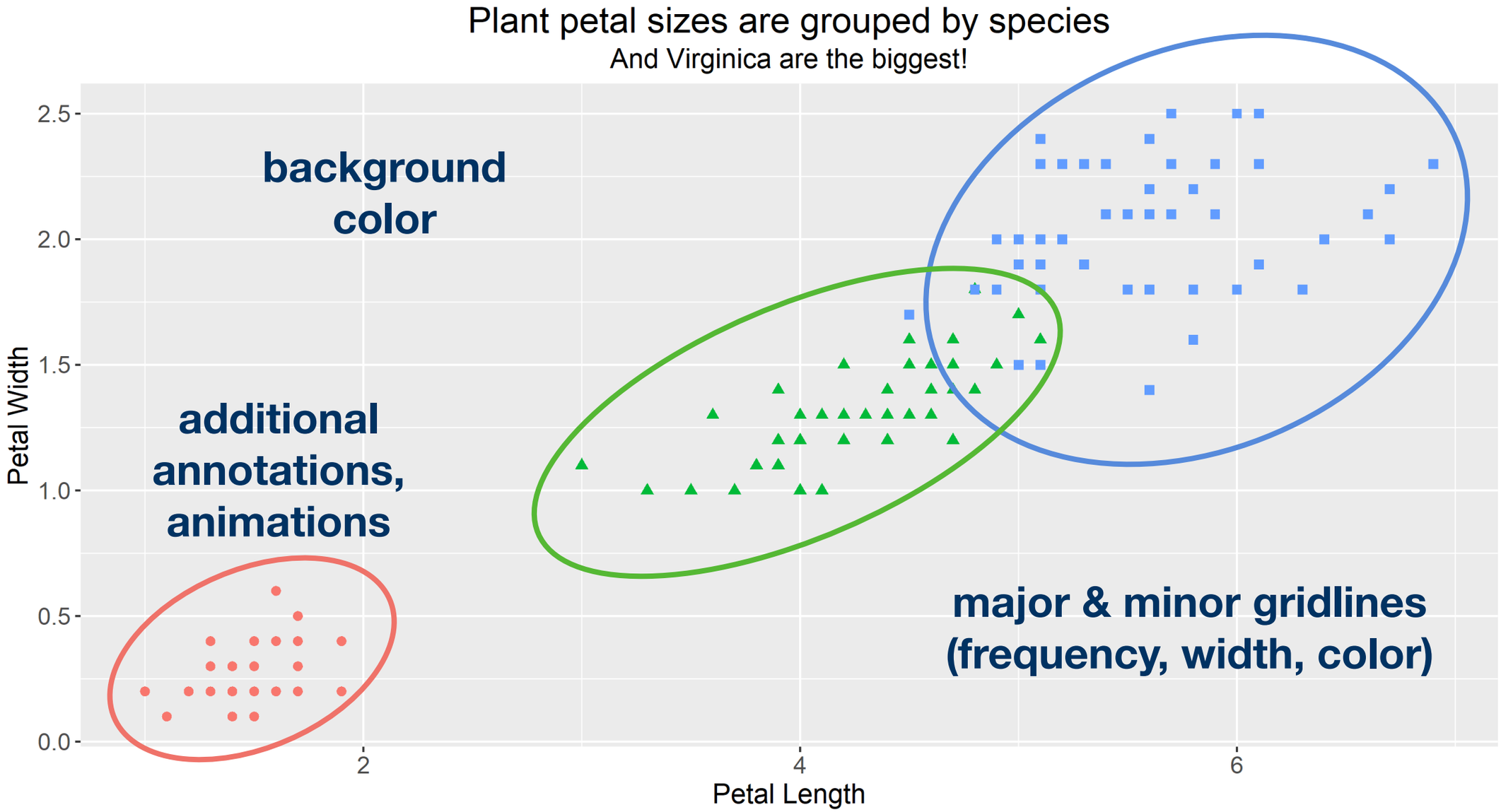
major & minor gridlines
(frequency, width, color)

X & Y axis labels
(size, font, color, angle)

X & Y tick marks
(frequency, size, font, color, angle)

title & subtitle
(size, font, color, location)

plot type & aesthetics
(colors, sizes, shapes, opacity)



background color

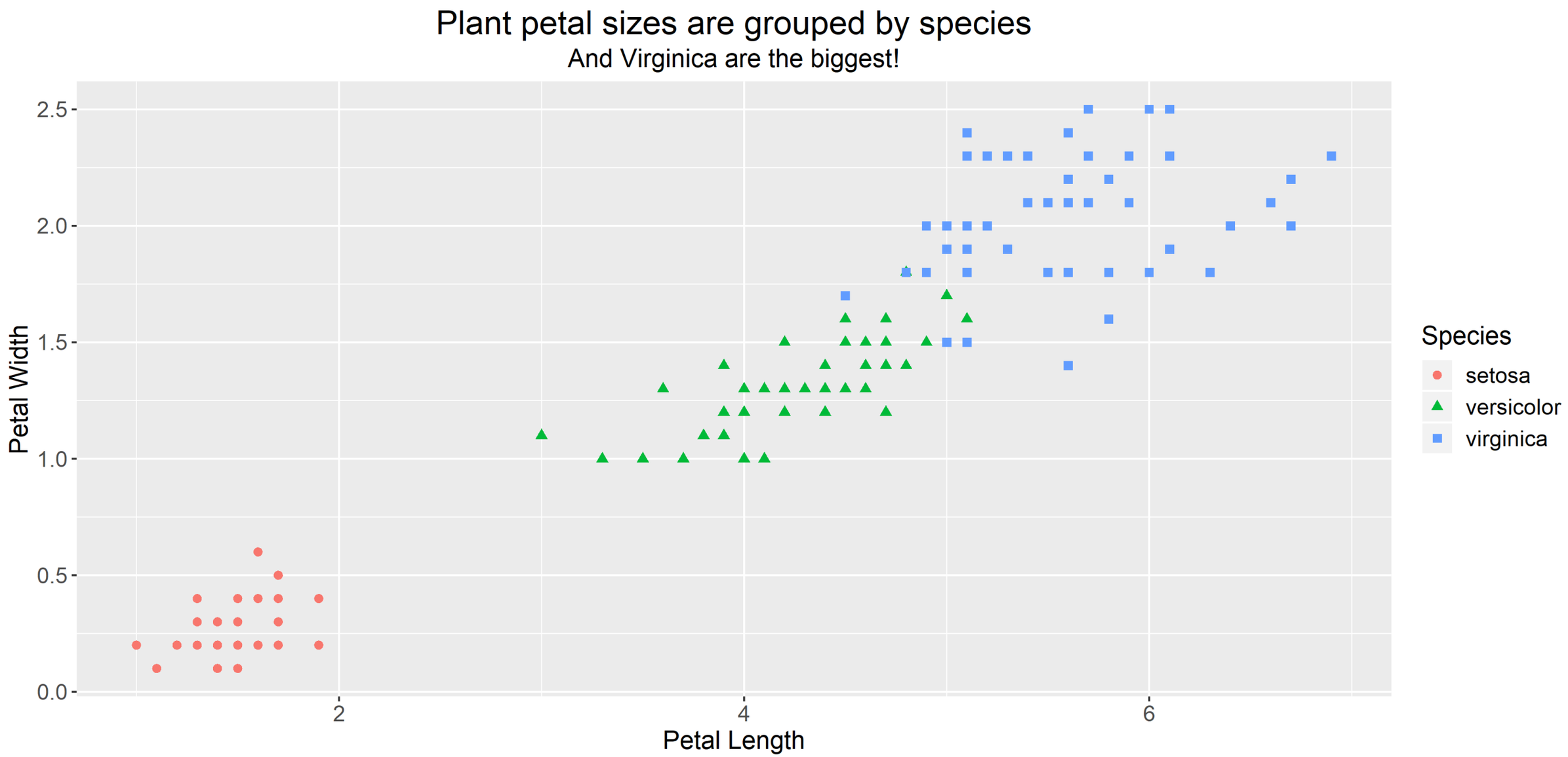
additional annotations, animations

major & minor gridlines
(frequency, width, color)

legend & legend title
(location, size, font, color, existence)

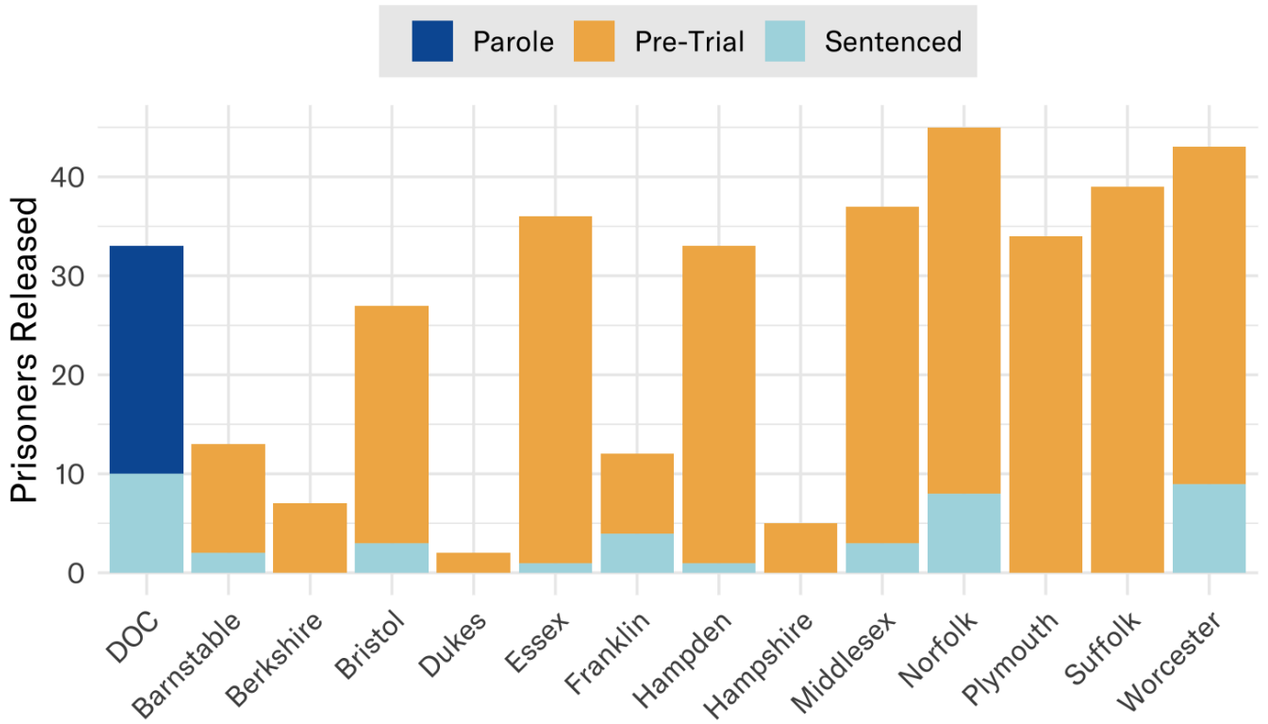
X & Y axis labels
(size, font, color, angle)

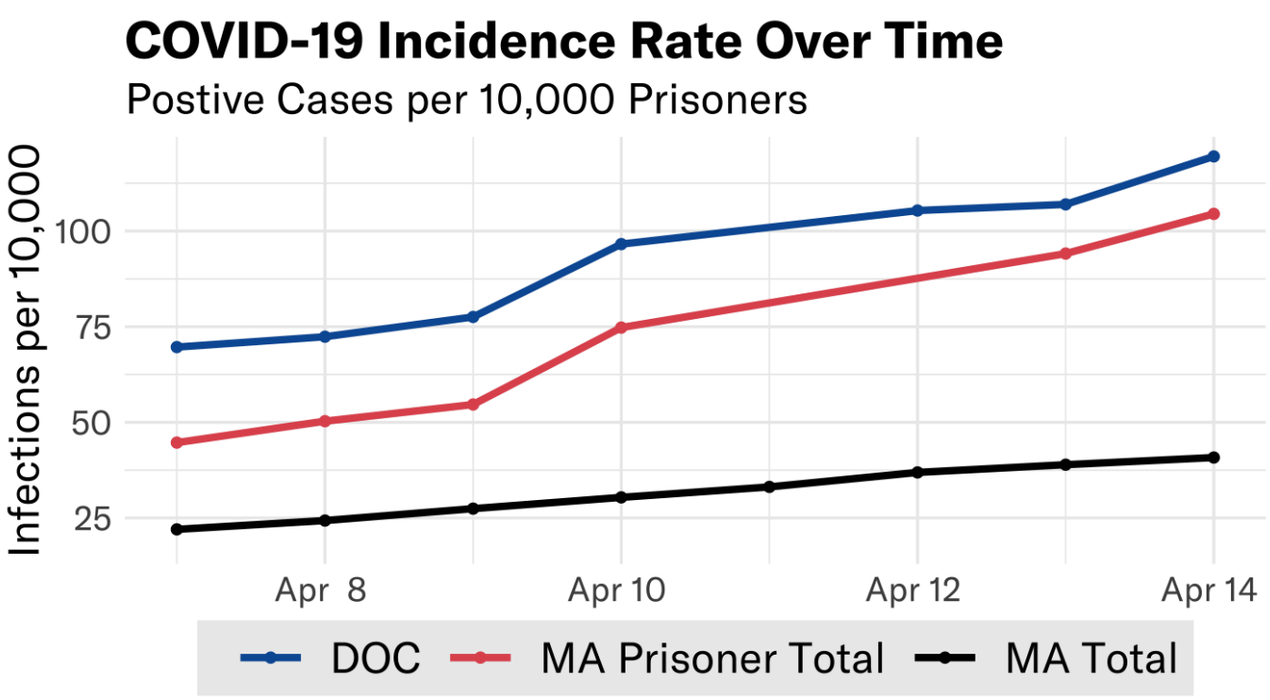
X & Y tick marks
(frequency, size, font, color, angle)

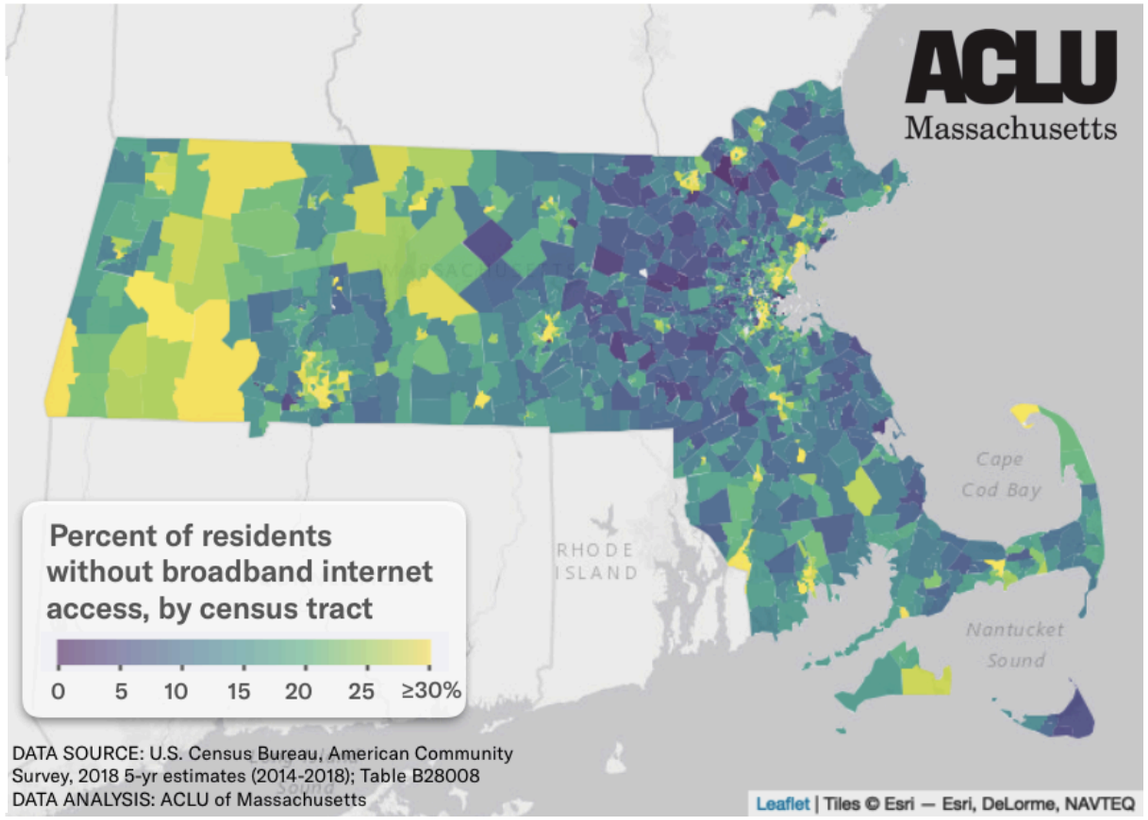

Plant petal sizes are grouped by species
And Virginica are the biggest!
Petal Width
2.5
2.0
1.5
1.0
0.5
0.0
2
4
6
Petal Length
Species
setosa
versicolor
virginica

Types of data visualization

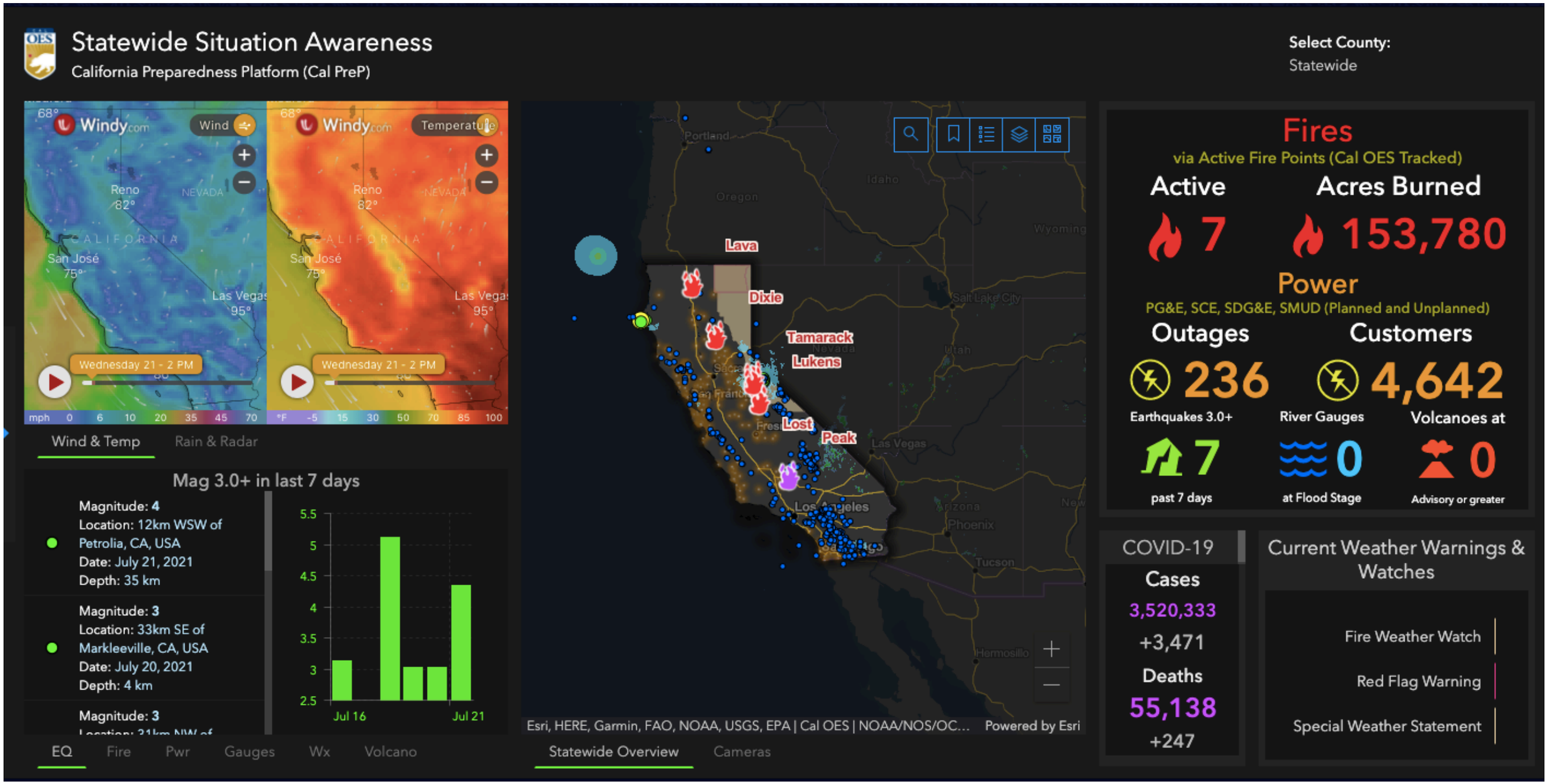
Rank	Boston Police Officer	FY20 Earnings	Change From FY19 Rank	Change From FY19 Earnings
1	Smith, Sean P.	$365,001	↑ +3	+$22,378
2	Lee, Waiman	$360,143	↑ +3	+$20,628
3	Brown, John M.	$345,758	↑ +12	+$28,388
4	Danilecki, John H.	$339,928	↓ -2	-$8,127
5	Barrett, Thomas E.	$339,353	↑ +2	+$7,275
6	McCormick, Kelley J.	$336,898	↑ +5	+$15,922
7	Connolly, Timothy	$334,077	↑ +18	+$31,109
8	Sweeney, Steve	$332,908	↑ +1	+$3,932
9	Cullity, Patrick J.	$329,101	↑ +3	+$9,032
10	Hughes, John C.	$322,637	↑ +31	+$31,203
11	Thomas, Terry J.	$319,001	↑ +9	+$11,134
12	Pusey, Daniel C.	$318,886	↑ +24	+$23,156
13	Webb, Keith A.	$318,141	↑ +1	-$200
14	Facey, Leighton B.	$317,506	↑ +29	+$28,844
15	Kervin, Timothy M.	$314,690	↓ -14	-$40,849
16	Harrington, Mark William	$314,443	↑ +29	+$27,098
17	Ciccolo, Robert W.	$314,026	↑ +9	+$12,894
18	Kozmiski, Therese M.	$313,656	↑ +53	+$35,552
19	Sullivan, Marc	$313,437	↓ -2	+$2,515
20	Gero, Jarrod A.	$311,210	↑ +19	+$17,573





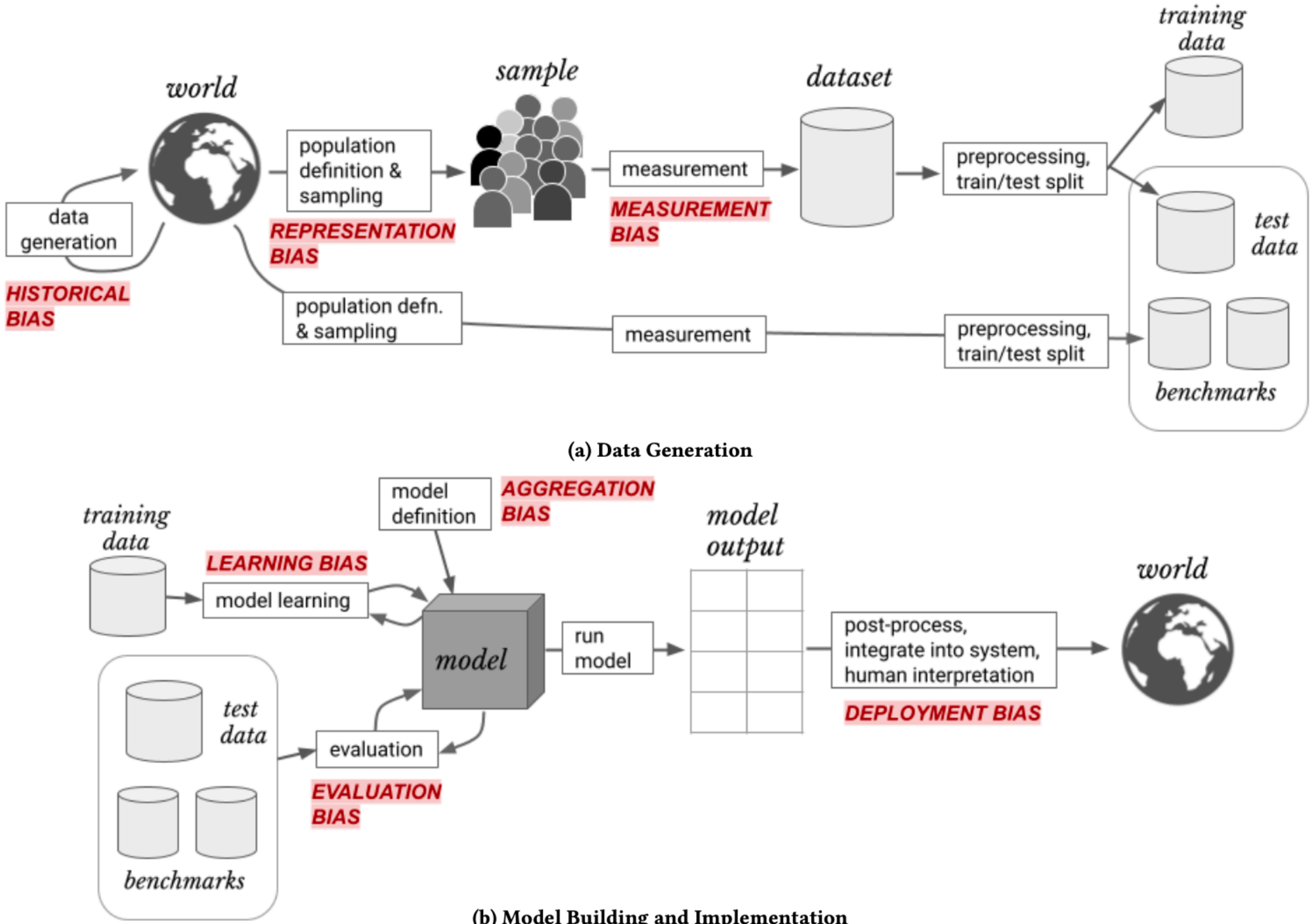


Compound visualizations: dashboards



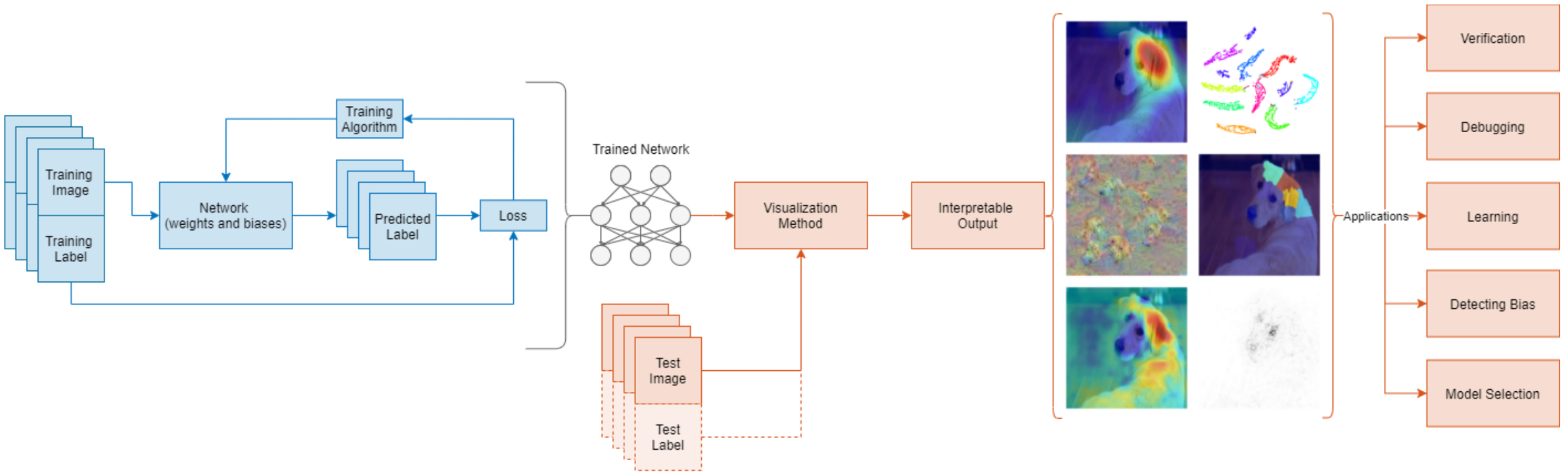
Source: California Governor's Office of Emergency Services Preparedness Platform

Visualizing models: flowcharts



Source: Suresh & Guttag 2021

Visualizing models: flowcharts



Source: MATLAB

aps
ASSOCIATION FOR PSYCHOLOGICAL SCIENCE

Psychological Science in the Public Interest
2021, Vol. 22(3) 110–161
© The Author(s) 2021
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/15291006211051956
www.psychologicalscience.org/PSPI
SAGE

The Science of Visual Data Communication: What Works

Steven L. Franconeri[1], Lace M. Padilla[2], Priti Shah[3], Jeffrey M. Zacks[4], and Jessica Hullman[5]

[1]Department of Psychology, Northwestern University; [2]Department of Cognitive and Information Sciences, University of California, Merced; [3]Department of Psychology, University of Michigan; [4]Department of Psychological & Brain Sciences, Washington University in St. Louis; and [5]Department of Computer Science, Northwestern University

Abstract

Effectively designed data visualizations allow viewers to use their powerful visual systems to understand patterns in data across science, education, health, and public policy. But ineffectively designed visualizations can cause confusion, misunderstanding, or even distrust—especially among viewers with low graphical literacy. We review research-backed guidelines for creating effective and intuitive visualizations oriented toward communicating data to students, coworkers, and the general public. We describe how the visual system can quickly extract broad statistics from a display, whereas poorly designed displays can lead to misperceptions and illusions. Extracting global statistics is fast, but comparing between subsets of values is slow. Effective graphics avoid taxing working memory, guide attention, and respect familiar conventions. Data visualizations can play a critical role in teaching and communication, provided that designers tailor those visualizations to their audience.

Keywords

visual communication, graph comprehension, reasoning, statistical cognition, uncertainty communication, data visualization

This report presents research-backed guidelines for creating powerful and intuitive visualizations oriented toward communicating data to students, coworkers, and the general public. We begin by reviewing guide-

visualizations that communicate clearly and suggest resources for readers interested in learning more.

Data visualizations range from simple graphs in elementary school classrooms, to depictions of uncertainty

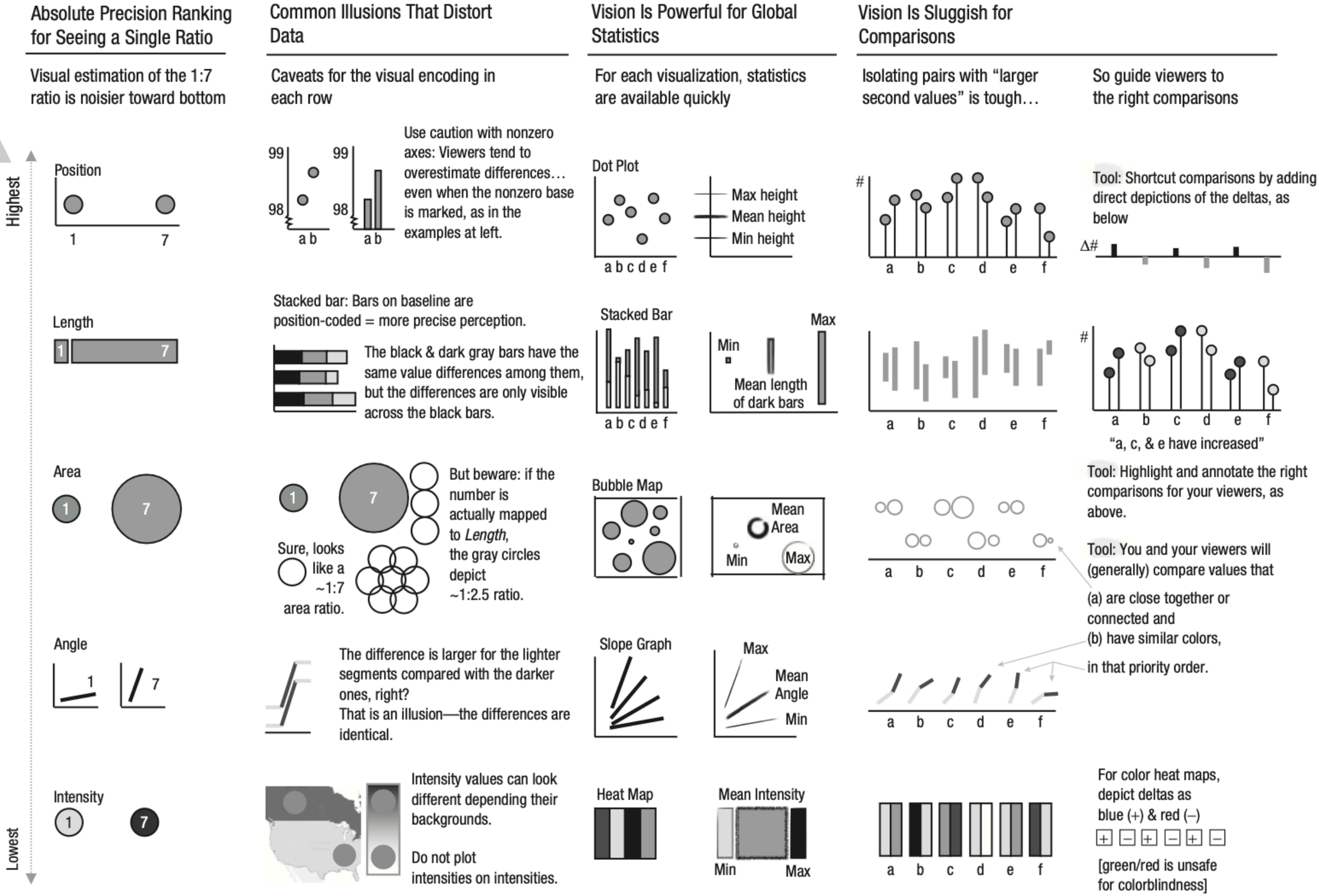
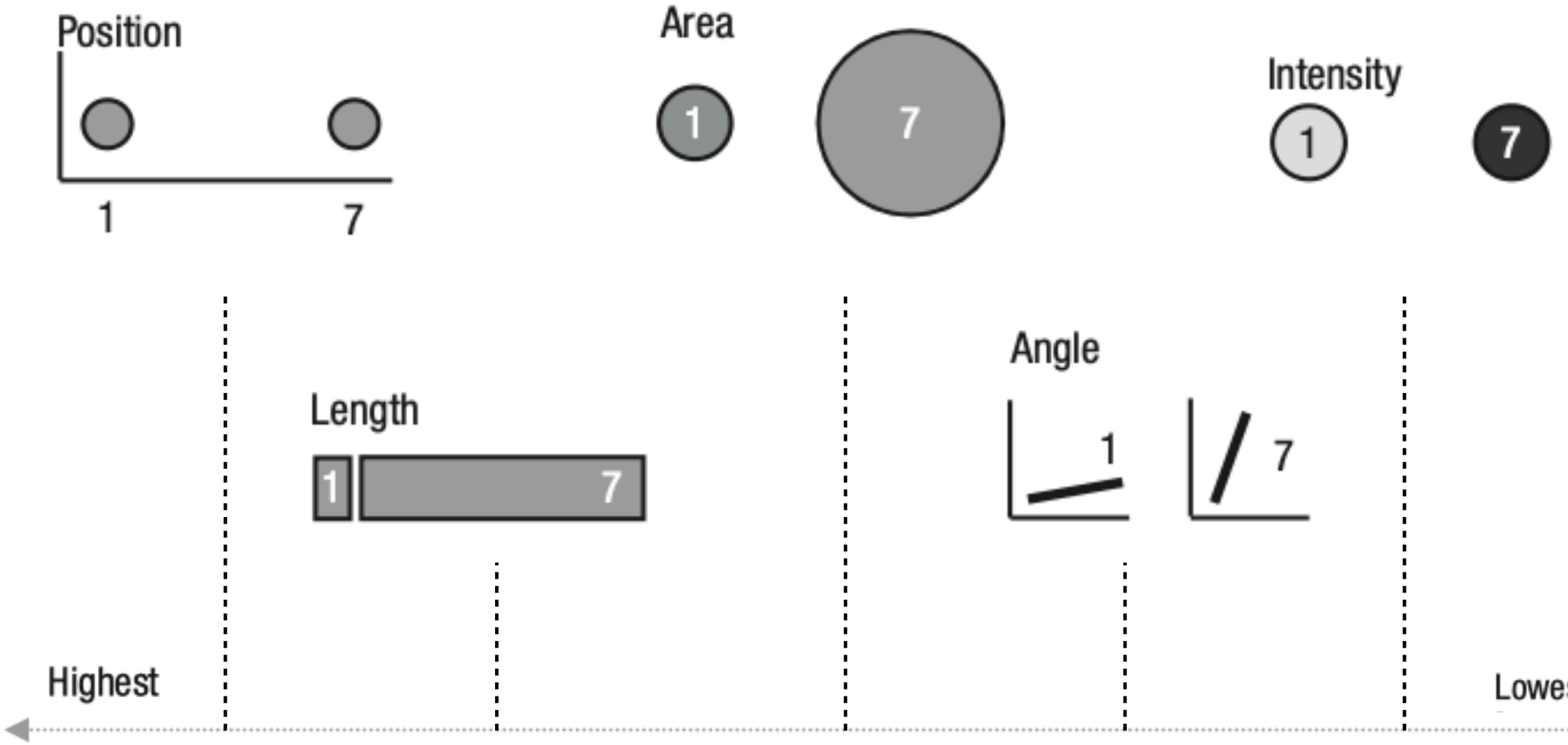


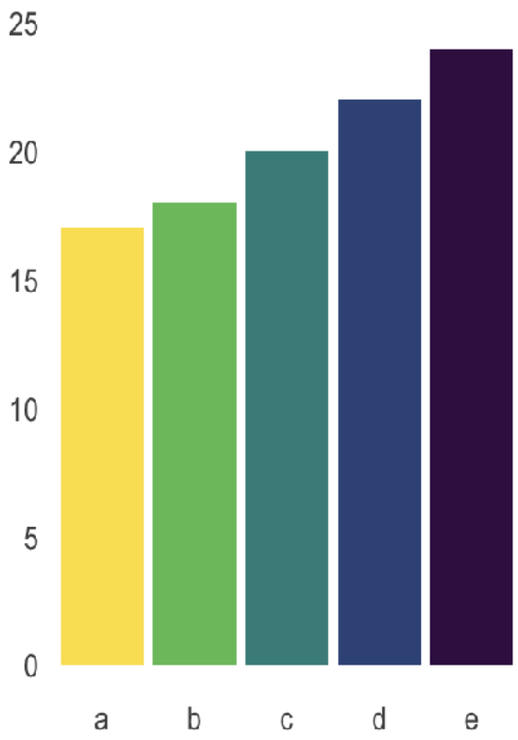
Fig. 2. A summary of the power and limits of visual data processing. The two columns on the left show a quick reference guide to channels that can depict data visually and common illusions for each channel. The column in the center presents a summary of how visual statistics are powerful. The two columns on the right illustrate how comparisons are severely limited and present a set of design techniques that focus viewers on the “right” ones.

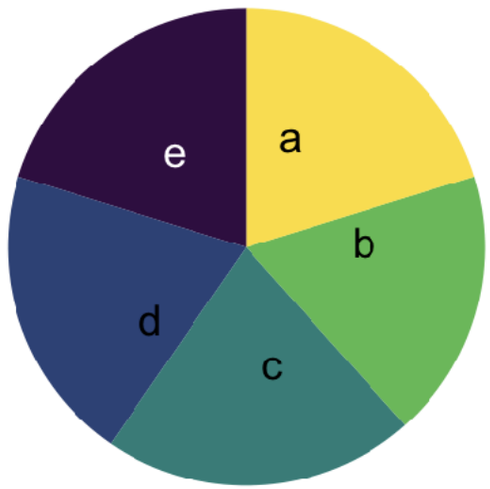
Absolute Precision Ranking for Seeing a Single Ratio

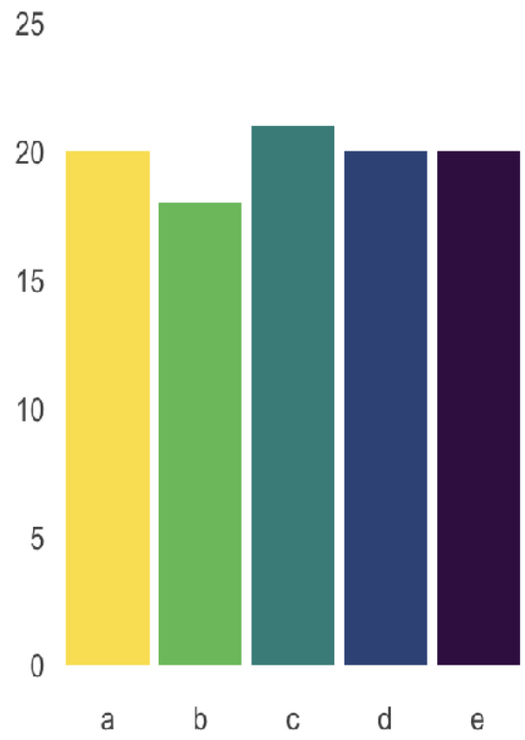
Visual estimation of the 1:7 ratio is noisier toward bottom

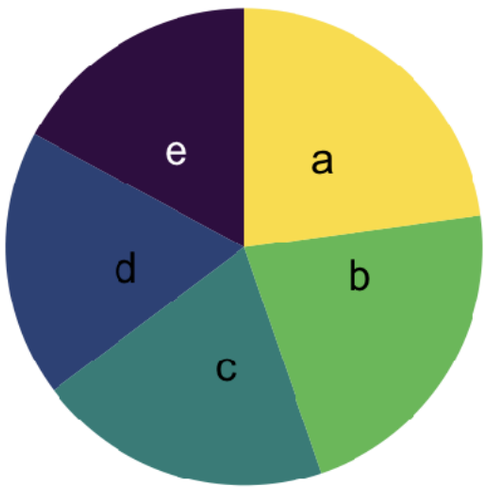


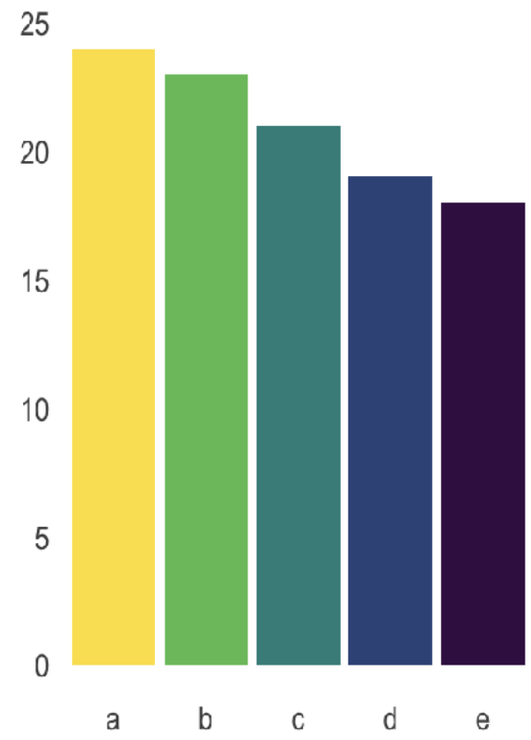








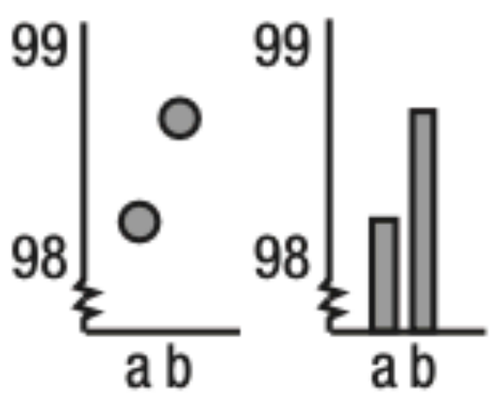




Source: "The Issue with Pie Chart," Data to Viz

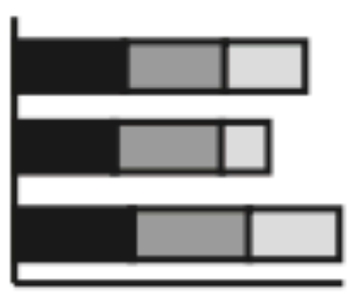
Common Illusions That Distort Data

Caveats for the visual encoding in each row



Use caution with nonzero axes: Viewers tend to overestimate differences… even when the nonzero base is marked, as in the examples at left.

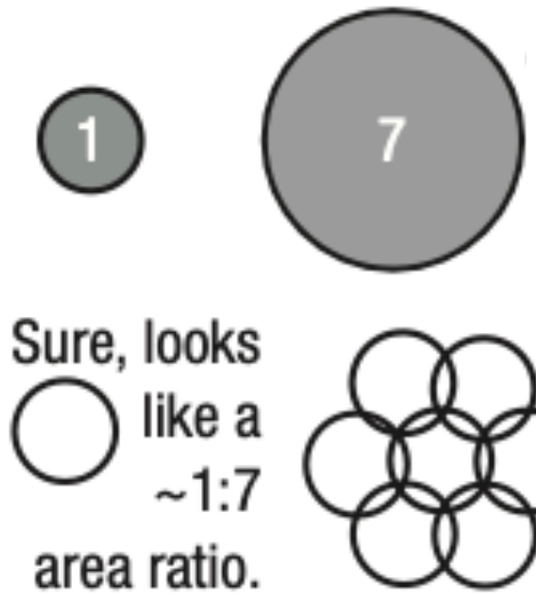
Stacked bar: Bars on baseline are position-coded = more precise perception.



The black & dark gray bars have the same value differences among them, but the differences are only visible across the black bars.

Common Illusions That Distort Data

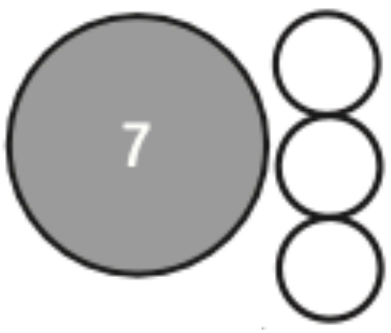
Caveats for the visual encoding in each row



Sure, looks like a ~1:7 area ratio.



But beware: if the number is actually mapped to *Length*, the gray circles depict ~1:2.5 ratio.





Intensity values can look different depending their backgrounds.

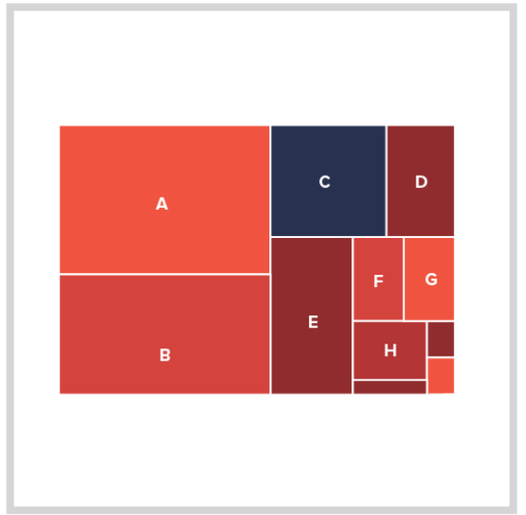
Do not plot intensities on intensities.

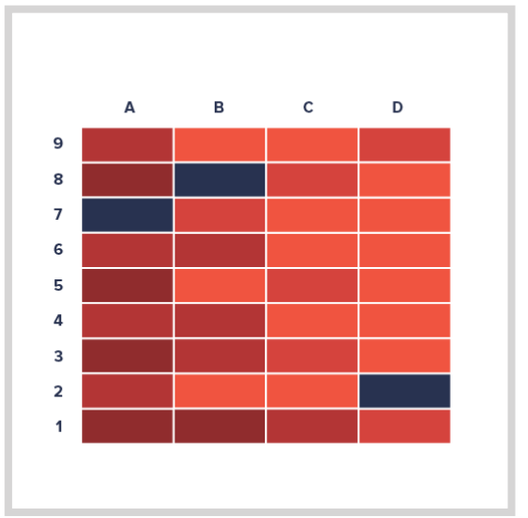
And...

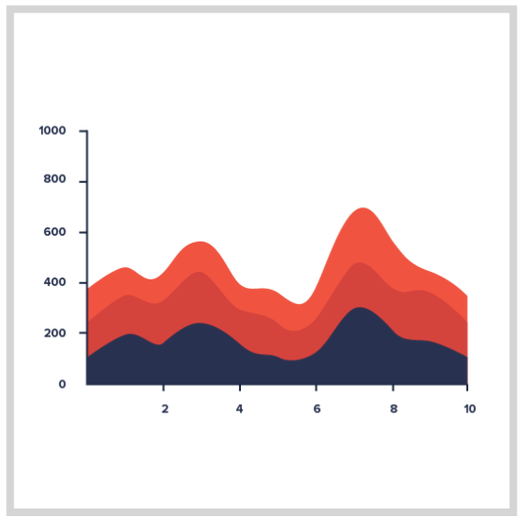
- Pictorial charts
- Treemaps
- 2-D histogram / heat map
- Area chart
- Alluvial/Sankey diagram
- Dumbbell plot

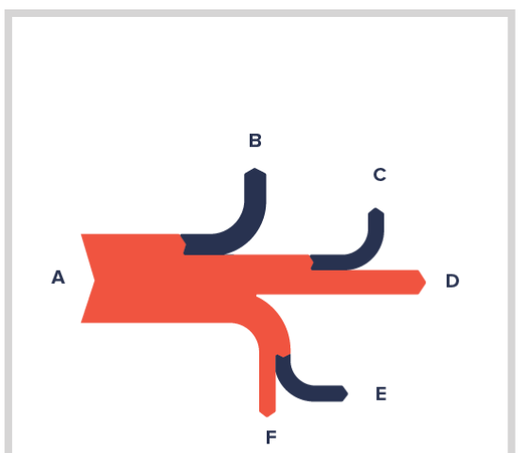
For more, see: ***datavizproject.com***

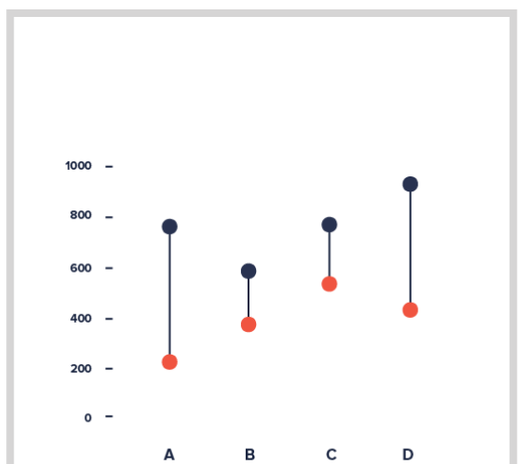












Source: Data Viz Project

matplotlib Cheat sheet

Version 3.2

Quick start

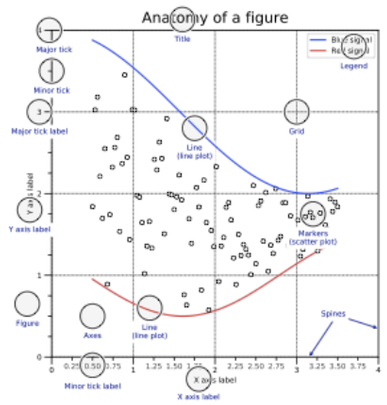
```
import numpy as np
import matplotlib as mpl
import matplotlib.pyplot as plt


X = np.linspace(0, 2*np.pi, 100)
Y = np.cos(X)

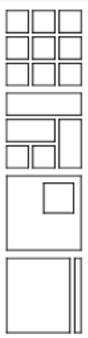
fig, ax = plt.subplots()
ax.plot(X,Y,color='C1')

fig.savefig("figure.pdf")
fig.show()
```

Anatomy of a figure



Subplots layout



```
subplot[s](cols,rows,…)
fig, axs = plt.subplots(3,3)

G = gridspec(cols,rows,…)
ax = G[0,:]

ax.inset_axes(extent)

d=make_axes_locatable(ax)
ax=d.new_horizontal('10%')
```

Getting help

- matplotlib.org
- github.com/matplotlib/matplotlib/issues
- discourse.matplotlib.org
- stackoverflow.com/matplotlib
- gitter.im/matplotlib
- twitter.com/matplotlib
- Matplotlib users mailing list

Basic plots



```
plot([X],Y,[fmt],…)
```
X, **Y**, fmt, color, marker, linestyle



```
scatter(X,Y,…)
```
X, **Y**, [s]izes, [c]olors, markers, cmap



```
bar[h](x,height,…)
```
x, **height**, width, bottom, align, color



```
imshow(Z,[cmap],…)
```
Z, cmap, interpolation, extent, origin



```
contour[f]([X],[Y],Z,…)
```
X, Y, **Z**, levels, colors, extent, origin



```
quiver([X],[Y],U,V,…)
```
X, Y, **U**, **V**, C, units, angles



```
pie(X,[explode],…)
```
Z, explode, labels, colors, radius




```
text(x,y,text,…)
```
x, **y**, **text**, va, ha, size, weight, transform

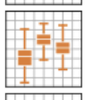


```
fill[_between][x]( … )
```
X, Y1, Y2, color, where

Advanced plots



```
step(X,Y,[fmt],…)
```
X, **Y**, fmt, color, marker, where



```
boxplot(X,…)
```
X, notch, sym, bootstrap, widths



```
errorbar(X,Y,xerr,yerr,…)
```
X, **Y**, xerr, yerr, fmt



```
hist(X, bins, …)
```
X, bins, range, density, weights



```
violinplot(D,…)
```
D, positions, widths, vert



```
barbs([X],[Y], U, V, …)
```
X, Y, **U**, **V**, C, length, pivot, sizes



```
eventplot(positions,…)
```
positions, orientation, lineoffsets



```
hexbin(X,Y,C,…)
```
X, **Y**, C, gridsize, bins



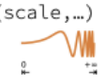
```
xcorr(X,Y,…)
```
X, **Y**, normed, detrend

Scales

```
ax.set_[xy]scale(scale,…)
```



linear any values



log values > 0



symlog any values



logit 0 < values < 1

Projections

```
subplot(…,projection=p)
```



p='polar'



p='3d'



```
p=Orthographic()
from cartopy.crs import Cartographic
```

Lines

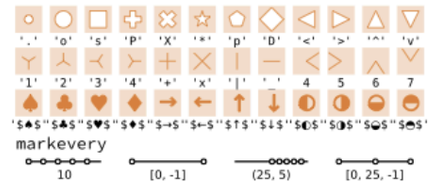
```
linestyle or ls
```




```
capstyle or dash_capstyle
```



Markers

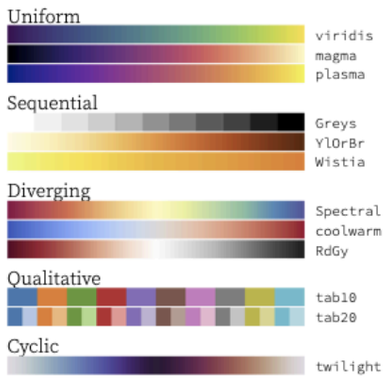


Colors



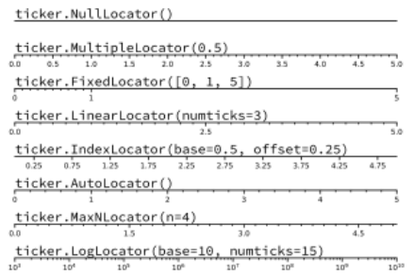
Colormaps

```
plt.get_cmap(name)
```



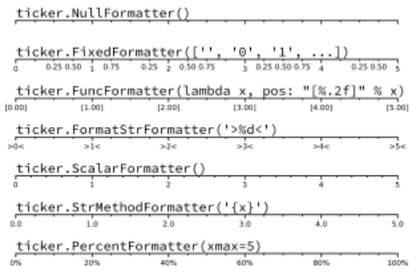
Tick locators

```
from matplotlib import ticker
ax.[xy]axis.set_[minor|major]_locator(locator)
```



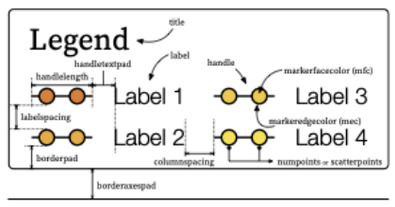
Tick formatters

```
from matplotlib import ticker
ax.[xy]axis.set_[minor|major]_formatter(formatter)
```

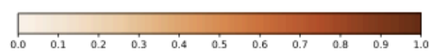


Ornaments

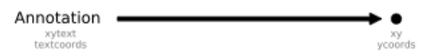
```
ax.legend(…)
```
handles, labels, loc, title, frameon




```
ax.colorbar(…)
```
mappable, ax, cax, orientation



```
ax.annotate(…)
```
text, **xy**, **xytext**, xycoords, textcoords, arrowprops



Event handling

```
fig, ax = plt.subplots()
def on_click(event):
  print(event)
fig.canvas.mpl_connect(
  'button_press_event', on_click)
```

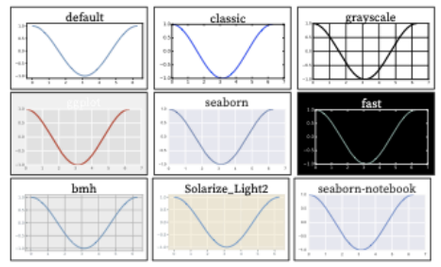
Animation

```
import matplotlib.animation as mpla

T = np.linspace(0,2*np.pi,100)
S = np.sin(T)
line, = plt.plot(T, S)
def animate(i):
  line.set_ydata(np.sin(T+i/50))
anim = mpla.FuncAnimation(
  plt.gcf(), animate, interval=5)
plt.show()
```

Styles

```
plt.style.use(style)
```



Quick reminder

```
ax.grid()
ax.patch.set_alpha(0)
ax.set_[xy]lim(vmin, vmax)
ax.set_[xy]label(label)
ax.set_[xy]ticks(list)
ax.set_[xy]ticklabels(list)
ax.set_[sup]title(title)
ax.tick_params(width=10, …)
ax.set_axis_[on|off]()

ax.tight_layout()
plt.gcf(), plt.gca()
mpl.rc('axes', linewidth=1, …)
fig.patch.set_alpha(0)
text=r'$\frac{-e^{i\pi}}{2^n}$'
```

Keyboard shortcuts

ctrl + s Save	ctrl + w Close plot
r Reset view	f Fullscreen 0/1
f View forward	b View back
p Pan view	o Zoom to rect
x X pan/zoom	y Y pan/zoom
g Minor grid 0/1	G Major grid 0/1
l X axis log/linear	L Y axis log/linear

Ten Simple Rules

READ

1. Know Your Audience
2. Identify Your Message
3. Adapt the Figure
4. Captions Are Not Optional
5. Do Not Trust the Defaults
6. Use Color Effectively
7. Do Not Mislead the Reader
8. Avoid "Chartjunk"
9. Message Trumps Beauty
10. Get the Right Tool

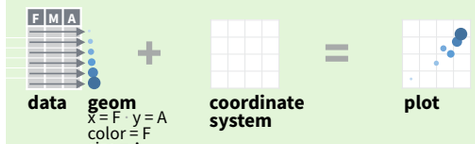
Data Visualization with ggplot2 : : CHEAT SHEET

Basics

ggplot2 is based on the **grammar of graphics**, the idea that you can build every graph from the same components: a **data** set, a **coordinate system**, and geoms—visual marks that represent data points.



To display values, map variables in the data to visual properties of the geom (**aesthetics**) like **size**, **color**, and **x** and **y** locations.



Complete the template below to build a graph.

```
ggplot (data = <DATA>) +                                  required
<GEOM_FUNCTION> (mapping = aes( <MAPPINGS> ),
 stat = <STAT>, position = <POSITION> ) +                 Not required, sensible defaults supplied
<COORDINATE_FUNCTION> +
<FACET_FUNCTION> +
<SCALE_FUNCTION> +
<THEME_FUNCTION>
```

ggplot(data = mpg, **aes(**x = cty, y = hwy**))** Begins a plot that you finish by adding layers to. Add one geom function per layer.

aesthetic mappings · data · geom

qplot(x = cty, y = hwy, data = mpg, geom = "point"**)** Creates a complete plot with given data, geom, and mappings. Supplies many useful defaults.

last_plot() Returns the last plot

ggsave("plot.png", width = 5, height = 5) Saves last plot as 5' x 5' file named "plot.png" in working directory. Matches file type to file extension.

Geoms

Use a geom function to represent data points, use the geom's aesthetic properties to represent variables. Each function returns a layer.

GRAPHICAL PRIMITIVES

a <- ggplot(economics, aes(date, unemploy))
b <- ggplot(seals, aes(x = long, y = lat))

a + geom_blank()
(Useful for expanding limits)

b + geom_curve(aes(yend = lat + 1, xend=long+1),curvature=1**)** - x, xend, y, yend, alpha, angle, color, curvature, linetype, size

a + geom_path(lineend="butt", linejoin="round", linemitre=1**)**
x, y, alpha, color, group, linetype, size

a + geom_polygon(aes(group = group)**)**
x, y, alpha, color, fill, group, linetype, size

b + geom_rect(aes(xmin = long, ymin=lat, xmax= long + 1, ymax = lat + 1)**)** - xmax, xmin, ymax, ymin, alpha, color, fill, linetype, size

a + geom_ribbon(aes(ymin=unemploy - 900, ymax=unemploy + 900)**)** - x, ymax, ymin, alpha, color, fill, group, linetype, size

LINE SEGMENTS

common aesthetics: x, y, alpha, color, linetype, size

b + geom_abline(aes(intercept=0, slope=1)**)**
b + geom_hline(aes(yintercept = lat)**)**
b + geom_vline(aes(xintercept = long)**)**

b + geom_segment(aes(yend=lat+1, xend=long+1)**)**
b + geom_spoke(aes(angle = 1:1155, radius = 1)**)**

ONE VARIABLE continuous

c <- ggplot(mpg, aes(hwy)); c2 <- ggplot(mpg)

c + geom_area(stat = "bin")
x, y, alpha, color, fill, linetype, size

c + geom_density(kernel = "gaussian"**)**
x, y, alpha, color, fill, group, linetype, size, weight

c + geom_dotplot()
x, y, alpha, color, fill

c + geom_freqpoly() x, y, alpha, color, group, linetype, size

c + geom_histogram(binwidth = 5**)** x, y, alpha, color, fill, linetype, size, weight

c2 + geom_qq(aes(sample = hwy)**)** x, y, alpha, color, fill, linetype, size, weight

discrete

d <- ggplot(mpg, aes(fl))

d + geom_bar()
x, alpha, color, fill, linetype, size, weight

TWO VARIABLES

continuous x , continuous y

e <- ggplot(mpg, aes(cty, hwy))

e + geom_label(aes(label = cty), nudge_x = 1, nudge_y = 1, check_overlap = TRUE**)** x, y, label, alpha, angle, color, family, fontface, hjust, lineheight, size, vjust

e + geom_jitter(height = 2, width = 2**)**
x, y, alpha, color, fill, shape, size

e + geom_point(), x, y, alpha, color, fill, shape, size, stroke

e + geom_quantile(), x, y, alpha, color, group, linetype, size, weight

e + geom_rug(sides = "bl"**)**, x, y, alpha, color, linetype, size

e + geom_smooth(method = lm**)**, x, y, alpha, color, fill, group, linetype, size, weight

e + geom_text(aes(label = cty), nudge_x = 1, nudge_y = 1, check_overlap = TRUE**)**, x, y, label, alpha, angle, color, family, fontface, hjust, lineheight, size, vjust

discrete x , continuous y

f <- ggplot(mpg, aes(class, hwy))

f + geom_col(), x, y, alpha, color, fill, group, linetype, size

f + geom_boxplot(), x, y, lower, middle, upper, ymax, ymin, alpha, color, fill, group, linetype, shape, size, weight

f + geom_dotplot(binaxis = "y", stackdir = "center"**)**, x, y, alpha, color, fill, group

f + geom_violin(scale = "area"**)**, x, y, alpha, color, fill, group, linetype, size, weight

discrete x , discrete y

g <- ggplot(diamonds, aes(cut, color))

g + geom_count(), x, y, alpha, color, fill, shape, size, stroke

continuous bivariate distribution

h <- ggplot(diamonds, aes(carat, price))

h + geom_bin2d(binwidth = c(0.25, 500)**)**
x, y, alpha, color, fill, linetype, size, weight

h + geom_density2d()
x, y, alpha, colour, group, linetype, size

h + geom_hex()
x, y, alpha, colour, fill, size

continuous function

i <- ggplot(economics, aes(date, unemploy))

i + geom_area()
x, y, alpha, color, fill, linetype, size

i + geom_line()
x, y, alpha, color, group, linetype, size

i + geom_step(direction = "hv")
x, y, alpha, color, group, linetype, size

visualizing error

df <- data.frame(grp = c("A", "B"), fit = 4:5, se = 1:2)
j <- ggplot(df, aes(grp, fit, ymin = fit-se, ymax = fit+se))

j + geom_crossbar(fatten = 2**)**
x, y, ymax, ymin, alpha, color, fill, group, linetype, size

j + geom_errorbar(), x, ymax, ymin, alpha, color, group, linetype, size, width (also **geom_errorbarh()**)

j + geom_linerange()
x, ymin, ymax, alpha, color, group, linetype, size

j + geom_pointrange()
x, y, ymin, ymax, alpha, color, fill, group, linetype, shape, size

maps

data <- data.frame(murder = USArrests$Murder, state = tolower(rownames(USArrests)))
map <- map_data("state")
k <- ggplot(data, aes(fill = murder))

k + geom_map(aes(map_id = state), map = map**)**
+ expand_limits(x = map$long, y = map$lat**)**,
map_id, alpha, color, fill, linetype, size

THREE VARIABLES

seals$z <- with(seals, sqrt(delta_long^2 + delta_lat^2)); l <- ggplot(seals, aes(long, lat))

l + geom_contour(aes(z = z))
x, y, z, alpha, colour, group, linetype, size, weight

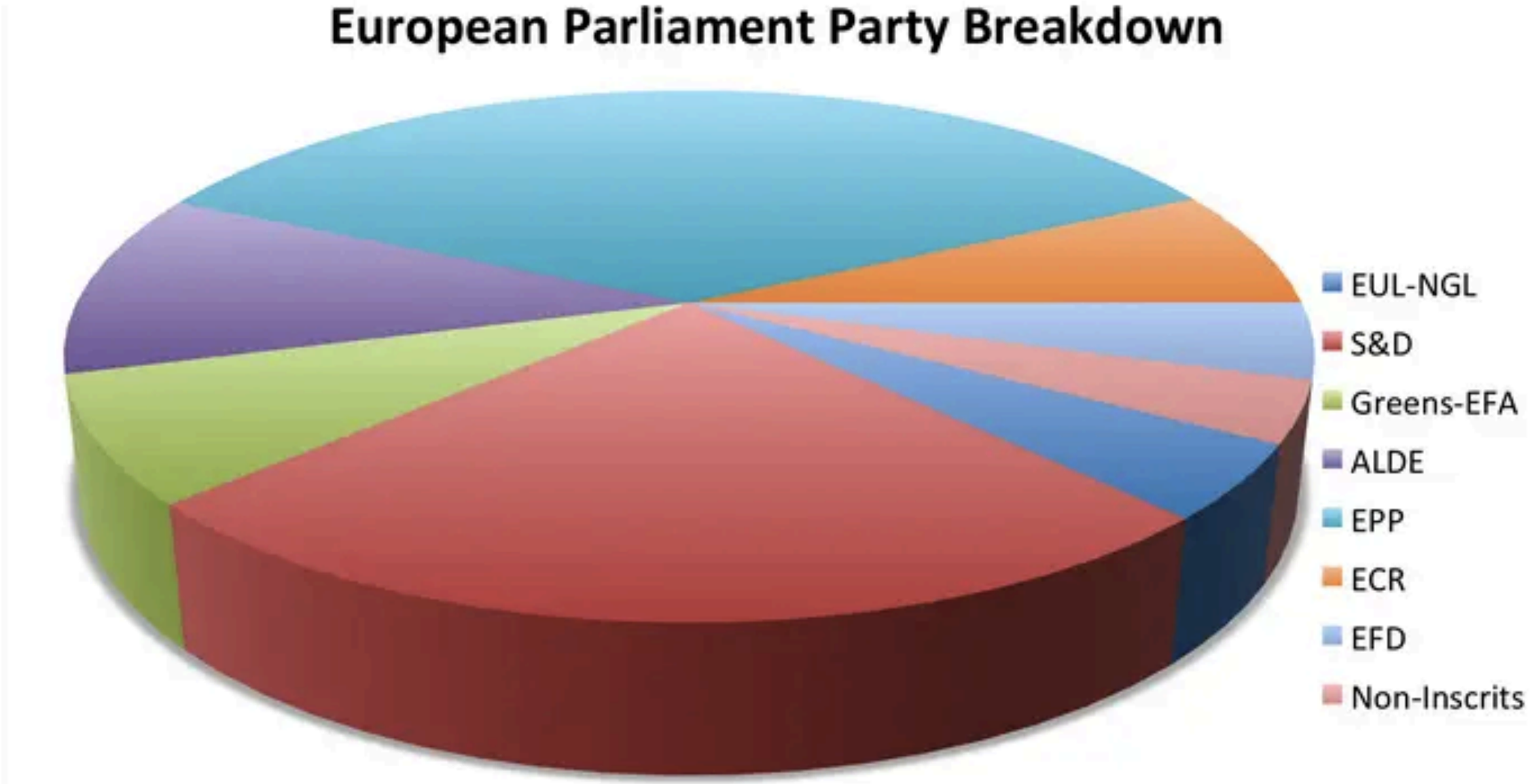
l + geom_raster(aes(fill = z), hjust=0.5, vjust=0.5, interpolate=FALSE**)**
x, y, alpha, fill

l + geom_tile(aes(fill = z)**)**, x, y, alpha, color, fill, linetype, size, width

RStudio® is a trademark of RStudio, Inc. • CC BY SA RStudio • info@rstudio.com • 844-448-1212 • rstudio.com • Learn more at **http://ggplot2.tidyverse.org** • ggplot2 3.1.0 • Updated: 2018-12

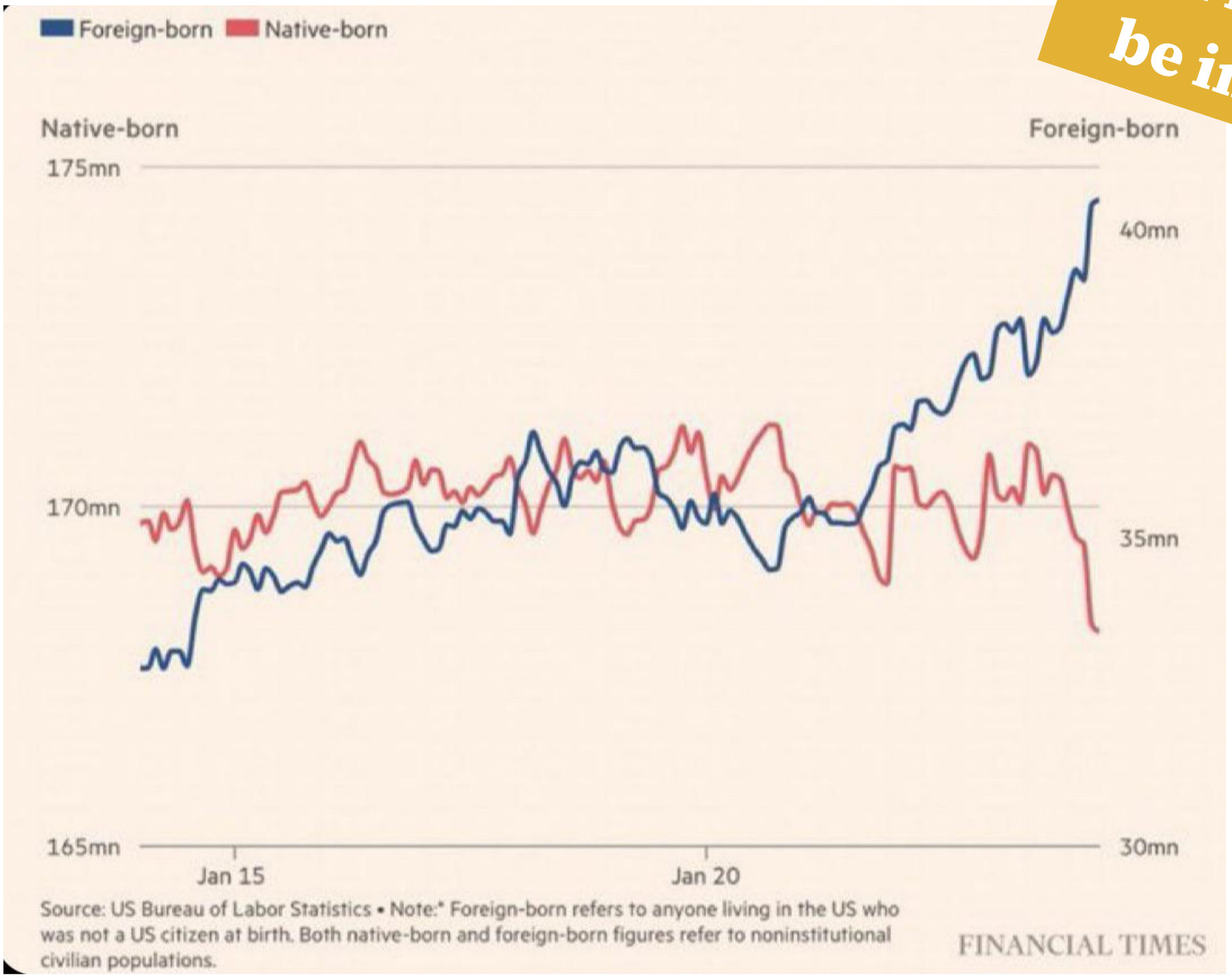
Roast! This! Viz!

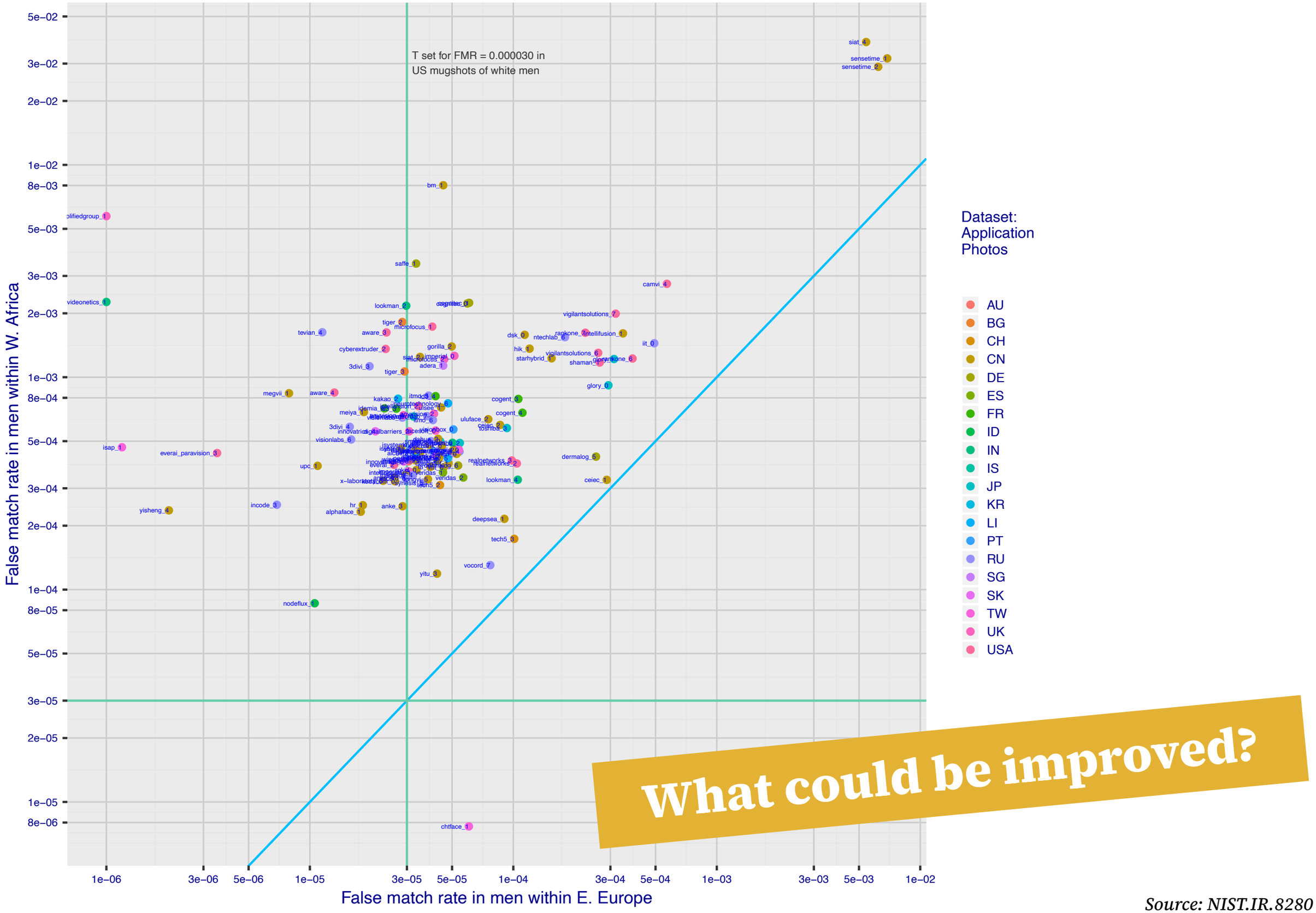
What could be improved?



Source: Walter Hickey, Business Insider

What could be improved?





What could be improved?

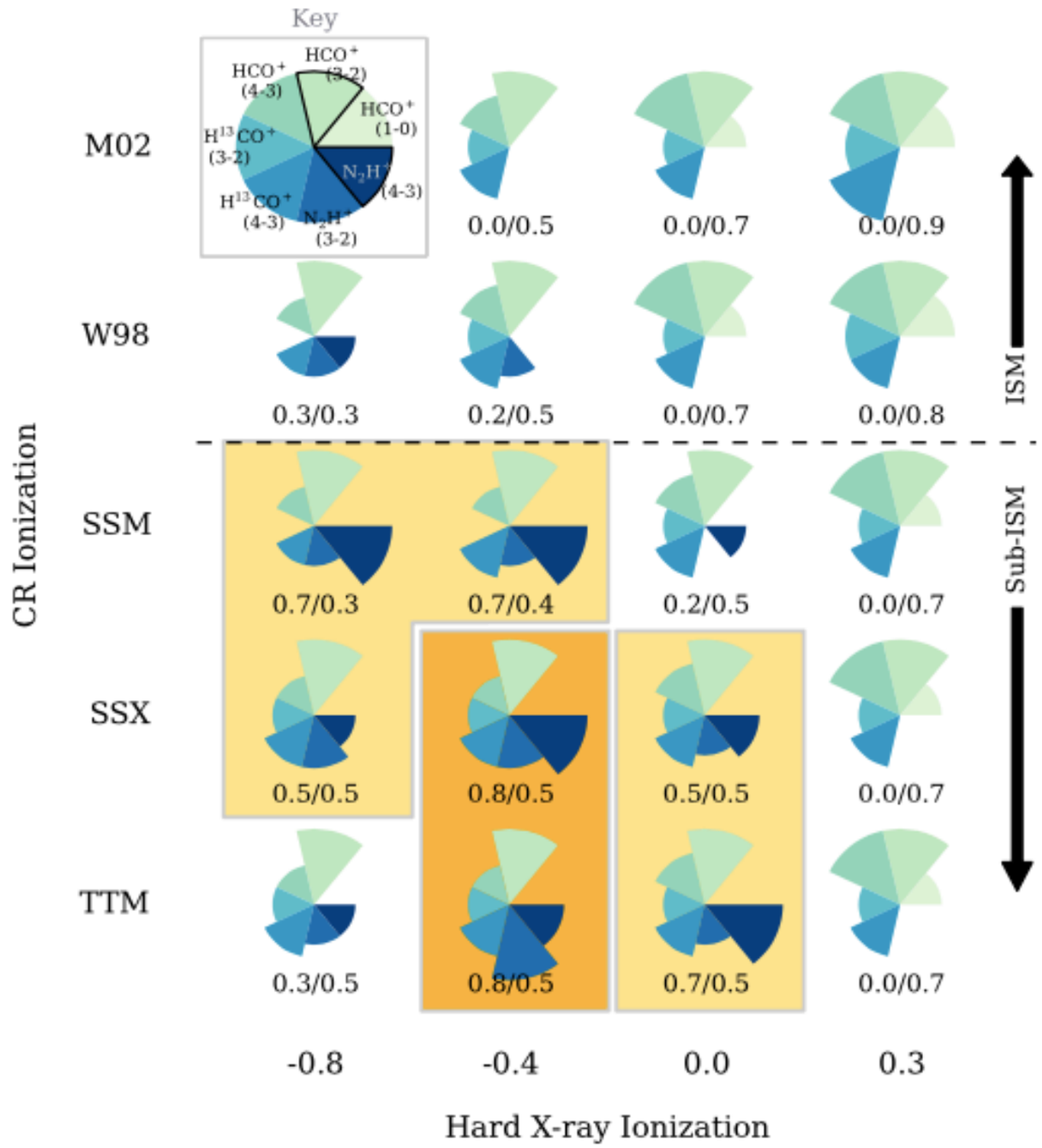
Source: NIST.IR.8280

10 Best Practices for Effective Data Visualization

1. **Know your audience!**
   - For public vizzes, this usually requires bar, line, and scatter plots.
   - Basically never use undefined acronyms or symbols.
2. **Question the defaults.** This means fonts, colors, sizes, shapes, and more.
3. **Use descriptive titles:** *tell* your viewer what they should see, and don't be redundant.
4. **Be selective in data and design.** Consider your data-to-ink ratio and remove distractions.
5. **Use color sparingly, intentionally, and consistently.**
   - Keep branding in mind
   - Remember colorblindness
6. **Use reasonable axis tick** formats and frequencies, not just the default.
7. **Consider creative alternatives to legends.**
8. **Don't be afraid of annotations.** Use whatever software you prefer to add them.
9. **Make your text bigger** than you'd think. Nobody likes to squint!
10. **Use the blink test.**

1. Know your audience!

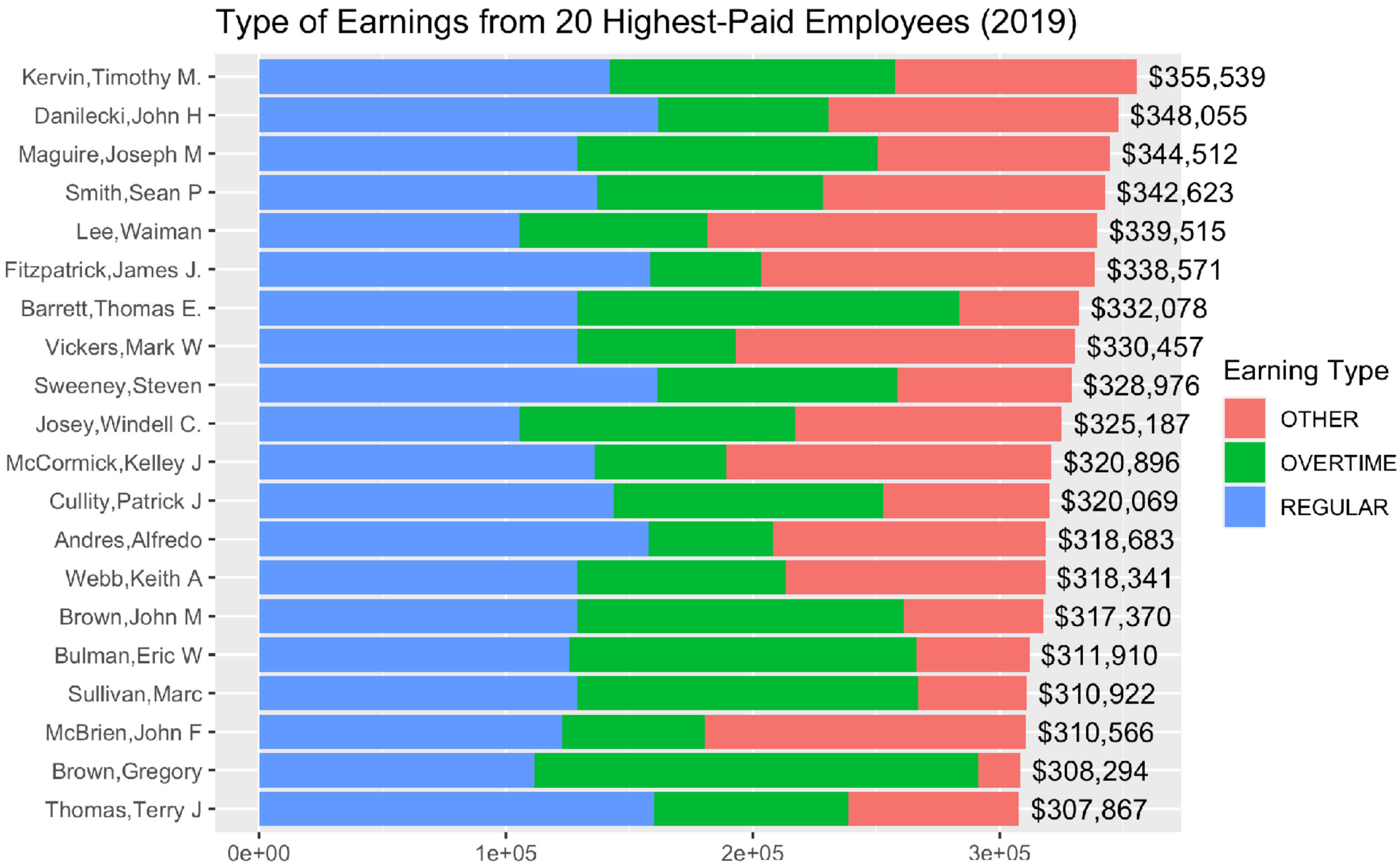
- For public vizzes, this usually requires bar, line, and scatter plots.
- Basically never use undefined acronyms or symbols.



Source: "Constraining the X-ray and Cosmic Ray Ionization Chemistry of the TW Hya Protoplanetary Disk: Evidence for a Sub-interstellar Cosmic Ray Rate," Cleeves et al., 2015

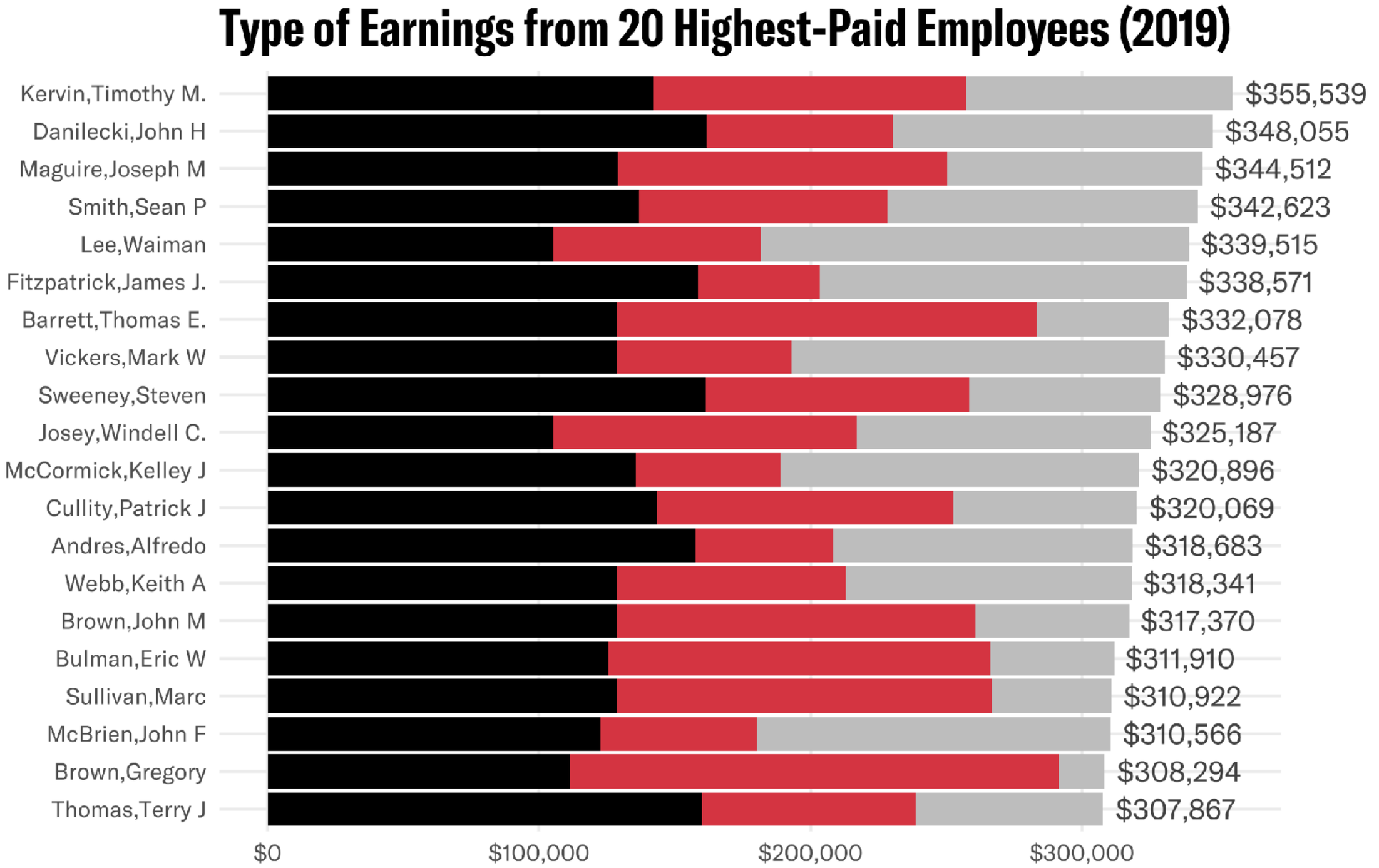
2. **Question the defaults.** This means fonts, colors, sizes, shapes, and more.

2. **Question the defaults.** This means fonts, colors, sizes, shapes, and more.



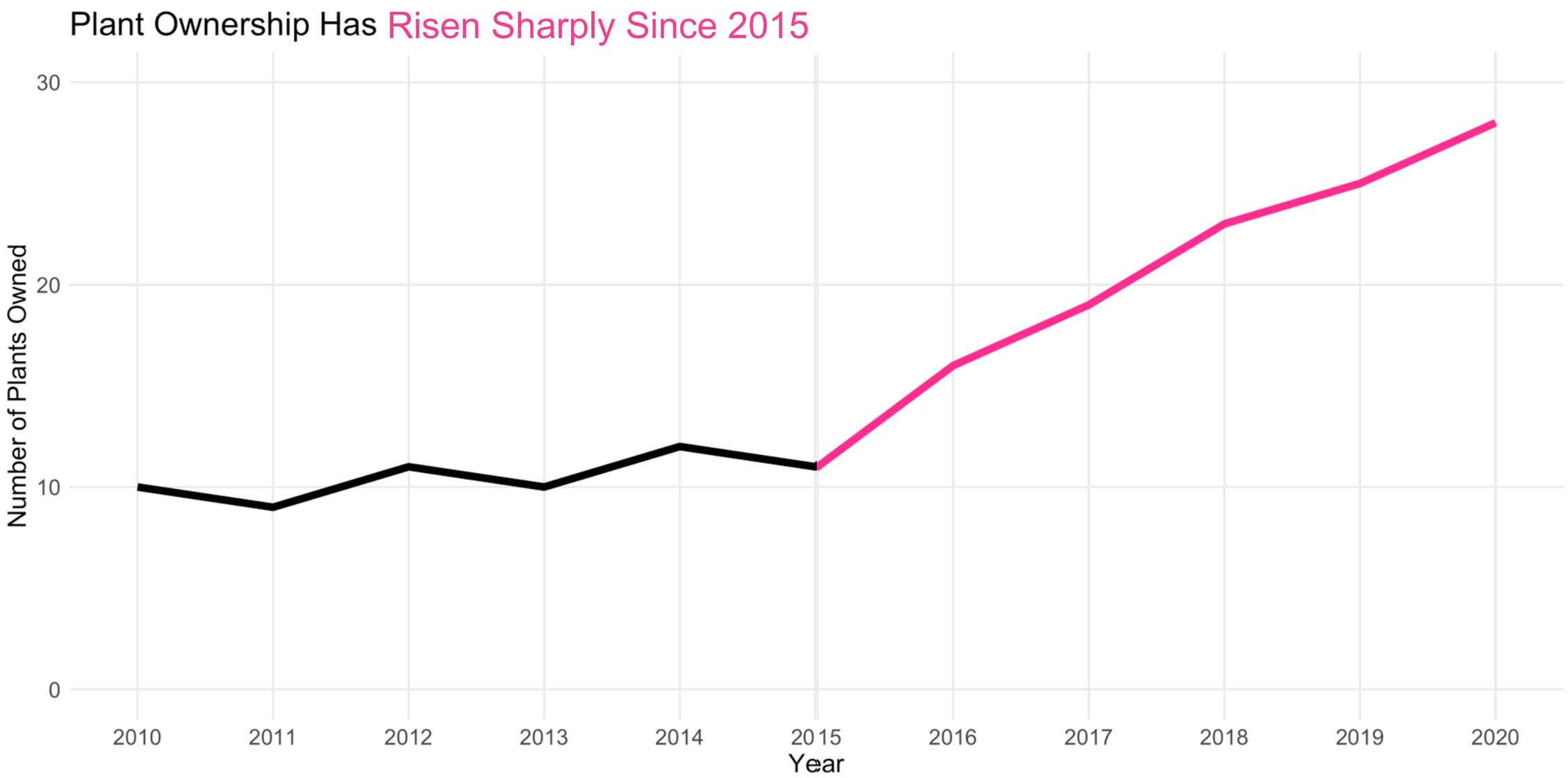
Source: "Unpacking the Boston Police Budget," ACLU of Massachusetts Data for Justice Project

2. **Question the defaults.** This means fonts, colors, sizes, shapes, and more.



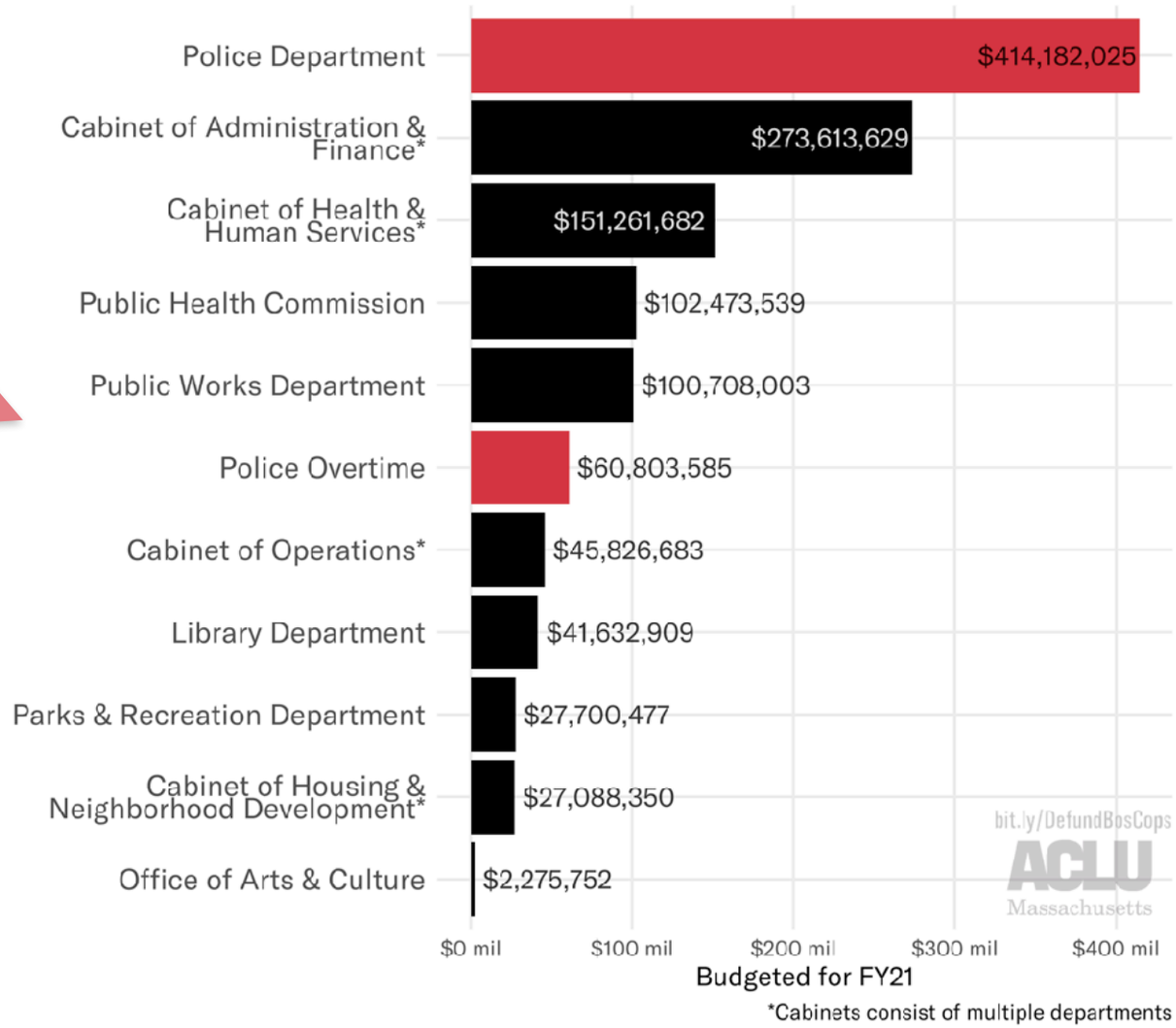
Source: "Unpacking the Boston Police Budget," ACLU of Massachusetts Data for Justice Project

3. **Use descriptive titles:** *tell* your viewer what they should see, and don't be redundant.



4. **Be selective in data and design.** Consider your data-to-ink ratio and remove distractions.

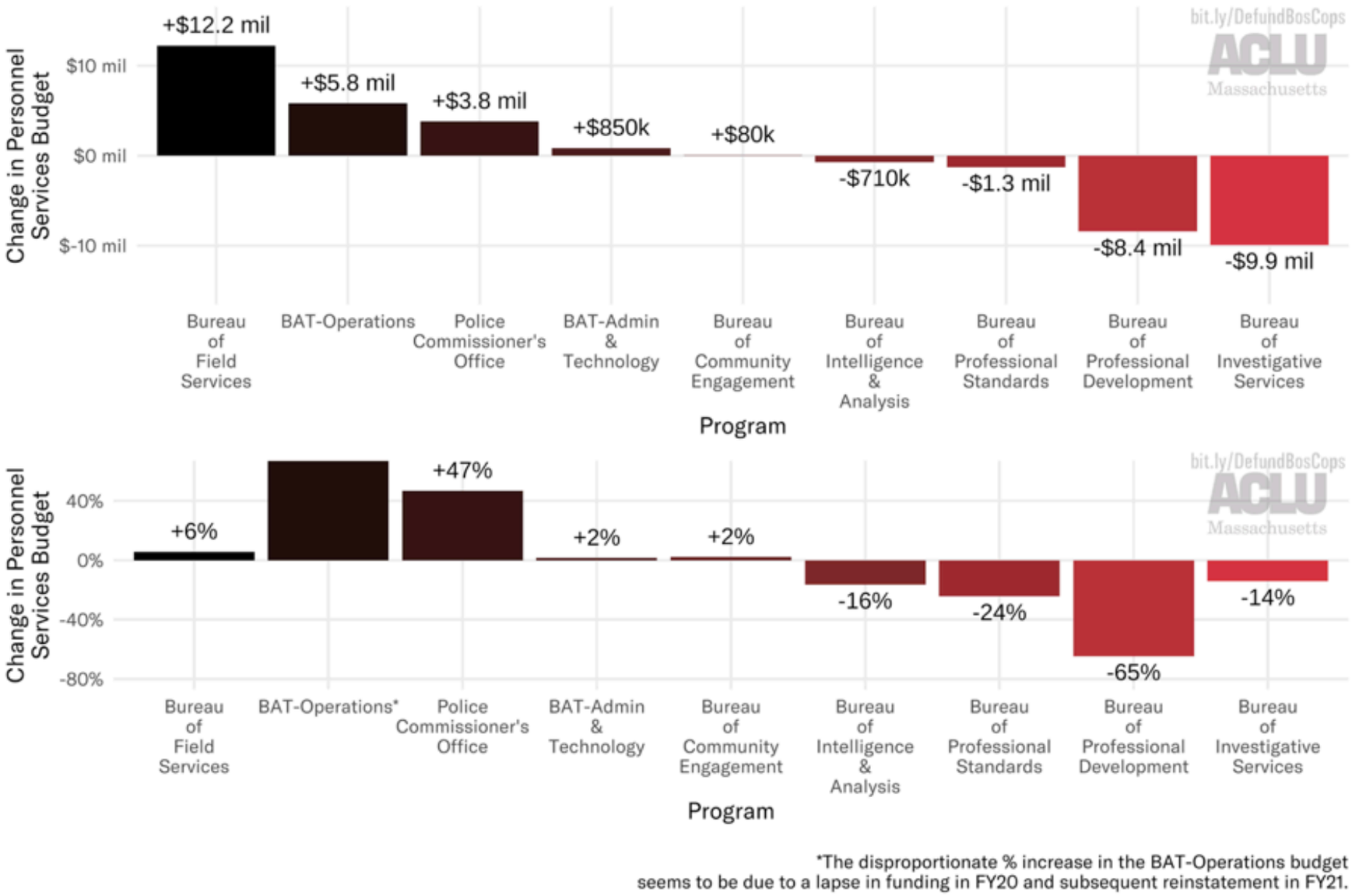
Cabinet	Department		FY 17 Expenditure	FY 18 Expenditure	FY20 Appropriation	FY21 Appropriation	21 vs 20
Mayor's Office	Mayor's Office		4,178,112	4,356,277	5,331,454	5,439,392	107,938
	Election Department		3,905,877	4,205,689	5,524,296	5,262,101	-262,095
	Intergovernmental Relations		1,197,436	1,291,712	1,580,011	1,517,657	-62,353
	Human Rights Commission					250,746	250,746
	Law Department		6,503,424	6,479,021	8,377,366	8,533,708	156,342
	Women's Advancement		242,700	257,236	346,188	462,817	116,629
		Total	**16,027,548**	**16,789,935**	**21,159,175**	**21,466,381**	**307,206**
Operations	Property Management Department		17,927,980	19,616,267	18,795,900	17,590,170	-1,205,730
	Public Facilities Department		5,449,174	6,554,871	7,169,689	7,245,344	75,655
	Inspectional Services Department		18,566,595	19,170,451	18,840,155	20,990,369	2,050,214
		Total	**41,943,748**	**45,341,589**	**44,905,744**	**45,826,883**	**920,939**
Civic Engagement	Neighborhood Services		3,061,883	3,044,743	3,718,338	3,887,589	169,252
		Total	**3,061,883**	**3,044,743**	**3,718,338**	**3,887,589**	**169,252**
Arts & Culture	Office of Arts & Culture		1,147,105	1,307,925	2,051,646	2,225,752	224,086
	Library Department		35,958,490	38,663,859	40,534,902	41,632,909	1,098,007
		Total	**37,105,595**	**39,971,834**	**42,586,568**	**43,908,662**	**1,322,093**
Economic Development	Office of Economic Development		2,319,130	3,506,139	3,446,210	3,403,600	-42,610
	Consumer Affairs & Licensing		1,151,755	1,222,148	1,353,313	1,515,705	162,192
	Office of Tourism		1,134,559	1,333,810	1,545,852	1,651,972	106,120
		Total	**4,625,444**	**6,062,297**	**6,345,374**	**6,571,277**	**225,703**
Education	Boston Public Schools		1,093,289,520	1,126,676,079	1,178,564,205	1,258,633,065	80,068,860
		Total	**1,093,289,520**	**1,126,676,079**	**1,178,564,205**	**1,258,633,065**	**80,068,860**
Environment, Energy & Open Space	Environment Department		2,470,122	4,779,357	3,197,886	3,383,879	185,993
	Parks & Recreation Department		26,322,942	25,988,612	26,666,627	27,700,477	1,033,850
		Total	**28,793,064**	**30,767,969**	**29,864,513**	**31,084,356**	**1,219,843**
Administration & Finance	Administration & Finance		1,205,918	757,317	1,737,927	1,395,228	-342,698
	Assessing Department		7,390,085	7,150,574	7,746,316	7,878,310	131,995
	Auditing Department		2,680,026	2,735,310	2,947,260	2,944,254	-3,006
	Budget Management		2,811,863	2,702,651	3,542,745	3,629,157	87,412
	Execution of Courts		19,513,268	18,233,939	5,000,000	5,000,000	0
	Health Insurance		210,986,298	212,029,318	221,181,299	219,396,323	-1,784,976
	Human Resources		4,257,960	4,694,941	5,849,713	5,970,600	120,888
	Medicare		9,815,432	10,573,357	11,200,000	11,200,000	0
	Office of Labor Relations		1,250,130	1,122,845	1,465,905	1,498,507	32,601
	Pensions & Annuities - City		3,636,293	3,699,684	4,100,000	4,100,000	0
	Pensions & Annuities - County		30,329	13,147	100,000	100,000	0
	Procurement		1,658,911	1,736,215	1,888,611	1,889,549	938
	Registry Division		957,564	1,116,384	1,109,488	1,132,415	22,927
	Treasury Department		5,216,098	20,169,845	19,800,156	4,729,285	-15,070,901
	Unemployment Compensation		0	0	350,000	350,000	0
	Workers' Compensation Fund		1,385,868	1,858,544	2,200,000	2,200,000	0
		Total	**272,589,842**	**288,453,828**	**290,418,439**	**273,613,629**	**-16,804,810**
Health & Human Services	Office of Health & Human Services		0	0	0	1,655,798	1,655,798
	Boston Center for Youth & Families		29,351,768	27,468,082	28,605,788	28,801,676	-804,112
	Commission For Persons W/Disabilities		428,328	400,491	482,540	510,616	28,076
	Age Strong		3,568,631	3,723,832	3,985,685	4,186,087	200,402
	Fair Housing & Equity		258,864	290,704	318,386	317,514	-852
	Office of Immigrant Advancement		463,901	411,934	627,767	875,549	247,782
	Public Health Commission		79,563,339	87,967,402	93,405,428	102,473,529	9,068,101
	Boston VETS		3,231,627	2,837,051	4,705,679	4,612,875	-92,804
	Youth Engagement & Employment		6,336,200	6,091,324	6,890,580	7,818,028	927,448
		Total	**122,984,259**	**129,290,540**	**140,106,814**	**151,261,682**	**11,154,868**
Housing & Neighborhood Development	Neighborhood Development		13,685,868	14,203,732	20,577,614	27,088,350	6,510,736
		Total	**13,685,868**	**14,203,732**	**20,577,614**	**27,088,350**	**6,510,736**
Information & Technology	Department of Innovation and Technology		33,816,198	39,059,083	34,360,307	35,353,549	993,242
		Total	**33,816,198**	**39,059,083**	**34,360,307**	**35,353,549**	**993,242**
Public Safety	Emergency Management		650,751	684,252	885,163	985,440	100,277
	Fire Department		233,418,618	259,809,035	271,616,063	275,059,399	3,443,336
	Police Department		389,924,488	416,762,368	414,306,878	414,182,026	-124,853
		Total	**653,994,857**	**677,255,656**	**686,808,104**	**690,226,864**	**3,418,760**
Streets	Central Fleet Management		2,829,169	2,993,212	3,110,484	3,047,315	-63,169
	Office of Streets		1,788,257	2,131,409	2,277,107	2,122,824	-94,283
	Public Works Department		82,903,493	85,363,039	100,267,573	100,708,003	440,430
	Snow Removal		28,168,776	22,762,894	24,788,144	25,067,583	279,440
	Transportation Department		35,559,546	38,038,731	39,276,378	39,699,764	423,386
		Total	**151,319,351**	**151,289,285**	**169,659,686**	**170,645,489**	**985,803**
Non-Mayoral Departments	City Clerk		1,234,257	1,233,660	1,346,251	1,392,267	46,017
	City Council		5,248,445	5,442,471	5,721,300	5,736,400	15,100
	Finance Commission		271,026	276,453	294,446	298,784	5,338
		Total	**6,753,728**	**6,952,584**	**7,361,996**	**7,427,451**	**65,455**
		Grand Total	**2,459,990,506**	**2,575,159,233**	**2,676,437,077**	**2,766,996,027**	**90,558,949**



Source: "Unpacking the Boston Police Budget," ACLU of Massachusetts Data for Justice Project

5. Use color sparingly, intentionally, and consistently.

- Keep branding in mind
- Remember colorblindness



Source: "Unpacking the Boston Police Budget," ACLU of Massachusetts Data for Justice Project

5. Use color sparingly, intentionally, and consistently.

- Keep branding in mind
- Remember colorblindness

STSci | SPACE TELESCOPE SCIENCE INSTITUTE

OSWALD

Overpass



ACLU Massachusetts

GT America

Century Schoolbook



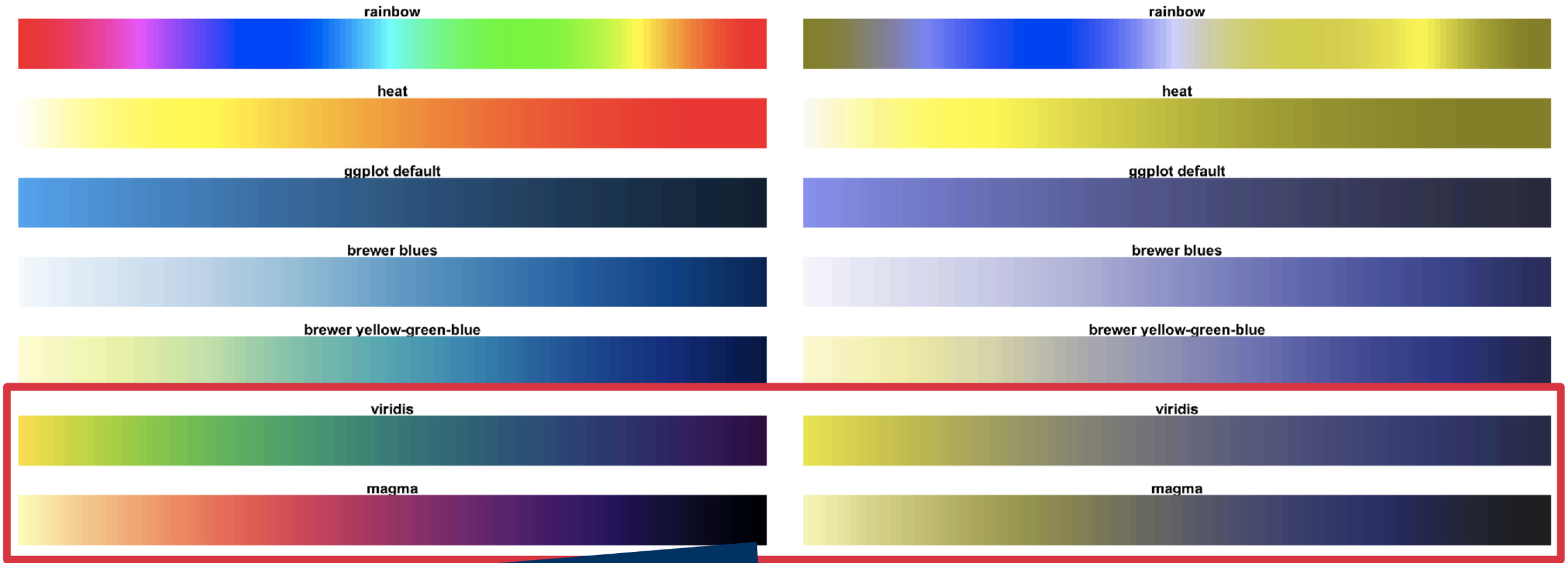
Source Serif

Inter



5. Use color sparingly, intentionally, and consistently.

- Keep branding in mind
- Remember colorblindness

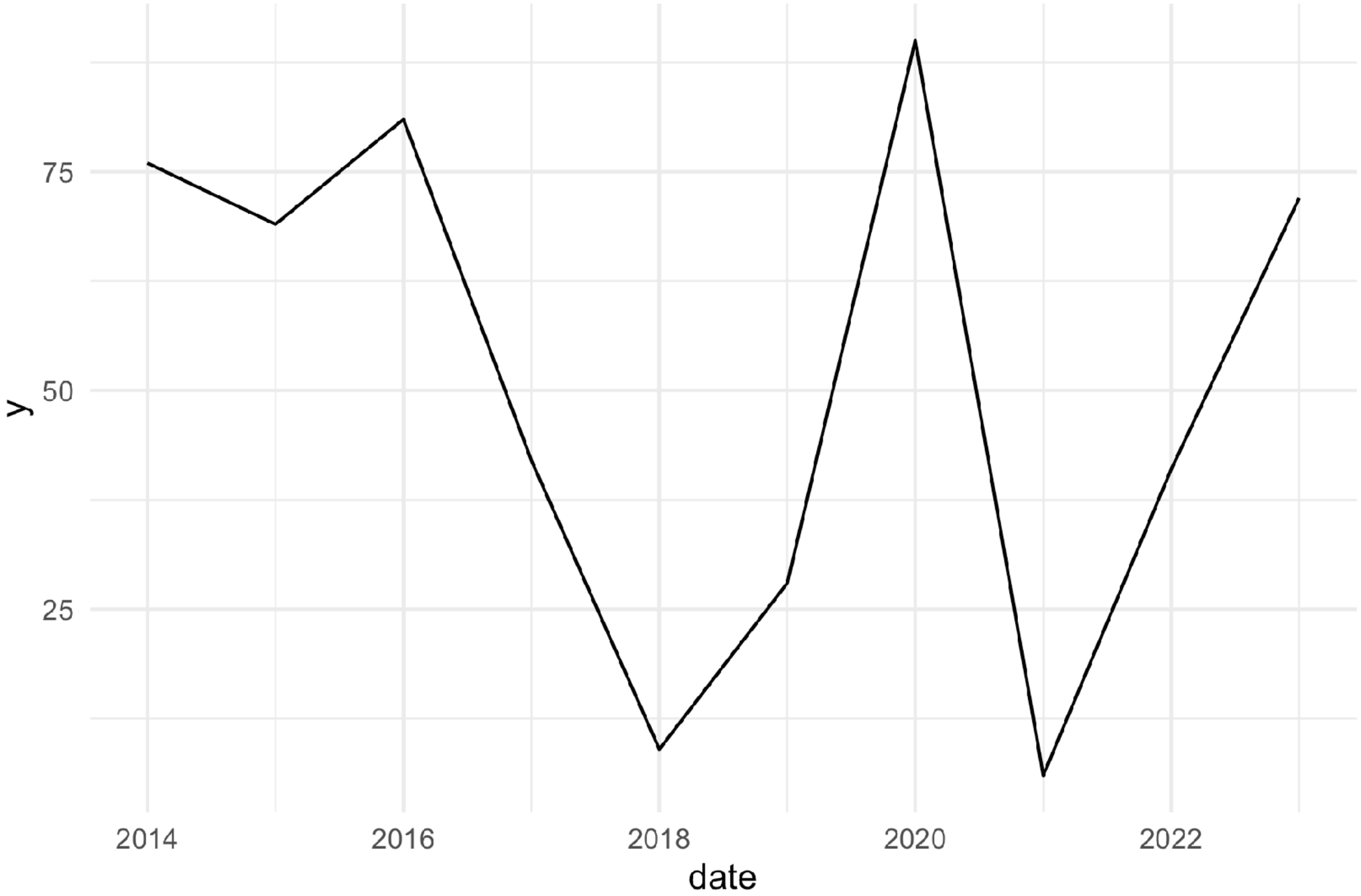


Paul Tol's Palettes

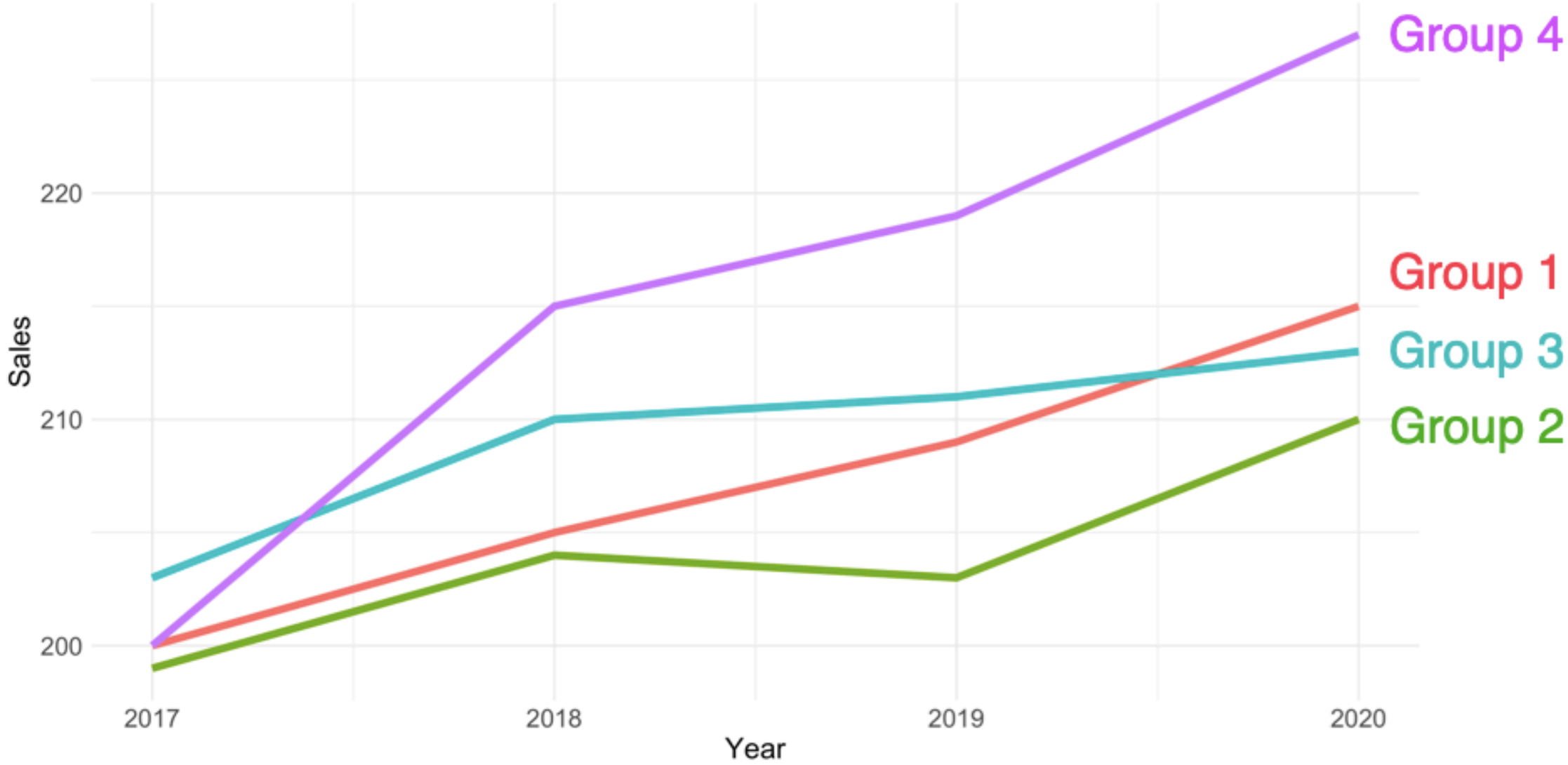
Deuteranopia

Source: "Introduction to the viridis color maps," Bob Rudis, Noam Ross and Simon Garnier, CRAN

6. **Use reasonable axis tick** formats and frequencies, not just the default.

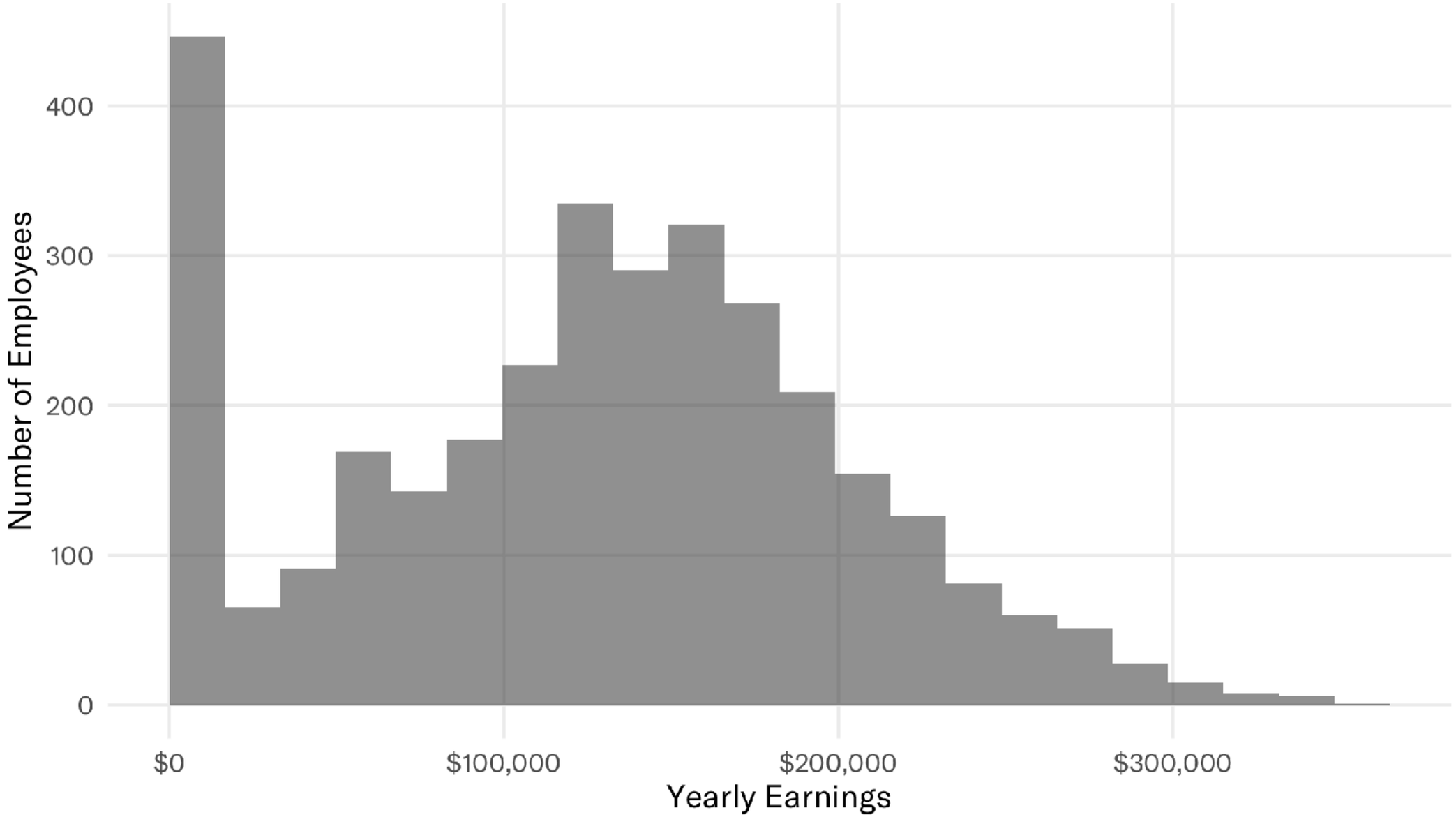


7. Consider creative alternatives to legends.



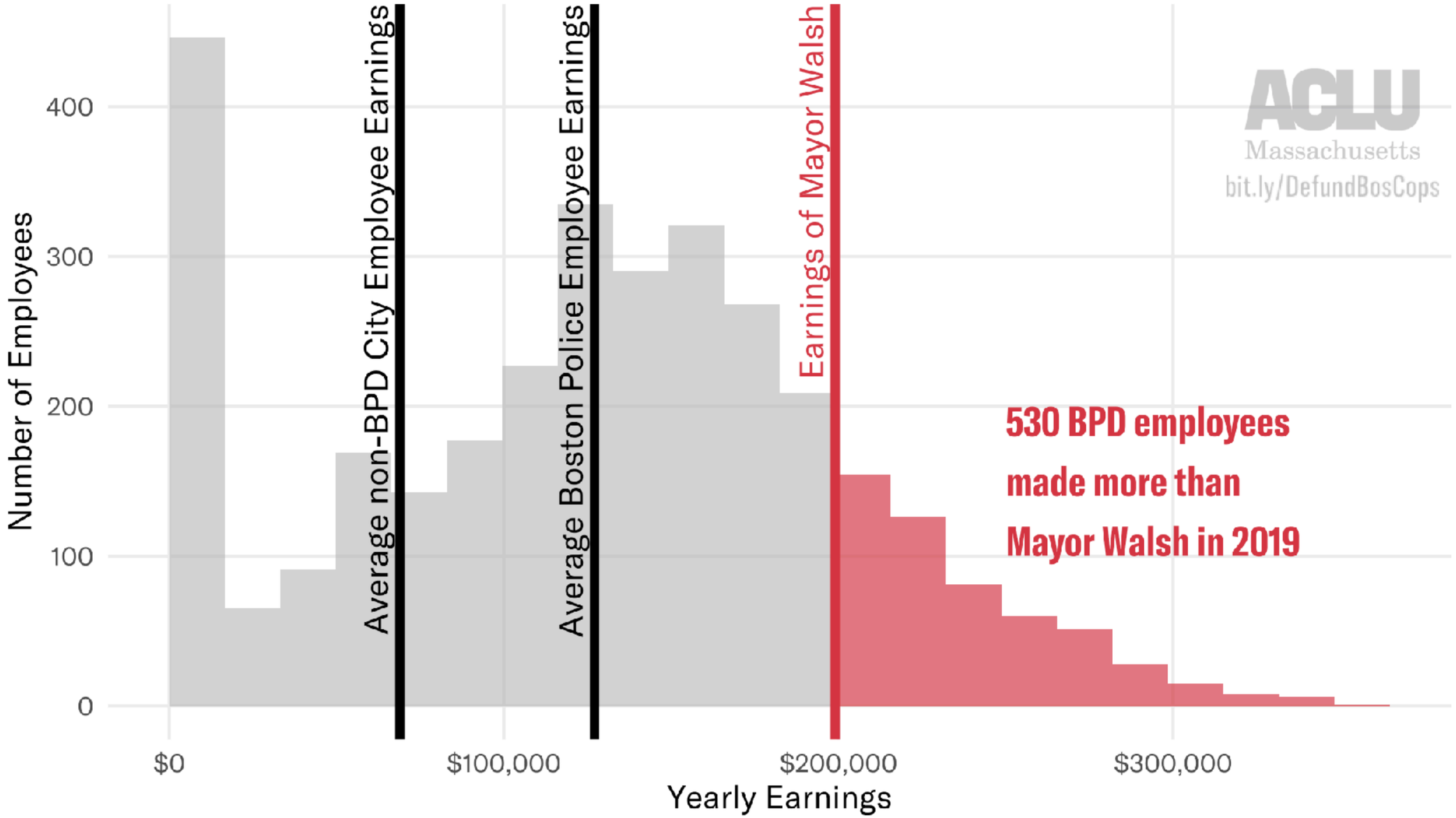
8. **Don’t be afraid of annotations.** Use whatever software you prefer to add them.

8. **Don’t be afraid of annotations.** Use whatever software you prefer to add them.



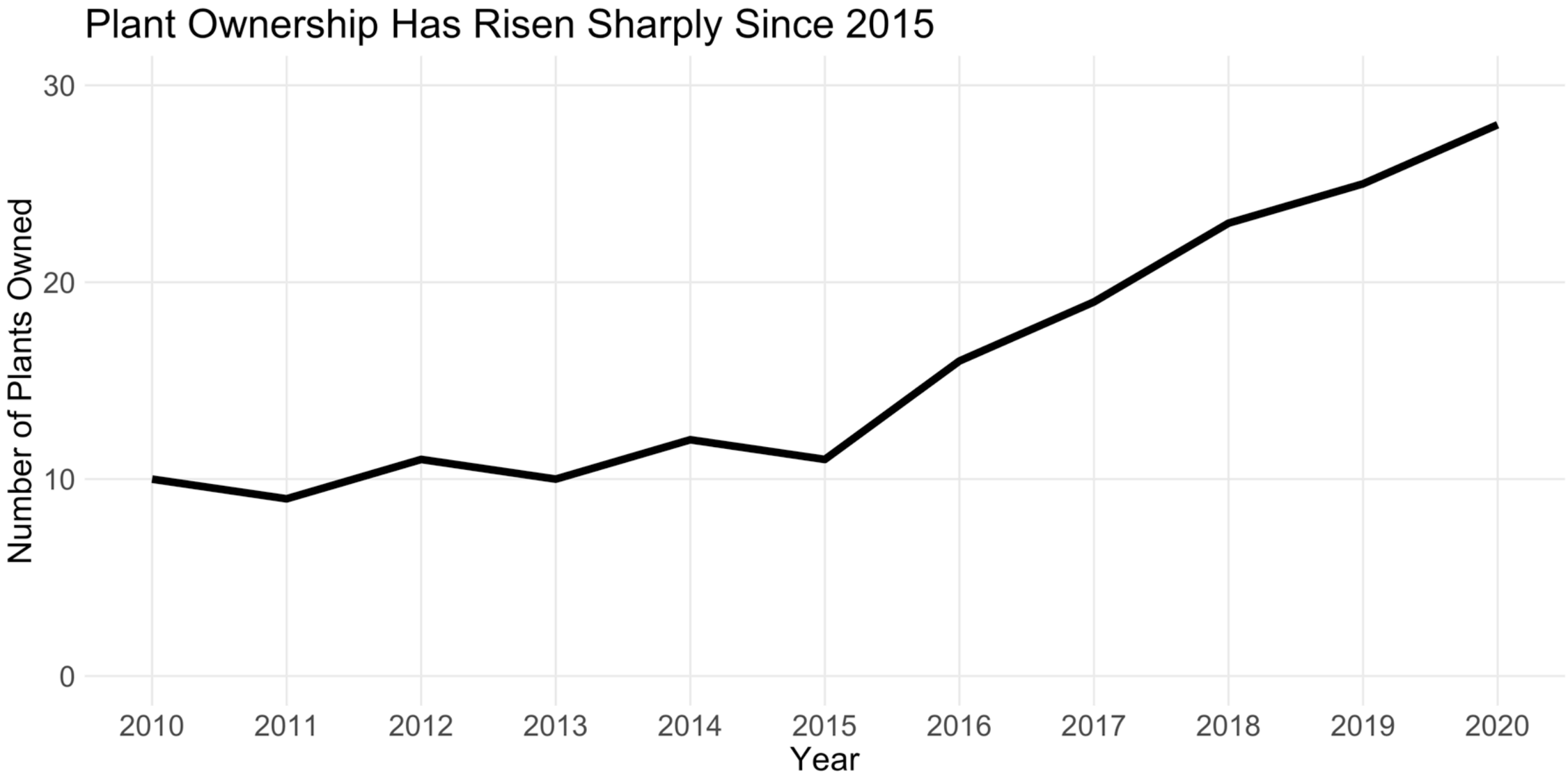
Source: “Unpacking the Boston Police Budget,” ACLU of Massachusetts Data for Justice Project

8. Don't be afraid of annotations. Use whatever software you prefer to add them.

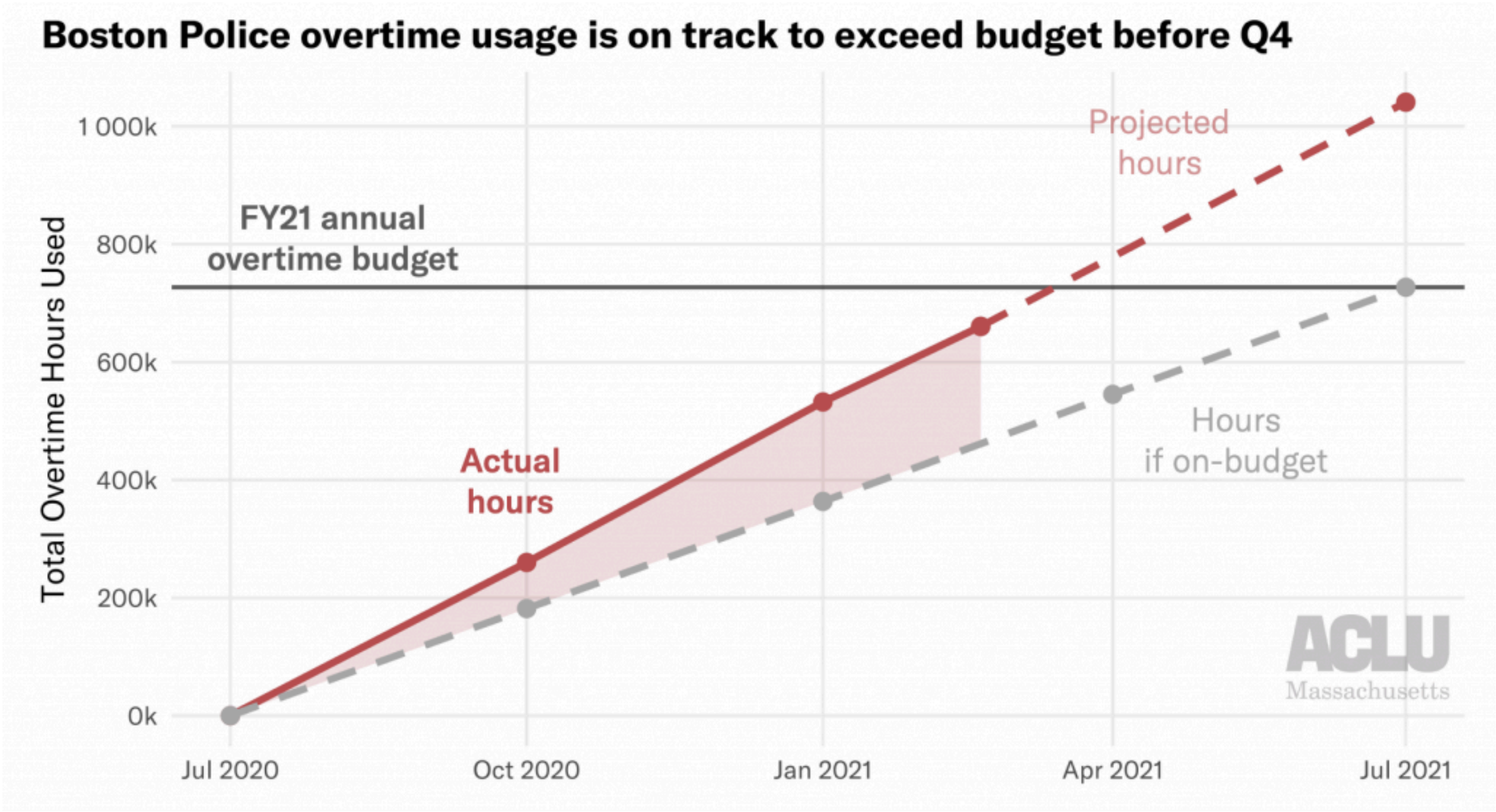


Source: "Unpacking the Boston Police Budget," ACLU of Massachusetts Data for Justice Project

9. Make your text bigger than you'd think. Nobody likes to squint!

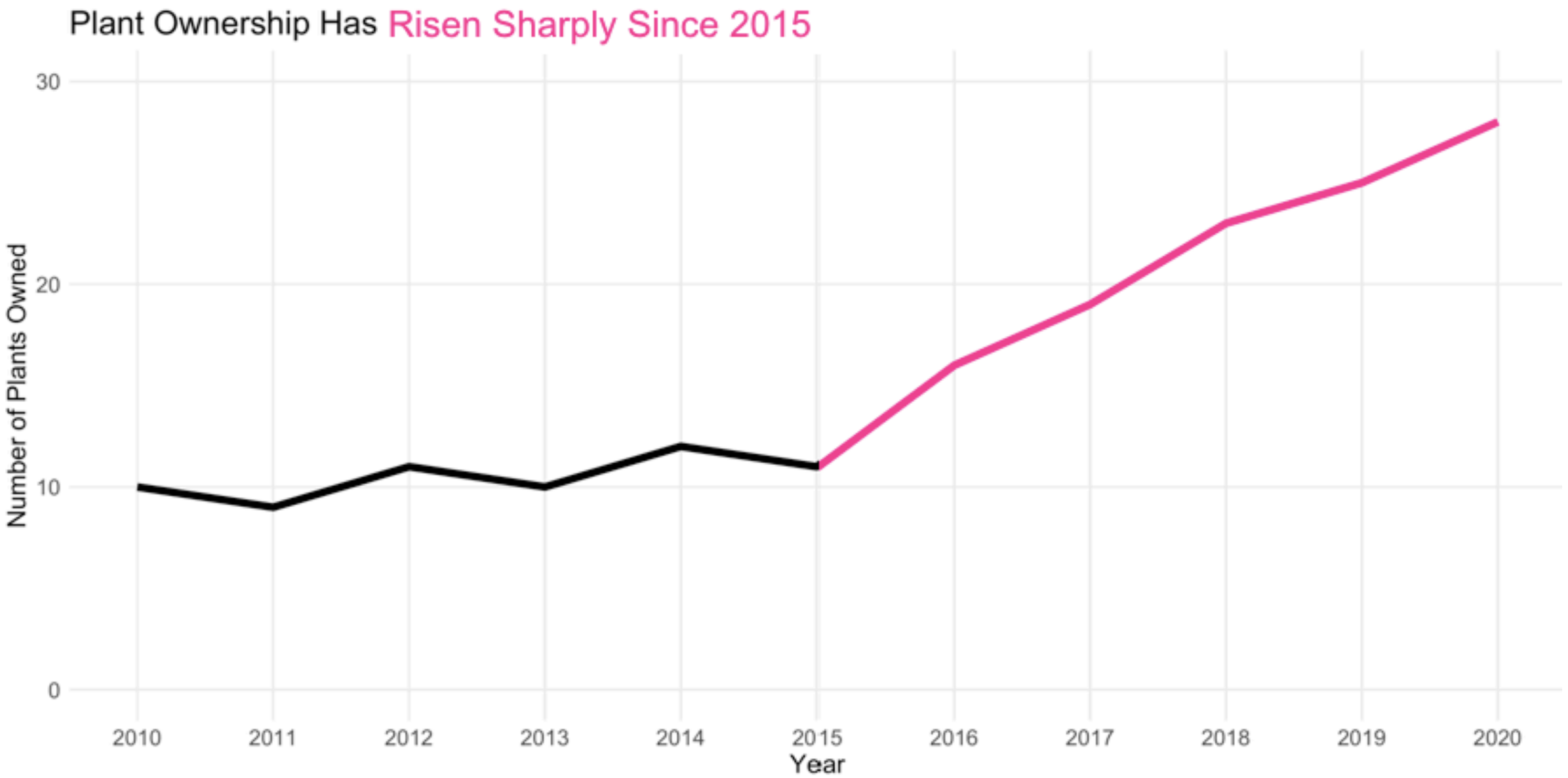


10. Use the blink test.



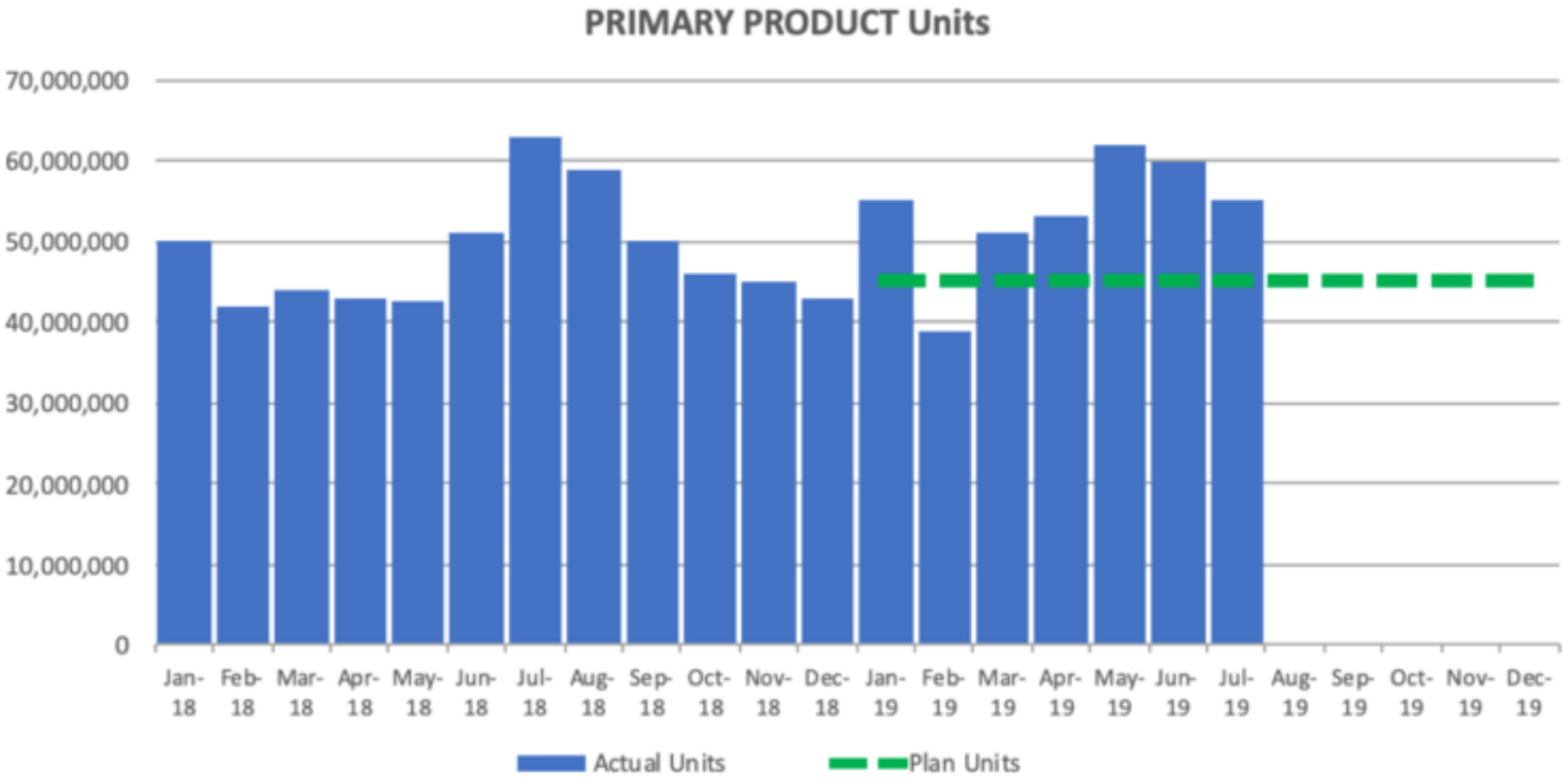
Source: "A Mayor's Roadmap to Curb Boston Police Overtime," ACLU of Massachusetts Data for Justice Project

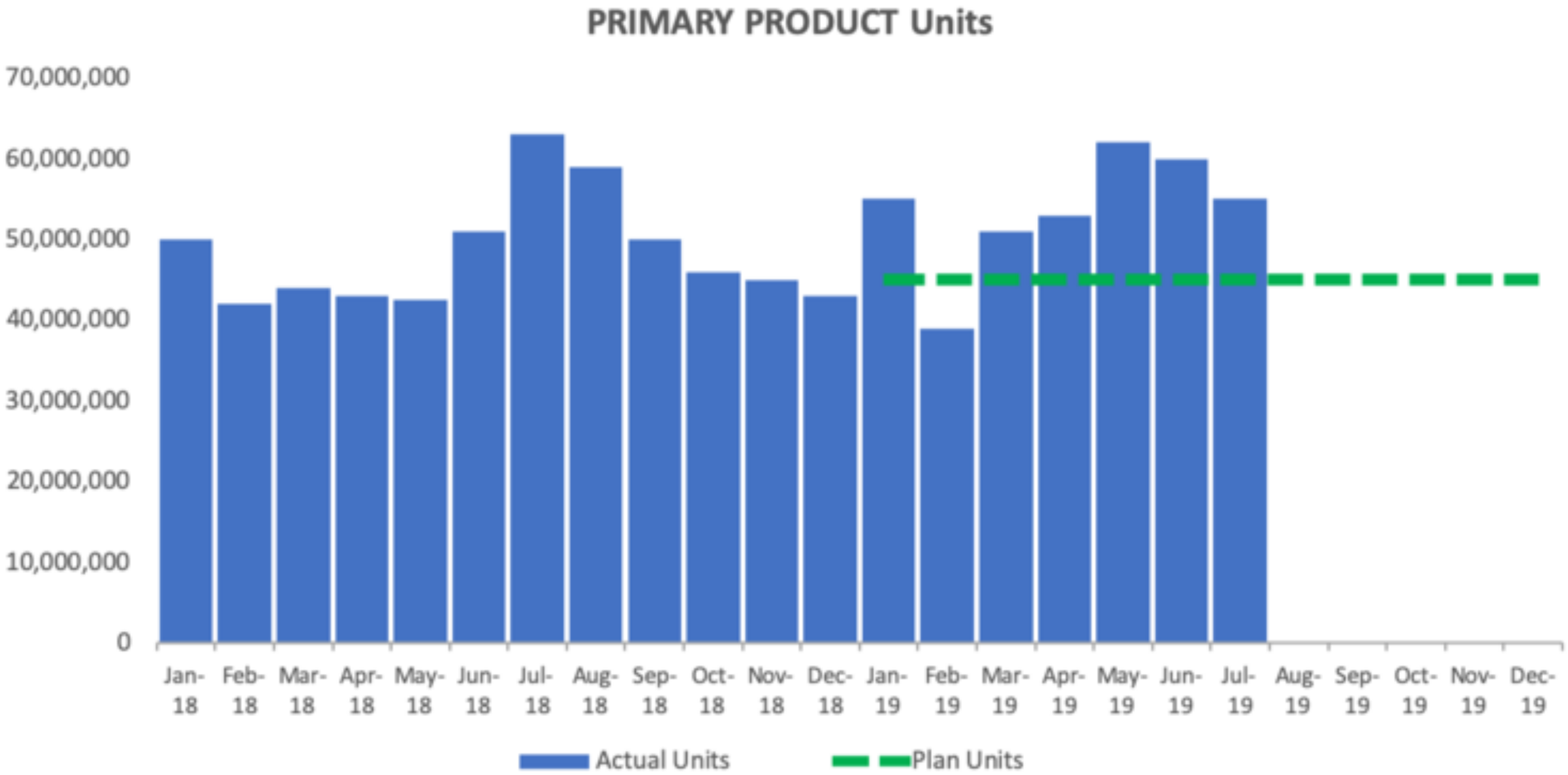
10. Use the blink test.

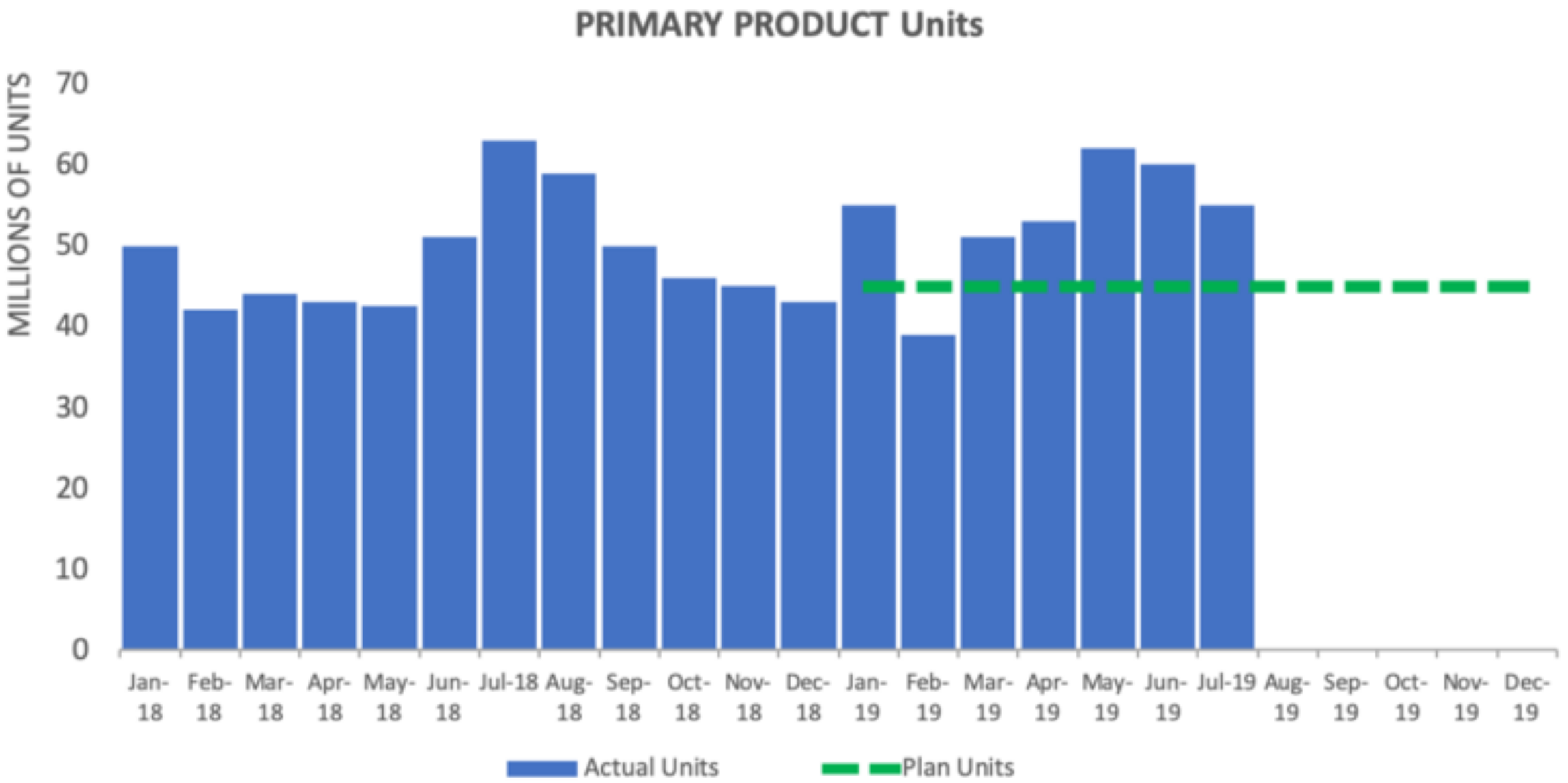


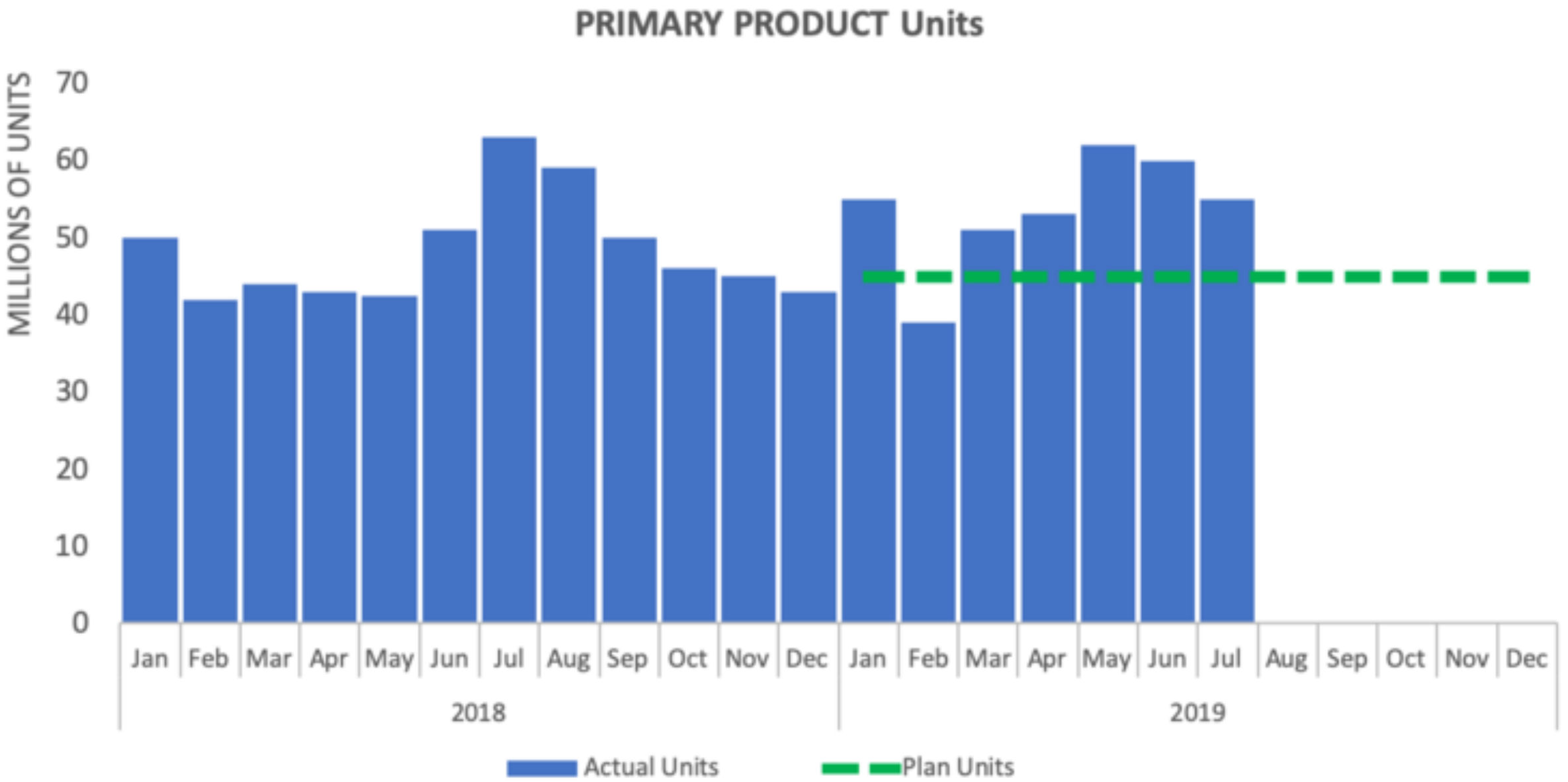
1. **Know your audience!**
   - For public vizzes, this usually requires bar, line, and scatter plots.
   - Basically never use undefined acronyms or symbols.
2. **Question the defaults.** This means fonts, colors, sizes, shapes, and more.
3. **Use descriptive titles:** *tell* your viewer what they should see, and don't be redundant.
4. **Be selective in data and design.** Consider your data-to-ink ratio and remove distractions.
5. **Use color sparingly, intentionally, and consistently.**
   - Keep branding in mind
   - Remember colorblindness
6. **Use reasonable axis tick** formats and frequencies, not just the default.
7. **Consider creative alternatives to legends.**
8. **Don't be afraid of annotations.** Use whatever software you prefer to add them.
9. **Make your text bigger** than you'd think. Nobody likes to squint!
10. **Use the blink test.**

Improve! This! Viz!

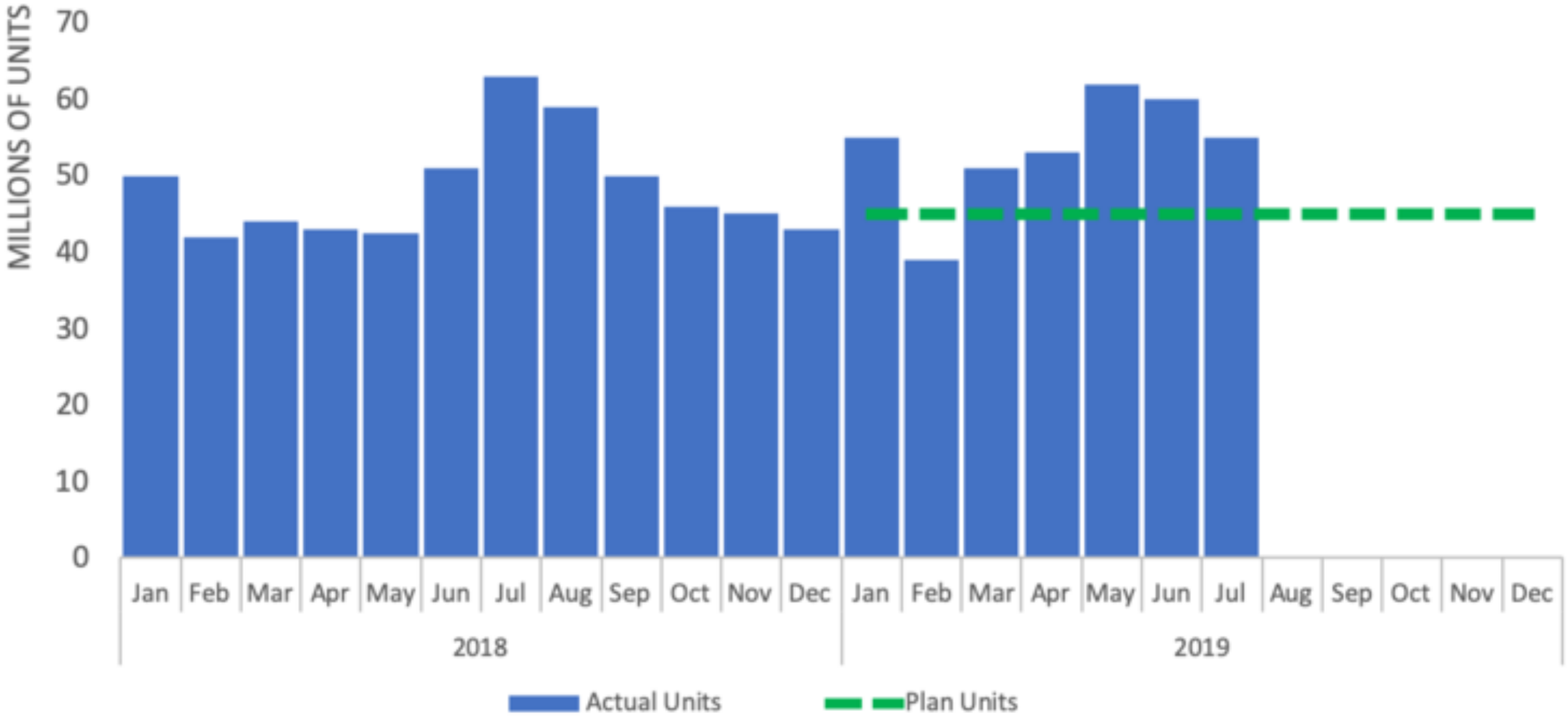

PRIMARY PRODUCT Units
70,000,000
60,000,000
50,000,000
40,000,000
30,000,000
20,000,000
10,000,000
0
Jan-18
Feb-18
Mar-18
Apr-18
May-18
Jun-18
Jul-18
Aug-18
Sep-18
Oct-18
Nov-18
Dec-18
Jan-19
Feb-19
Mar-19
Apr-19
May-19
Jun-19
Jul-19
Aug-19
Sep-19
Oct-19
Nov-19
Dec-19
Actual Units
Plan Units


PRIMARY PRODUCT Units
70,000,000
60,000,000
50,000,000
40,000,000
30,000,000
20,000,000
10,000,000
0
Jan-18
Feb-18
Mar-18
Apr-18
May-18
Jun-18
Jul-18
Aug-18
Sep-18
Oct-18
Nov-18
Dec-18
Jan-19
Feb-19
Mar-19
Apr-19
May-19
Jun-19
Jul-19
Aug-19
Sep-19
Oct-19
Nov-19
Dec-19
Actual Units
Plan Units

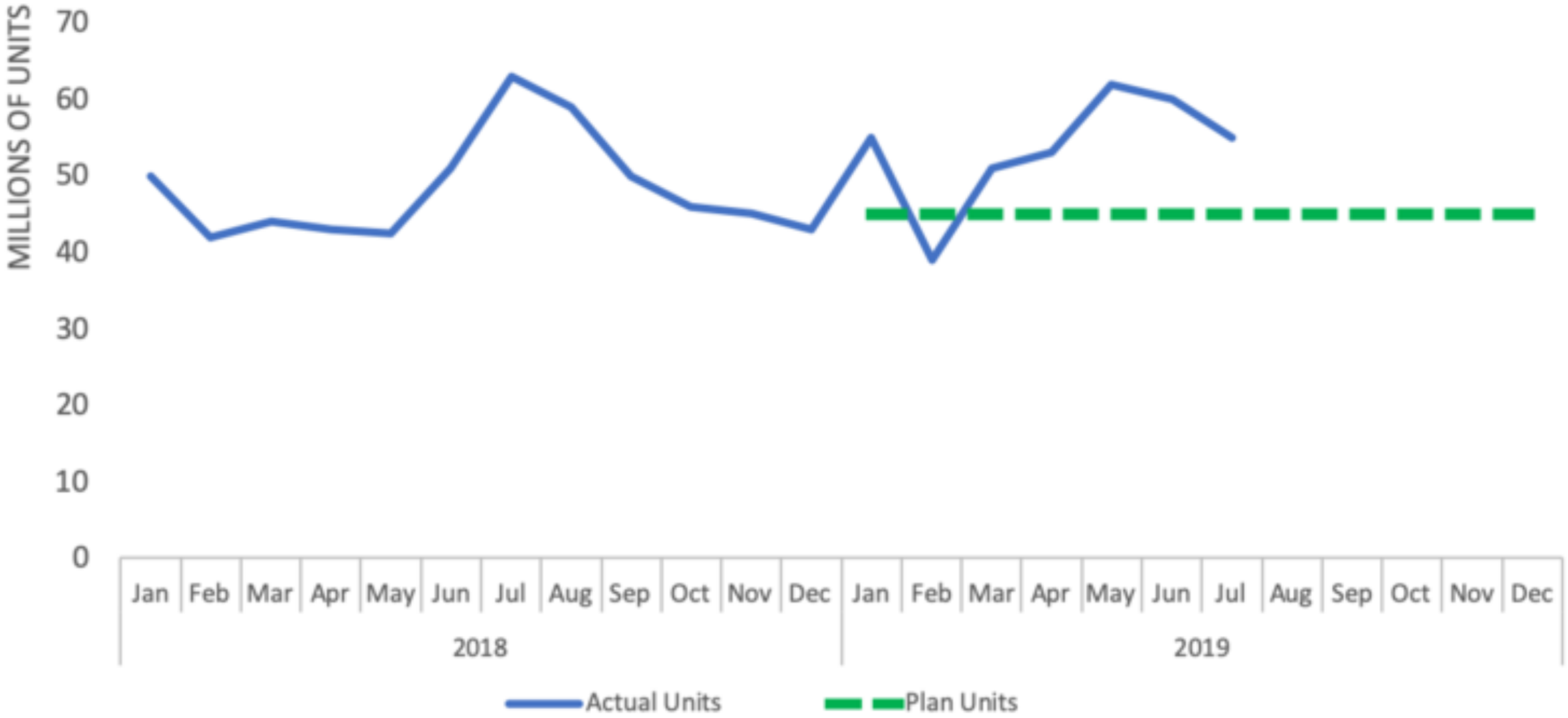

PRIMARY PRODUCT Units
MILLIONS OF UNITS
70
60
50
40
30
20
10
0
Jan-18
Feb-18
Mar-18
Apr-18
May-18
Jun-18
Jul-18
Aug-18
Sep-18
Oct-18
Nov-18
Dec-18
Jan-19
Feb-19
Mar-19
Apr-19
May-19
Jun-19
Jul-19
Aug-19
Sep-19
Oct-19
Nov-19
Dec-19
Actual Units
Plan Units


PRIMARY PRODUCT Units
MILLIONS OF UNITS
70
60
50
40
30
20
10
0
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
2018
2019
Actual Units
Plan Units

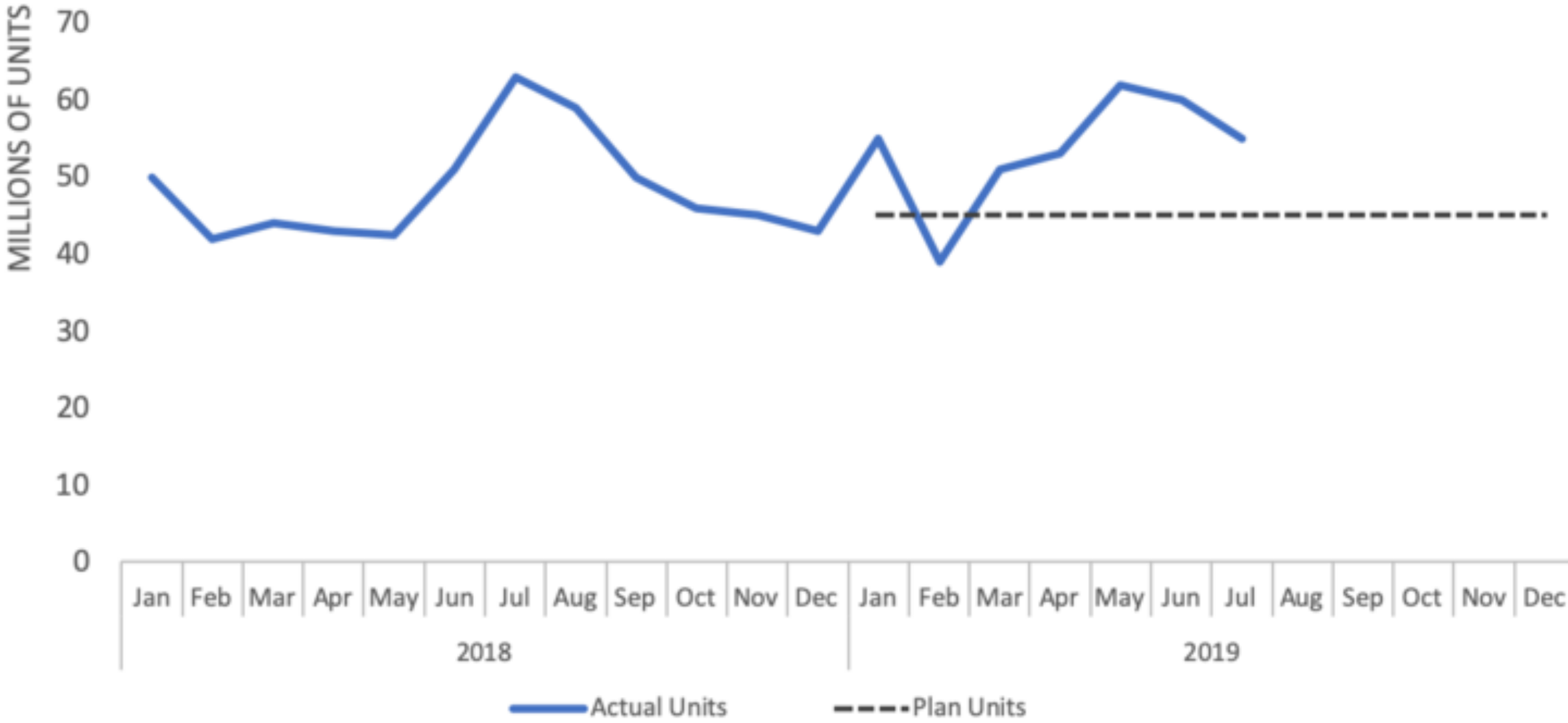
PRIMARY PRODUCT: unit sales



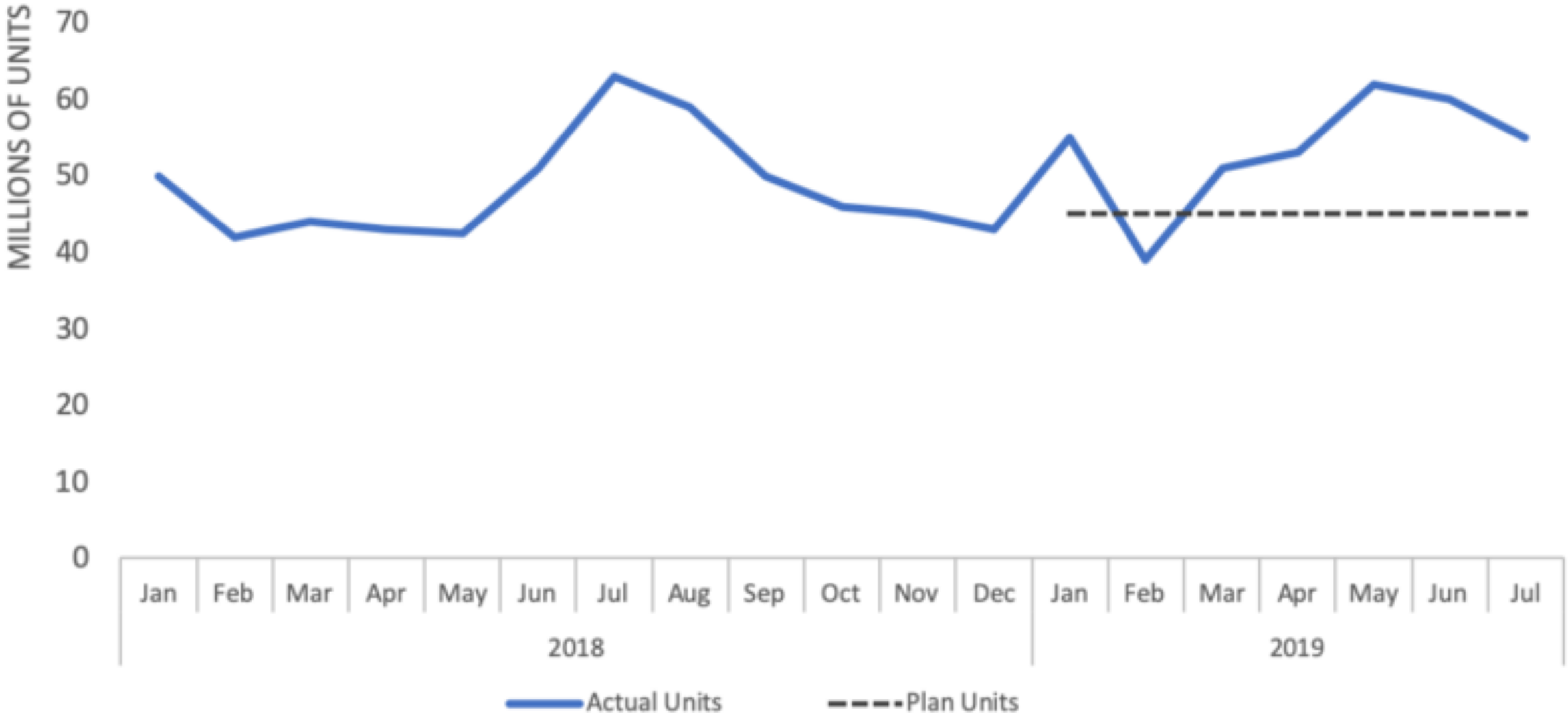
PRIMARY PRODUCT: unit sales



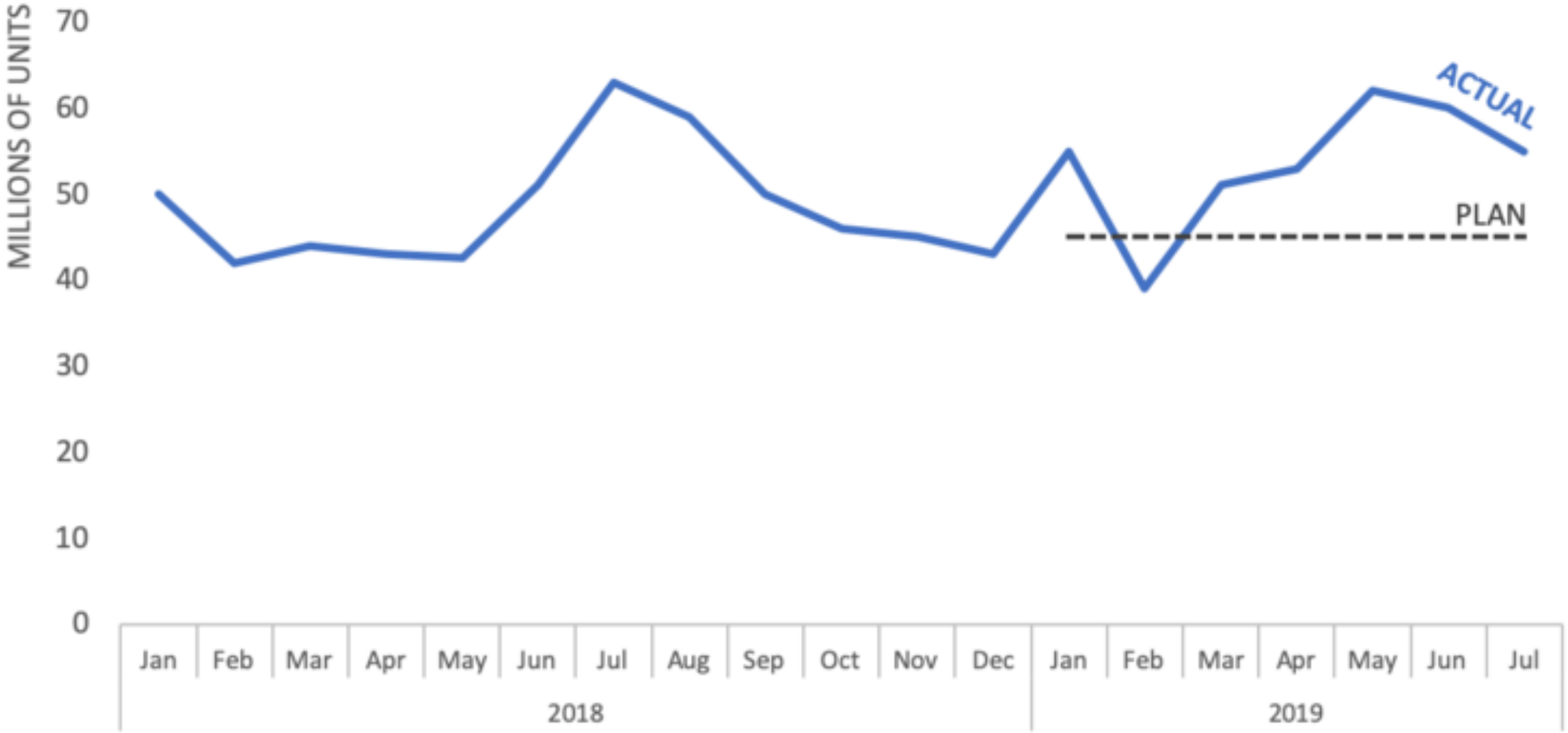
PRIMARY PRODUCT: unit sales



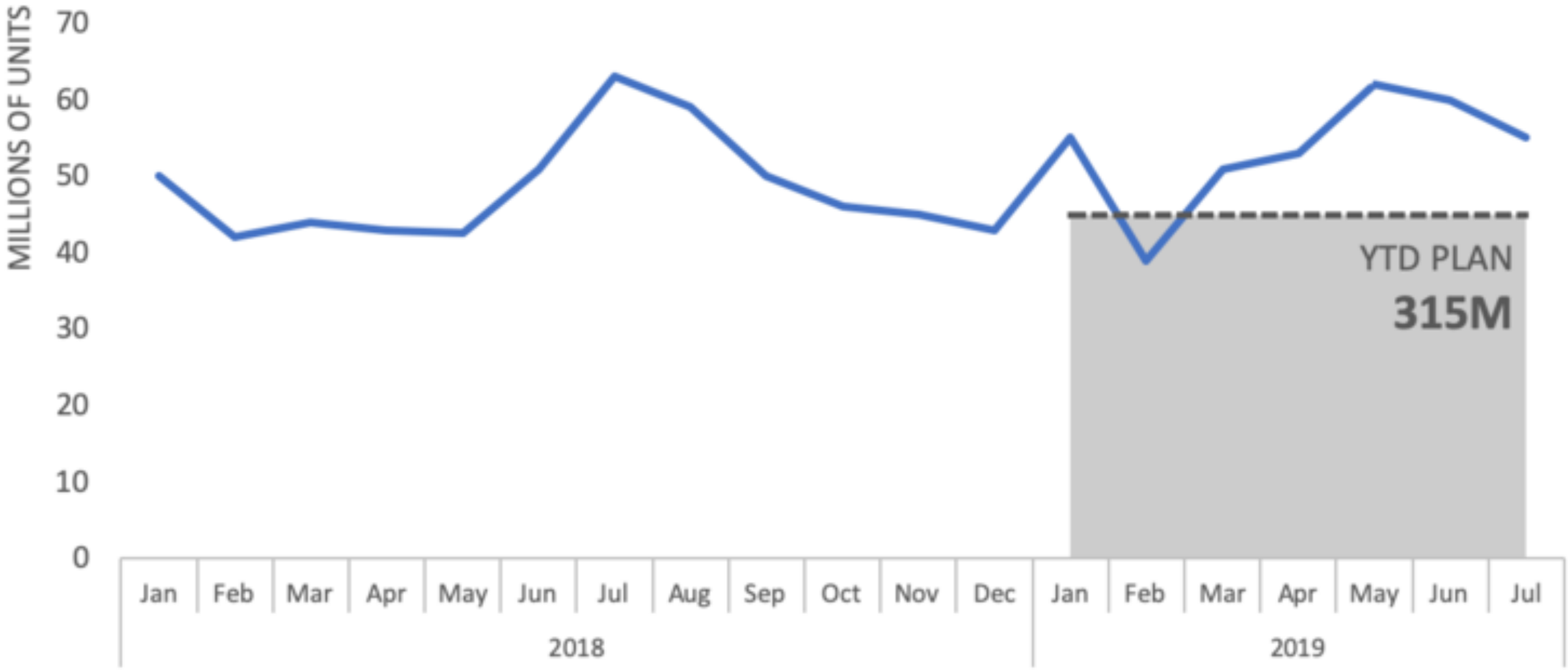
PRIMARY PRODUCT: unit sales



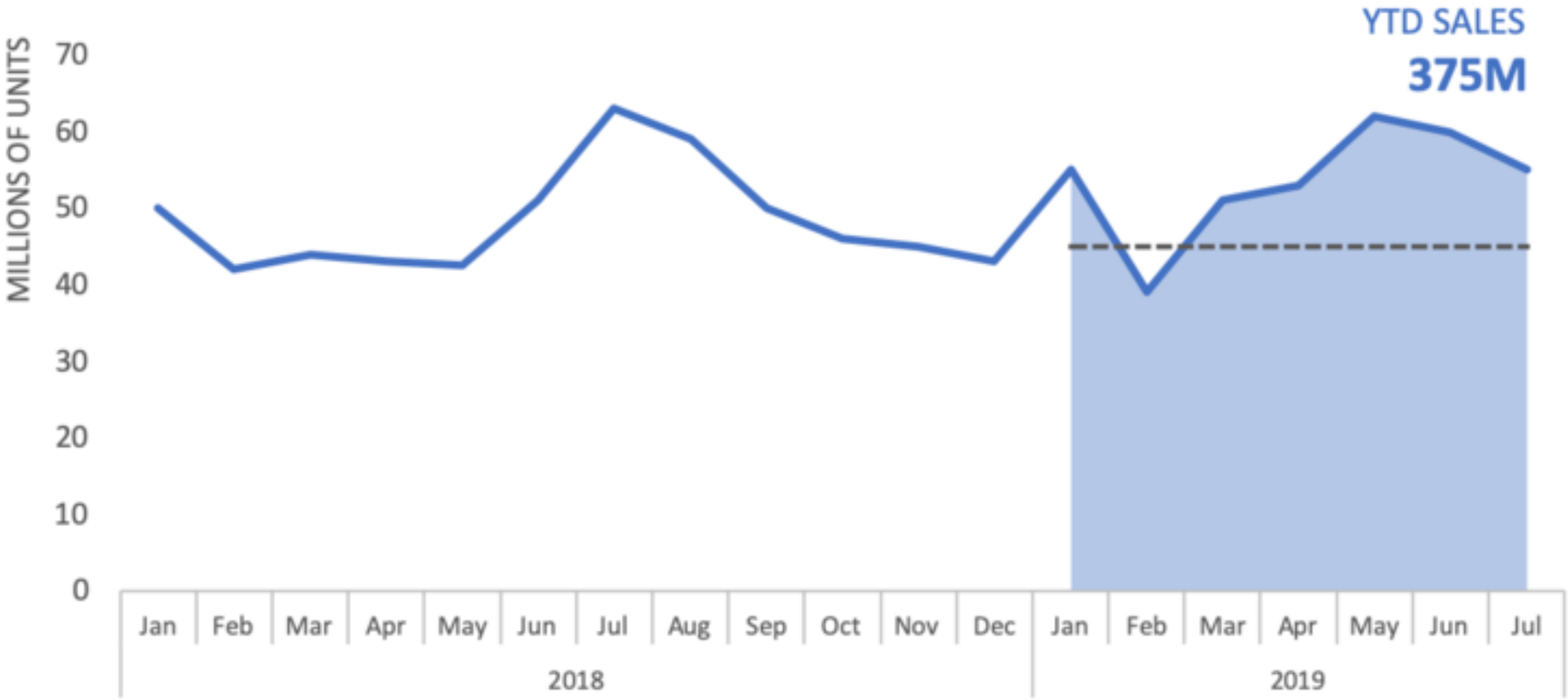
PRIMARY PRODUCT: unit sales



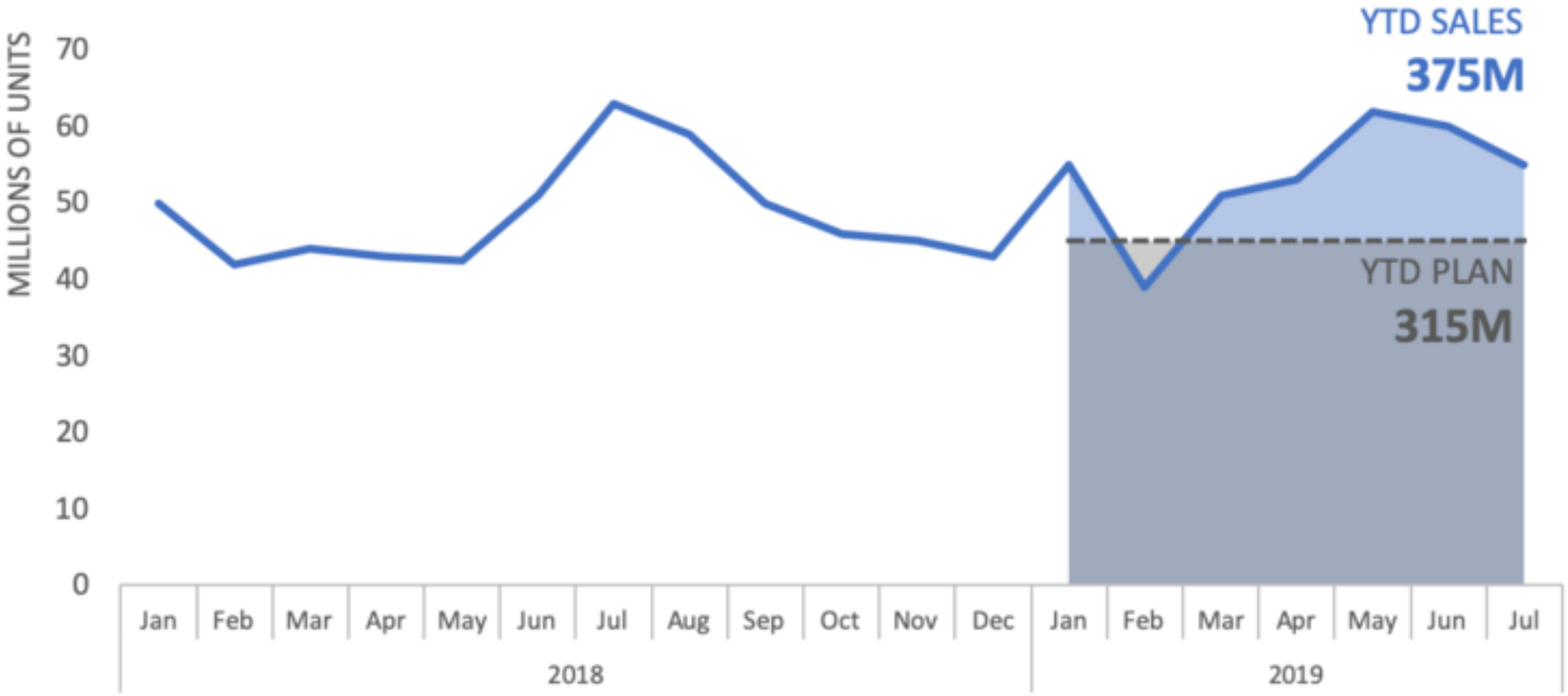
PRIMARY PRODUCT: unit sales



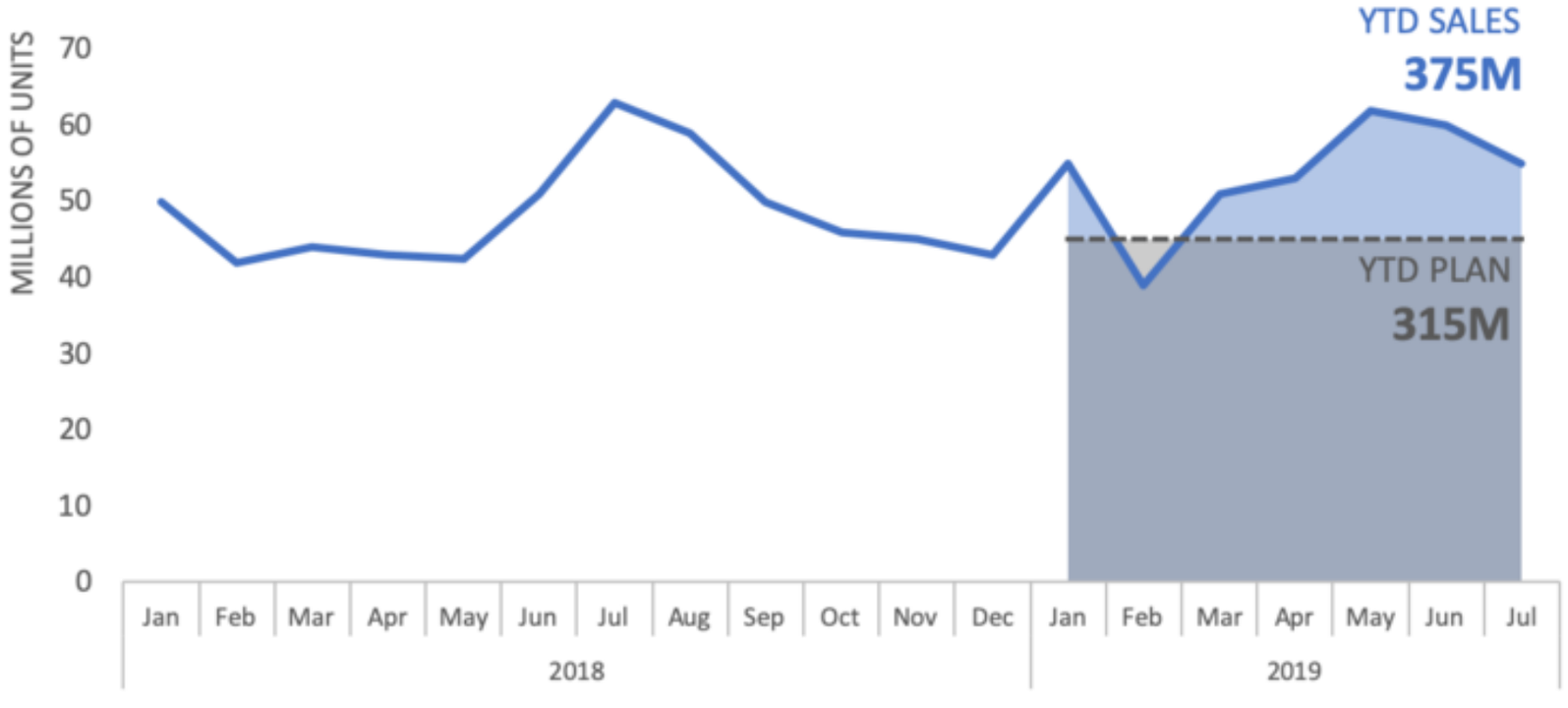
PRIMARY PRODUCT: unit sales

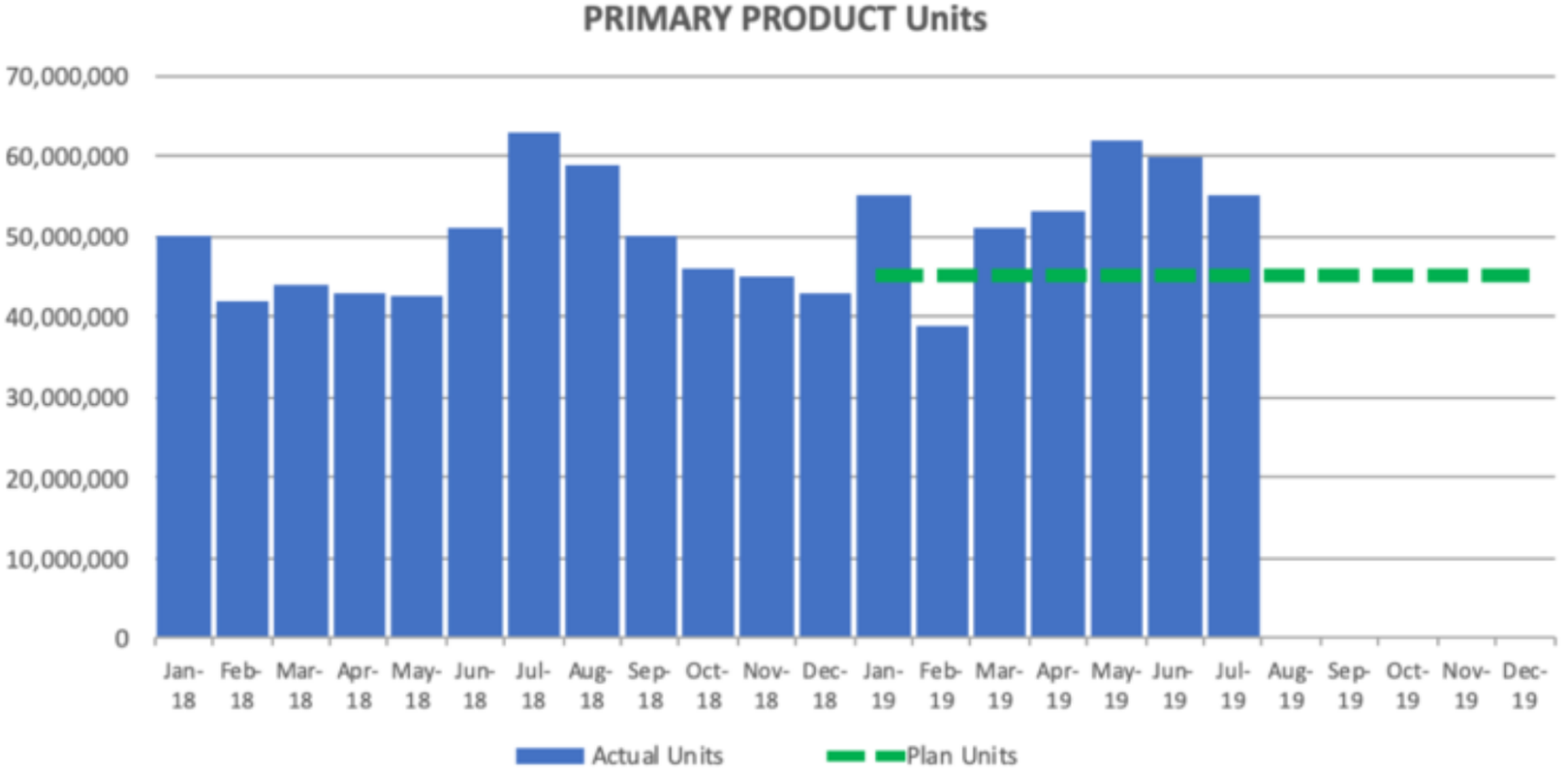


PRIMARY PRODUCT: **unit sales exceed plan by nearly 20% year to date**



PRIMARY PRODUCT: **unit sales exceed plan by nearly 20% year to date**





Group Exercise:

Write the headline

http://tiny.cc/INFO290_headlines

Individual Exercise:

Visualizing tabular data by hand

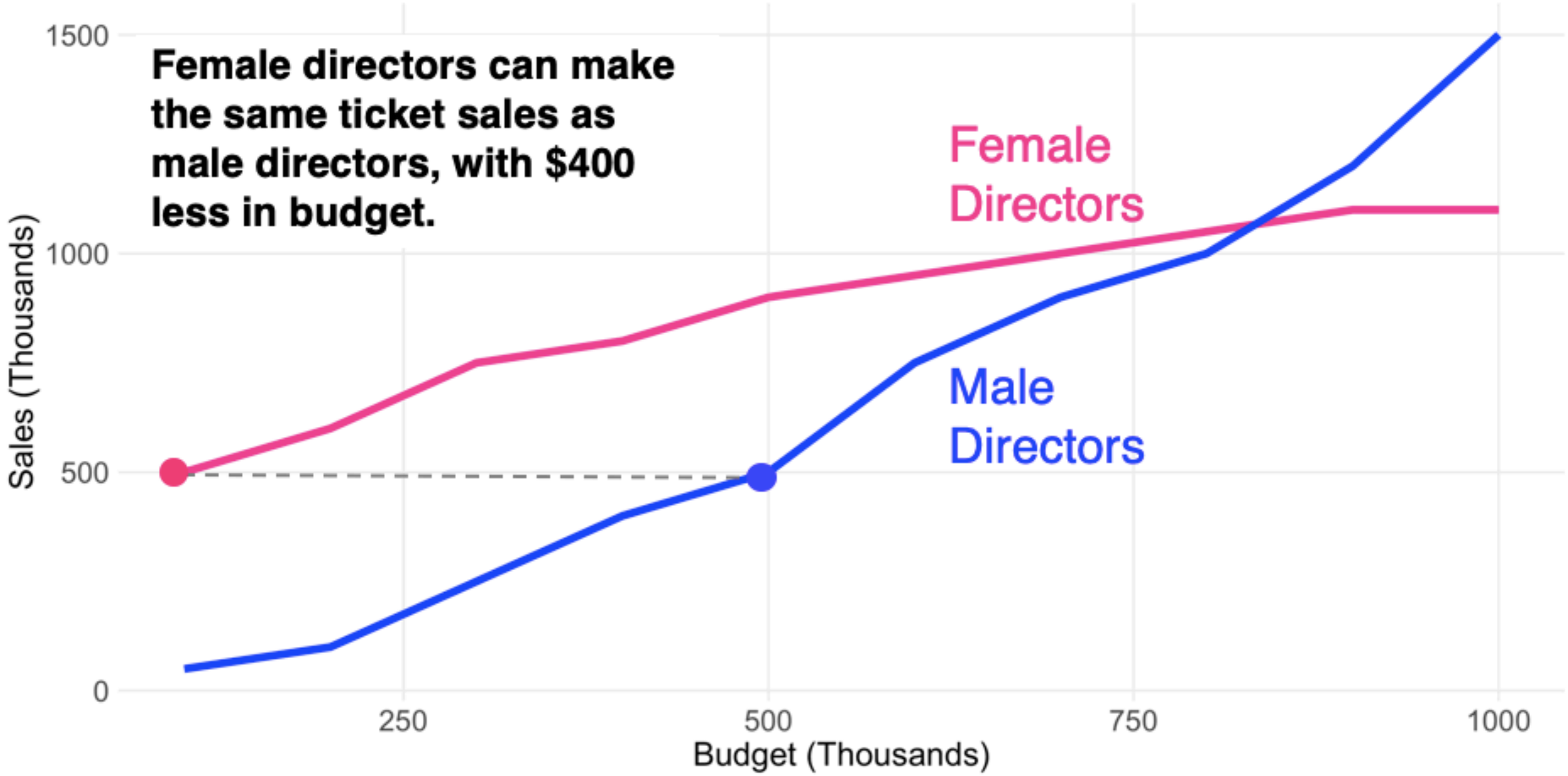
Individual Exercise:

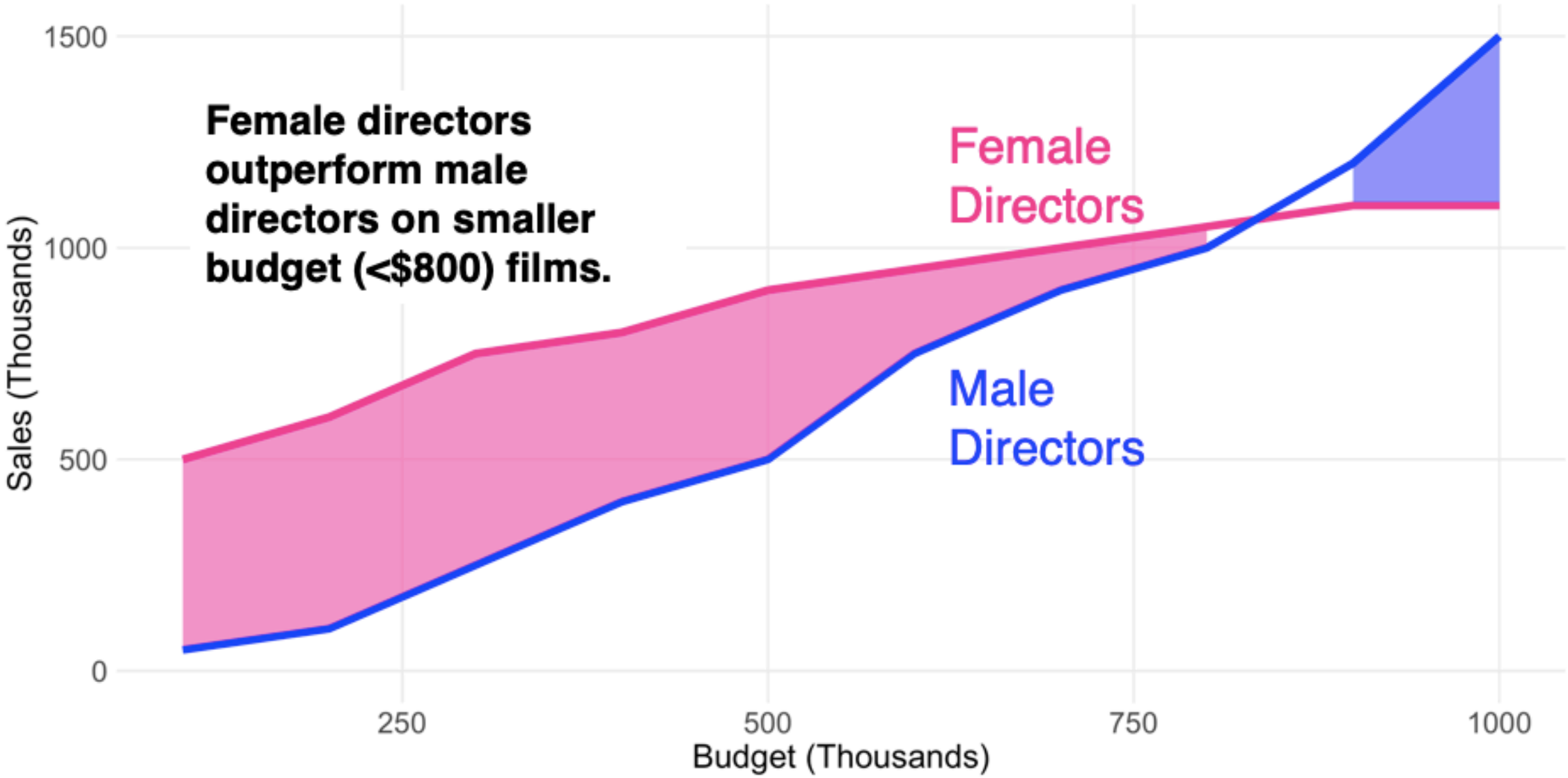
Visualizing tabular data by hand

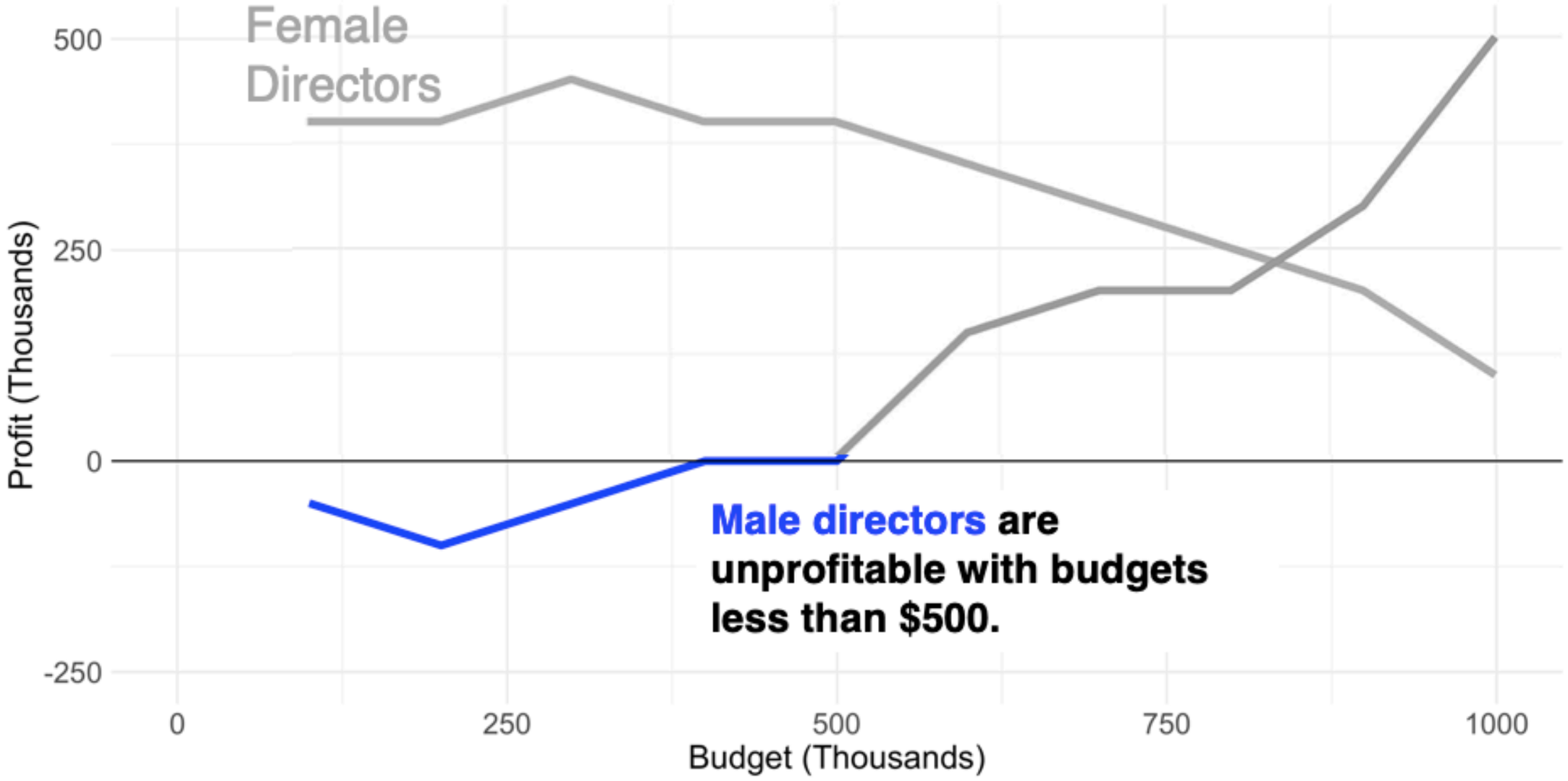
Take 7 minutes to draw, by hand, a visualization of the below data:

Budget (thousands)	Director Gender	Ticket Sales (thousands)
100	Female	500
200	Female	600
300	Female	750
400	Female	800
500	Female	900
600	Female	950
700	Female	1,000
800	Female	1,050
900	Female	1,100
1,000	Female	1,100

Budget (thousands)	Director Gender	Ticket Sales (thousands)
100	Male	50
200	Male	100
300	Male	250
400	Male	400
500	Male	500
600	Male	750
700	Male	900
800	Male	1,000
900	Male	1,200
1,000	Male	1,500


Female directors can make the same ticket sales as male directors, with $400 less in budget.
Female Directors
Male Directors
Sales (Thousands)
1500
1000
500
0
250
500
750
1000
Budget (Thousands)


Female directors outperform male directors on smaller budget (<$800) films.
Female Directors
Male Directors
Sales (Thousands)
1500
1000
500
0
250
500
750
1000
Budget (Thousands)


Female
Directors
Male directors are
unprofitable with budgets
less than $500.
500
250
0
-250
Profit (Thousands)
0
250
500
750
1000
Budget (Thousands)

Going interactive and beyond


+ableau
SOFTWARE


Leaflet


plotly


bokeh

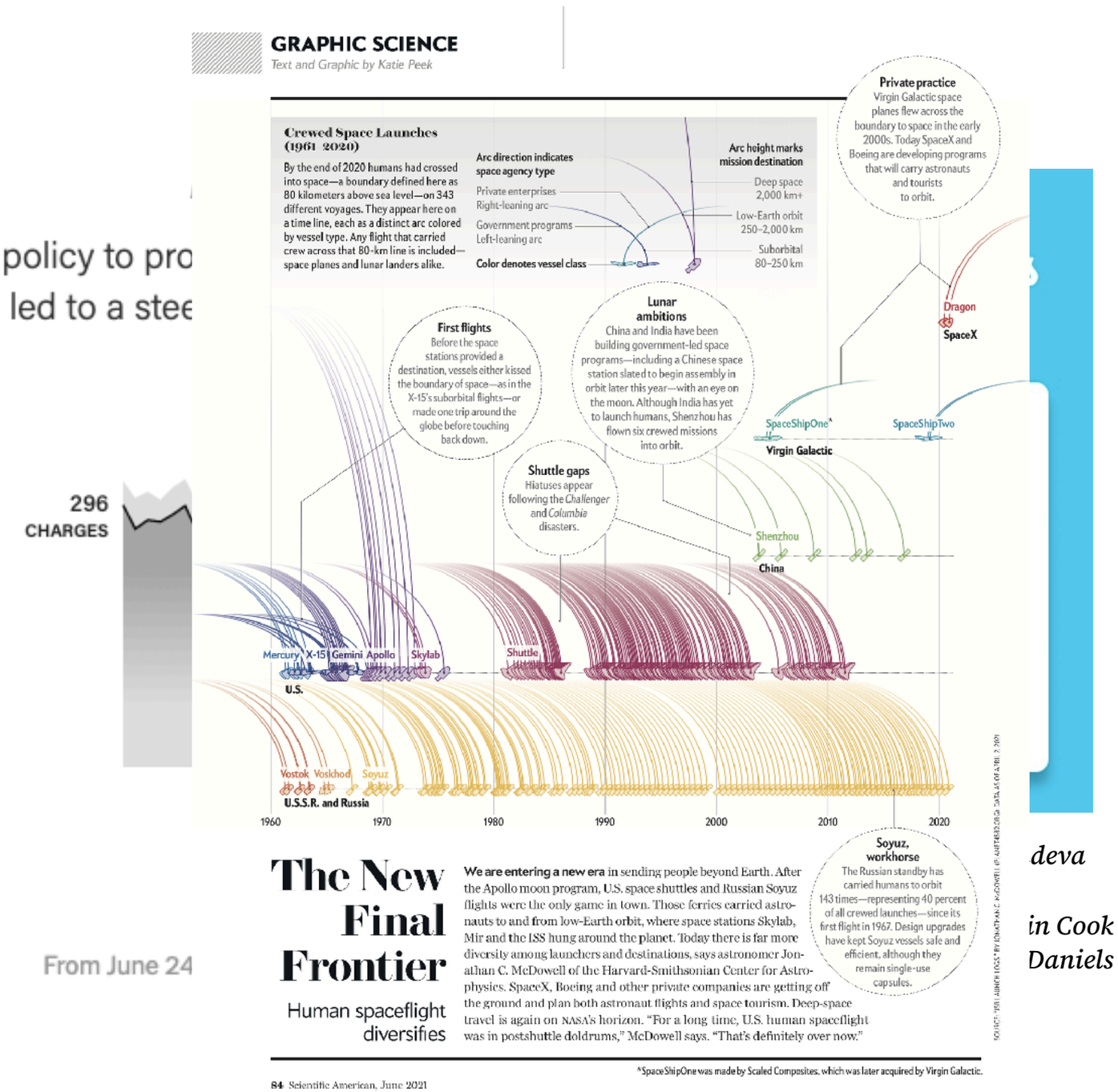


Examples

- Tableau: Massachusetts Trial Court
- Leaflet: Police Violence Happens Here
- Plotly: Drug Prosecution in MA
- Bokeh: COVID-19 in Portugal
- D3: Googling for birds

Where to find viz inspiration

- NYT's *The Upshot*
- *The Pudding*
- Tableau's public gallery
- Graphic science in *Scientific American*
- Awwwards



Foxx's policy to pro
$1,000 led to a stee

296
CHARGES

From June 24

deva

in Cook
Daniels

Helpful resources

- Paul Tol's colorblind-friendly categorical color palettes
- Data Viz Project: catalog of different viz types
- Gina Reynolds' ggplot flipbook
- Kira Tebbe's Effective Data Visualization workshop
- Free online book: Fundamentals of Data Visualization
- *Storytelling with Data* books

Find these resources:

https://github.com/laurenmarietta/effective-data-viz

lauren@ischool.berkeley.edu